

# Exact Rvachev Up-Function Polynomial Synthesis

Dictionaries, Lagrange cardinal loops, Legendre spectral closure,  
operator transmutation, and self-reconstruction

Consolidated frontier volume for Vladimir Reshetnikov's ProveIt project

30 August 2026

## Abstract

Let  $u = \text{up}$  be Rvachev's even  $C^\infty$  probability density supported on  $[-1, 1]$ . This volume consolidates thirteen independently developed frontier reports: three exact-polynomial synthesis packages, six Lagrange–Rvachev cardinal-loop packages, and four Legendre–Rvachev spectral/self-reconstruction packages. Their shared problem is to represent polynomials exactly on a compact interval by finite signed sums of shifted and scaled copies of  $u$ , then close the representation with interpolation or orthogonal analysis. Repeated foundations and equivalent matrix notations are stated once; genuinely different constructions and consequences are proved in a common notation.

*The common-scale dictionary.* For degree  $d$ , the minimal integer scale-to-shift ratio with Strang–Fix reproduction order  $d$  is  $M = 2^d$ . Moment deconvolution  $Q = M_u(MD)^{-1}P$  gives a global cardinal identity; on one cell its  $2M$  active atoms collapse, because the complete dyadic alias nullspace is governed by the  $q$ -integer/cyclotomic annihilator  $A_d(z) = \prod_{m=1}^d [2^m]_z$  of degree  $2M - d - 2$ . The one-cell up-spline space therefore has exact dimension  $d + 2$ ; a proved endpoint basis represents every  $P \in \Pi_d$  by  $d + 2$  common-scale atoms; no fixed dictionary of nontrivial affine up-atoms spanning  $\Pi_d$  can do better; and an output-sensitive algorithm with Bernoulli differential symbol  $H_d(z) = (z/2 \sinh(z/2))^d / M_u(z)$  computes the coefficients in  $\mathcal{O}(d^2)$  exact operations. Polynomiality of an arbitrary local up-spline is decided by a *single scalar*, which after one finite difference becomes a signed Thue–Morse block sum; the one-dimensional nonpolynomial defect has an explicit odd-frequency Fourier residue.

*The scale-ladder windows.* The refinement equation inverts to an antiderivative train  $Ju(x) = \sum_{k \geq 0} u((x - 2k - 1)/2)$ , whose  $m$ -fold iterate has restricted binary partition numbers  $p_m(N)$  as weights – the causal Green sequence of the finite Thue–Morse difference operator. Truncating by a support-edge invariance yields exact polynomial windows  $\Psi_{n,h}$  at scales  $2^{n+1}h$ : with  $h = (b-a)/2$ , every degree uses exactly *two* atoms, so  $\deg \leq d$  costs at most  $2d + 2$  atoms; smaller  $h$  trades atoms for a prescribed support collar. The coefficients are a triangular Appell transform with Bernoulli–Bell closed forms, the block basis is an extended complete Chebyshev system with Vandermonde interpolation determinants, and the construction is an explicit compactly supported  $C^\infty$  extension operator for polynomial data.

*The oversampled lattice.* A third construction detaches the dictionary from its special lattice: the reproduction order of the translate system at radius-to-spacing ratio  $m$  is exactly  $\nu_2(m)$ , the physical scale is free, and an explicit interval algorithm produces, for any prescribed atom radius  $\rho$ , an exact plateau supported in  $[A - 2\rho, B + 2\rho]$  described by only  $\mathcal{O}(d)$  field elements with  $(2m + 1)$ -local evaluation. Its real-space face is a ghost-atom antiderivative calculus and the exact dyadic derivative stencil  $D^d u((x - c)/2^d h) = h^{-d} 2^{-\binom{d}{2}} \sum_k \varepsilon_k u(\dots)$  – Thue–Morse signs, the convolution inverse of the ladder's binary-partition weights.

*Combined results.* The dictionary bound sharpens the ladder count: the minimal number  $N_d$  of up-atoms sufficient for all of  $\Pi_d$  on an interval satisfies  $N_0 = 2$  and  $N_d \leq d + 2$ , with  $N_d = d + 2$  conjectured. And the canonical defect is identified, with an explicit constant, as the first-failure trigonometric series of the shifted dyadic-comb quadrature of the companion volume *Dyadic-Comb Frontiers*: its special values are governed by the spectral Dirichlet values  $D(2r)$  proved there, giving the exact rational evaluations

$$\mathcal{E}_1(0) = \frac{1}{18}, \quad \mathcal{E}_3(0) = -\frac{23}{1350}, \quad \mathcal{E}_5(0) = \frac{1543}{119070}, \quad \mathcal{E}_7(0) = -\frac{1196113}{65063250},$$

two of which independently reconfirm exact phase-error values computed in that volume.

*Lagrange and Legendre closure.* Exact cardinal synthesis produces a canonical right inverse and oblique coefficient projector, a one-dimensional Thue–Morse defect completion, nested detail idempotents, signed-decoder and Lebesgue barriers, Appell–Vandermonde determinant growth, and exact Gaussian-binomial rows. Fourier–Legendre analysis produces finite biorthogonal up-atom blocks, reverse local closure, rational spectral identities, positive energy series, refinement null trains, and exact boundary jets. The deconvolved family  $Q_n^- = M_u(D)^{-1}P_n$  and the distinct smoothed family  $R_n = M_u(D)P_n$  carry dual pullback geometries with coordinates  $X_{\mp} = x \mp K'(D)$ ; their operator recurrences, Christoffel–Darboux kernels, Legendre–Appell connection determinant, central reflected-pair  $q$ -determinant, factorial/Lambert asymptotics, and non-flattening theorem are proved below. Provenance of every absorbed source and the exact-computation boundary are recorded in [section B](#).

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## Provenance and merge conventions

This volume is an editorial consolidation (30 August 2026) of thirteen source packages: the three polynomial-synthesis drafts already merged here, six Lagrange–Rvachev reports, and four Legendre–Rvachev reports. Shared foundations are stated once, each genuinely different construction keeps a coherent chapter, and overlapping Appell calculus, Thue–Morse products, rationality, projector identities, conditioning statements, and Lean roadmaps are deduplicated. [Section B](#) records the absorbed sources with SHA-256 hashes and the deduplication detail; their supporting files are under `assets/`.

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### Status discipline

Theorems, propositions, lemmas, and corollaries have complete proofs here or explicit citations to the repository’s primary exposition. The classical fact that Rvachev’s atomic-function spaces contain polynomials goes back to V. L. and V. A. Rvachev [3, 5]; novelty claims concern the explicit finite constructions, dimensions, algorithms, certificates, and identities below, relative to the audited corpus and a targeted literature search — never unqualified global priority. Conjectures and problems are labeled. Most of the finite dictionaries, local dimensions, algorithms, and atom-count results are not formalized in Lean. The focused 2026-08-30 module `CompositeMes hSharpness.lean` now formalizes only the natural-mesh, universally quantified shifted-comb threshold described below; its claim boundary is recorded after [theorem 3.1](#).

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### Notation

$u$  = up as in the corpus;  $\widehat{f}(\xi) = \int f e^{-2\pi i \xi x} dx$ ;  $\widehat{u}(\xi) = \prod_{j \geq 0} \text{sinc}_\pi(\xi/2^j)$  with  $\text{ord}_{\xi=n} \widehat{u} = 1 + \nu_2(n)$  at nonzero integers  $n$ . The normalized Fabius function on the unit interval is  $F_{\text{bd}}(t) := u(1 - t)$  for  $0 \leq t \leq 1$ ; hence  $u(x) = F_{\text{bd}}(1 - |x|)$  on  $[-1, 1]$ . Where an endpoint estimate evaluates  $F_{\text{bd}}(t)$  for  $t \geq 1$ , we use the explicit constant-extension convention  $F_{\text{bd}}(t) = 1$ . The probability model is  $Z = \sum_{j \geq 1} 2^{-j} V_j$ ,  $V_j$  i.i.d. uniform on  $[-1, 1]$ , with density  $u$ , moment generating function  $M_u(z) = \mathbb{E} e^{zZ} = \prod_{j \geq 1} \frac{\sinh(z/2^j)}{z/2^j}$ , even moments  $\mu_{2r}$  ( $\mu_2 = \frac{1}{9}$ ,  $\mu_4 = \frac{19}{675}$ ,  $\mu_6 = \frac{583}{59535}$ , ...), and cumulants  $\kappa_{2m} = B_{2m}/(2m(1 - 4^{-m}))$ . Two dual Appell families recur:

$$\begin{aligned} G_n(y) &:= \mathbb{E} (y - Z)^n = M_u(D) y^n = \sum_r \binom{n}{r} \mu_r y^{n-r}, \\ \mathcal{A}_n(y) &:= M_u(D)^{-1} y^n = \sum_r \binom{n}{r} \gamma_r y^{n-r}, \end{aligned} \tag{0.1}$$

the *moment family*  $G_n$  (the same polynomials as the Appell kernel of the companion volume’s dyadic evaluator) and the *deconvolution family*  $\mathcal{A}_n$  (the corpus’s Rvachev–Appell polynomials), where  $1/M_u(z) = \sum_r \gamma_r z^r / r!$  with

$$\gamma_2 = -\frac{1}{9}, \quad \gamma_4 = \frac{31}{675}, \quad \gamma_6 = -\frac{2347}{59535}, \quad \gamma_8 = \frac{1904369}{32531625}, \quad \gamma_{10} = -\frac{3309760579}{24405225075}, \tag{0.2}$$

odd coefficients zero, and  $\gamma_r = Y_r(-\kappa_1, \dots, -\kappa_r)$  (complete Bell polynomials in the negated cumulants). Thue–Morse signs are  $\varepsilon_r = (-1)^{\text{wt}_2(r)}$ , the finite product is  $\Theta_m(z) = \prod_{j < m} (1 - z^{2^j}) = \sum_{r < 2^m} \varepsilon_r z^r$ , and  $p_m(N)$  denotes the restricted binary partition numbers (2.3).  $\Pi_d$  is the space of real polynomials of degree at most  $d$ , and  $[q]_z = (1 - z^q)/(1 - z)$  the  $q$ -integer.

# 1 The synthesis problem and its boundary conditions

Given  $p \in \mathbb{R}[x]$  of degree at most  $d$  and a nondegenerate interval  $[a, b]$ , we seek a *finite* functional identity

$$p(x) = \sum_{r=1}^N c_r u\left(\frac{x - \tau_r}{\sigma_r}\right), \quad x \in [a, b], \quad (1.1)$$

with real amplitudes and shifts and positive scales. Three structural facts frame every construction.

(i) *Globality is impossible; locality is the right question.* Every finite sum (1.1) is compactly supported, so no nonzero polynomial admits a global identity on  $\mathbb{R}$ ; the achievable object is an exact *window* – polynomial on  $[a, b]$ , smoothly decaying to zero outside a finite support.

(ii) *Signed amplitudes are essential, and two atoms are needed even for constants.* Negative target polynomials force signed amplitudes in the unrestricted problem because every up-atom is nonnegative. Independently, by the refinement derivative  $u'(x) = 2u(2x+1) - 2u(2x-1)$ , the function  $u$  is strictly increasing on  $(-1, 0)$  and strictly decreasing on  $(0, 1)$ ; hence a single atom is never constant on a nondegenerate interval, and  $N \geq 2$  already for  $p \equiv 1$ . The degree-zero solution is the two-shift partition of unity

$$u\left(\frac{x-a}{L}\right) + u\left(\frac{x-b}{L}\right) = 1, \quad a \leq x \leq b, \quad L = b - a. \quad (1.2)$$

(iii) *One fixed scale cannot reproduce high degree globally.* Under the corpus convention the dual-lattice zeros of a single integer-shift system generated by  $u$  have minimum multiplicity one (at odd frequencies), so its global Strang–Fix reproduction order is zero beyond constants [8, 10]: any exact construction must either enlarge the scale-to-shift ratio (the dictionary route of section 3) or use several scales and boundary truncation (the ladder route of section 5).

Within the corpus, the nonduplication boundary is sharp: the audited inverse/sampling volume proves the *opposite orientation*

$$x^n = 2^{-N} \sum_{k \in \mathbb{Z}} \mathcal{A}_n\left(x - \frac{k + \theta}{2^N}\right) u\left(\frac{k + \theta}{2^N}\right) \quad (n \leq N), \quad (1.3)$$

where the translated functions are *polynomials* and the up-values are scalar weights. Here the translated and dilated copies of  $u$  are the functions and polynomial data become their amplitudes. Two other imported facts do heavy work below: the exact integer zero orders of  $\hat{u}$ , and the corpus’s non-elementarity theorem, which implies that no nonzero affine copy of  $u$  agrees with a polynomial on any open subinterval of the interior of its support – the fact that converts dimension *bounds* into dimension *equalities*.

## 2 Shared calculus: Appell pairs, Thue–Morse products, and binary partitions

### 2.1 The Appell pair

The two families (0.1) are formal inverses:  $M_u(D)\mathcal{A}_n = y^n$  and  $M_u(D)^{-1}G_n = y^n$ , both operators triangular with unit diagonal on polynomials. Their generating functions are  $\sum_n G_n(y)t^n/n! = e^{yt}M_u(t)$  and  $\sum_n \mathcal{A}_n(y)t^n/n! = e^{yt}/M_u(t)$ ; both satisfy the Appell derivative ladder  $G'_n = nG_{n-1}$ ,  $\mathcal{A}'_n = n\mathcal{A}_{n-1}$ ; and their coefficient sequences  $(\mu_r)$ ,  $(\gamma_r)$  are mutually inverse under binomial convolution, with

$$\gamma_0 = 1, \quad \gamma_n = - \sum_{r=1}^n \binom{n-1}{r-1} \kappa_r \gamma_{n-r}, \quad \mu_n = \sum_{r=1}^n \binom{n-1}{r-1} \kappa_r \mu_{n-r}. \quad (2.1)$$

The first moment family members are  $G_0 = 1$ ,  $G_1 = y$ ,  $G_2 = y^2 + \frac{1}{9}$ ,  $G_3 = y^3 + \frac{1}{3}y$ ,  $G_4 = y^4 + \frac{2}{3}y^2 + \frac{19}{675}$ ,  $G_5 = y^5 + \frac{10}{9}y^3 + \frac{19}{135}y$ .

## 2.2 Thue–Morse products, $q$ -integers, and their Green sequences

Three finite dyadic products organize everything discrete in this volume:

$$\Theta_m(z) = \prod_{j=0}^{m-1} (1 - z^{2^j}), \quad A_d(z) = \prod_{m=1}^d [2^m]_z = \prod_{m=1}^d \frac{1 - z^{2^m}}{1 - z}, \quad \frac{1}{\Theta_m(z)} = \sum_{N \geq 0} p_m(N) z^N. \quad (2.2)$$

**Definition 2.1** (Restricted binary partitions).  $p_m(N)$  counts representations  $N = \sum_{j=0}^{m-1} 2^j k_j$  with  $k_j \in \mathbb{N}_0$ ; equivalently (2.2) holds, and

$$p_m(N) = p_{m-1}(N) + p_m(N - 2^{m-1}), \quad p_m(N < 0) = 0, \quad p_m(0) = p_m(1) = 1. \quad (2.3)$$

**Proposition 2.2** (Green pair and quasipolynomial structure). *The partition sequence is the causal convolution inverse of the signed Thue–Morse block:  $\sum_r \varepsilon_r p_m(N-r) = \delta_{N0}$ , i.e.  $[\prod_{j < m} (I - S^{2^j})] p_m = \delta_0$  for the backward shift  $S$ . Moreover  $p_m(N)$  is a quasipolynomial in  $N$  of degree  $m-1$  and period dividing  $2^{m-1}$ , with residue-independent leading term*

$$p_m(N) = \frac{N^{m-1}}{(m-1)! 2^{m(m-1)/2}} + \mathcal{O}(N^{m-2}); \quad (2.4)$$

hence the normalized weights  $w_{n,N} = n! 2^{n(n+1)/2} p_{n+1}(N)$  are monic degree- $n$  quasipolynomials,  $w_{n,N} = N^n + \mathcal{O}(N^{n-1})$ .

*Proof.* The Green identity is  $\Theta_m(z) \cdot \Theta_m(z)^{-1} = 1$  read coefficientwise. The generating function is rational with all poles at  $2^{m-1}$ -th roots of unity, giving the quasipolynomial form; the pole at  $z = 1$  has order  $m$  with  $\Theta_m(z) = 2^{m(m-1)/2} (1-z)^m (1 + \mathcal{O}(1-z))$ , whose  $(1-z)^{-m}$  coefficient  $\binom{N+m-1}{m-1}$  gives (2.4).  $\square$

The annihilator  $A_d$  is the mirror object: where  $\Theta^{-1}$  integrates  $m$  dyadic differences (the ladder construction),  $A_d$  applies them after removing the pole (the dictionary construction). Its structure theorem:

**Theorem 2.3** ( $q$ -integer and cyclotomic factorization). *With  $M = 2^d$  and  $\omega = e^{2\pi i/M}$ ,*

$$A_d(z) = \prod_{j=1}^{M-1} (z - \omega^j)^{\nu_2(j)+1} = \prod_{\ell=1}^d (z^{2^{\ell-1}} + 1)^{d-\ell+1}, \quad (2.5)$$

monic, palindromic, with positive integer coefficients,  $\deg A_d = 2M - d - 2$ , and  $A_d(1) = 2^{d(d+1)/2}$ . Writing  $A_d(z) = \sum_{\ell} a_{\ell} z^{\ell}$ , the coefficient  $a_{\ell}$  counts tuples  $(r_1, \dots, r_d)$  with  $0 \leq r_m < 2^m$  and  $\sum r_m = \ell$ ; in particular  $(a_{\ell})$  is symmetric, unimodal, and log-concave, and  $a_{\ell}/A_d(1)$  is the law of a sum  $L = \sum_{m=1}^d U_m$  of independent  $U_m$  uniform on  $\{0, \dots, 2^m - 1\}$ .

*Proof.* The roots of  $z^{2^{\ell-1}} + 1$  are the  $\omega^j$  with  $\nu_2(j) = d - \ell$ , required with multiplicity  $d - \ell + 1$ ; since  $[2^m]_z = \prod_{\ell \leq m} (z^{2^{\ell-1}} + 1)$ , the multiplicities telescope to (2.5). Degree and mass follow by summing  $2^m - 1$  and multiplying  $2^m$ ; the coefficient statement is the expansion of the product of uniform blocks, each log-concave, and convolution preserves log-concavity.  $\square$

Finally, the two products meet in one factorization that will become the polynomiality certificate:

$$A_d(z) (z - 1)^{d+1} = (-1)^{d+1} (1 - z) \Theta_d(z^2) = (-1)^{d+1} (1 - z) \sum_{r=0}^{M-1} \varepsilon_r z^{2r}. \quad (2.6)$$

### 3 The common-scale dictionary

The first construction fixes *one* dyadic scale and lets the alias arithmetic of the sinc product do all the work. Normalize atoms for  $\rho > 0$  by

$$\psi_{\rho,k}(t) = \frac{1}{\rho} u\left(\frac{t-k}{\rho}\right), \quad \widehat{\psi_{\rho,k}}(\xi) = e^{-2\pi i k \xi} \widehat{u}(\rho \xi), \quad (3.1)$$

supported on  $[k - \rho, k + \rho]$  with unit integral.

#### 3.1 Why the minimal cardinal ratio is $2^d$

**Theorem 3.1** (Scale classification). *The integer translates  $\{\psi_{\rho,k}\}_{k \in \mathbb{Z}}$  form a partition of unity iff  $\rho \in \mathbb{N}_{>0}$ ; for positive integer  $\rho$  the Strang–Fix polynomial reproduction order of the system is exactly  $\nu_2(\rho)$ . Hence the smallest ratio reproducing all of  $\Pi_d$  is  $\rho = 2^d$ , and odd oversampling ( $\rho = 2^d q$ ,  $q$  odd) buys no order.*

*Proof.* The  $n$ -th Fourier coefficient of the periodized sum is  $\widehat{u}(\rho n)$ , which vanishes for all  $n \neq 0$  iff  $\rho n$  is always a nonzero integer, i.e.  $\rho \in \mathbb{N}_{>0}$ . For integer  $\rho$  and  $n \neq 0$ ,  $\text{ord}_{\xi=n} \widehat{u}(\rho \cdot) = 1 + \nu_2(\rho) + \nu_2(n)$ , minimal at odd  $n$ ; the classical Strang–Fix equivalence [8, 9] gives order  $\nu_2(\rho)$ .  $\square$

The exact Lean anchor for the natural-mesh universal slice is `rvachevCombExactThrough in CompositeMeshSharpness.lean`. The focused build exports that one definition and the seven theorems `exists_shift_tsum_shifted_monomial_ne_integral_nat_real`, `rvachevCombExactThrough_iff_padicValNat`, `rvachevCombExactThrough_iff_pow_two_dvd`, `rvachevCombExactThrough_two_pow`, `two_pow_le_of_rvachevCombExactThrough`, `isLeast_rvachevCombExactThrough`, and `isLeast_rvachevCombExactThrough_even`. They prove, for nonzero natural  $M$ , that shifted-comb exactness at every real shift for every real polynomial of degree at most  $d$  is equivalent to  $d \leq \nu_2(M)$ , equivalently  $2^d \mid M$ ; the monomial of degree  $\nu_2(M) + 1$  fails at some shift; and  $2^d$  is the least such universal mesh, with  $4^N$  least for the complete space through degree  $2N$ . This does *not* formalize the full theorem above for arbitrary real  $\rho$ , the Strang–Fix equivalence as such, an individual target polynomial, a Legendre mode or partial sum, the local dictionary dimension, or any atom-count minimality claim.

Fix henceforth  $M = 2^d$  and  $\psi_k = \psi_{M,k}$ .

#### 3.2 Global synthesis by moment deconvolution

**Lemma 3.2** (Polynomial-coefficient Poisson collapse). *For every  $Q \in \Pi_d$ ,*

$$\sum_{k \in \mathbb{Z}} Q(k) \psi_k(t) = \int_{-1}^1 Q(t - My) u(y) dy = M_u(MD) Q(t), \quad (3.2)$$

*a locally finite sum.*

*Proof.* Apply Poisson summation to  $x \mapsto Q(x) \psi_0(t - x + \cdot)$  – i.e. to  $f_t(x) = Q(x) \frac{1}{M} u((t - x)/M)$ , which is  $C_c^\infty$ . The zero-frequency term is the stated average. Each nonzero Fourier sample  $\widehat{f_t}(n)$  is a combination of derivatives of  $\widehat{u}(M\xi)$  through order  $d$  at  $\xi = n$ , and  $\text{ord} = 1 + \nu_2(Mn) = d + 1 + \nu_2(n) > d$ , so every alias vanishes. Evenness of  $u$  turns the average into  $M_u(MD)Q$ .  $\square$

**Theorem 3.3** (Global exact up-atom synthesis). *For  $P \in \Pi_d$  let*

$$Q = M_u(MD)^{-1} P = P - \frac{M^2}{18} P'' + \frac{31M^4}{16200} P^{(4)} - \frac{2347M^6}{42865200} P^{(6)} + \cdots = \sum_{n=0}^d \gamma_n M^n \frac{P^{(n)}}{n!}, \quad (3.3)$$



a terminating series. Then

$$\boxed{P(t) = \frac{1}{M} \sum_{k \in \mathbb{Z}} Q(k) u\left(\frac{t-k}{M}\right), \quad t \in \mathbb{R},} \quad (3.4)$$

and for  $P(t) = t^n$  the coefficient polynomial is the scaled deconvolution Appell polynomial  $Q(t) = M^n \mathcal{A}_n(t/M)$ .

*Proof.* On  $\Pi_d$  the constant-coefficient operator  $M_u(MD)$  is unipotent triangular in degree, so its inverse is the displayed terminating series. Apply lemma 3.2 to this  $Q$ :

$$\frac{1}{M} \sum_{k \in \mathbb{Z}} Q(k) u\left(\frac{t-k}{M}\right) = M_u(MD)Q(t) = P(t).$$

For  $P(t) = t^n$ , the defining Appell identity  $M_u(D)\mathcal{A}_n = t^n$  and the change of variable  $t \mapsto t/M$  give  $M_u(MD)(M^n \mathcal{A}_n(t/M)) = t^n$ , which proves the final assertion by uniqueness of the inverse image.  $\square$

Restricted to one normalized cell  $[0, 1]$ , only the  $2M$  centers  $-M+1 \leq k \leq M$  act. A first compression is the *partition pair* identity  $\psi_{r-M} + \psi_r = \frac{1}{M}$  on  $[0, 1]$  ( $1 \leq r \leq M$ ), which folds the  $2M$  atoms to  $M+1$ ; but the true dimension is exponentially smaller.

### 3.3 The alias nullspace and the local dimension theorem

**Theorem 3.4** (Complete dyadic alias family). *Let  $\omega = e^{2\pi i/M}$ . For every  $1 \leq j < M$  and  $0 \leq s \leq \nu_2(j)$ ,*

$$\sum_{k \in \mathbb{Z}} k^s \omega^{jk} \psi_k(t) = 0 \quad (t \in \mathbb{R}), \quad (3.5)$$

and these  $N_d^{\text{rel}} := \sum_{j=1}^{M-1} (\nu_2(j) + 1) = 2M - d - 2$  relations are linearly independent on any  $N_d^{\text{rel}}$  consecutive integers. The solution space of the recurrence  $A_d(E)n = 0$  (shift  $E$ ) is exactly their span, with  $A_d$  the annihilator of theorem 2.3.

*Proof.* The Fourier transform of the twisted comb  $\sum_k k^s \omega^{jk} \delta_k$  is carried, with derivatives through order  $s$ , by the coset  $j/M + \mathbb{Z}$ ; there the multiplier  $\hat{u}(M\xi)$  vanishes to order  $1 + \nu_2(j + Mn) = 1 + \nu_2(j) > s$ , since  $0 < j < M$ . Independence is the confluent Vandermonde theorem for the distinct roots  $\omega^j$  with multiplicities  $\nu_2(j) + 1$ ; the count is  $\sum_{r < d} 2^{d-r-1}(r+1) = 2M - d - 2$ .  $\square$

**Theorem 3.5** (Exact local dimension). *Let  $V_d = \text{span}\{\psi_k|_{[0,1]} : -M+1 \leq k \leq M\}$ . Then*

$$\boxed{\dim V_d = d + 2, \quad V_d \cap \mathbb{R}[t] = \Pi_d.} \quad (3.6)$$

The kernel of the synthesis map  $(q_k) \mapsto \sum q_k \psi_k|_{[0,1]}$  is exactly the restriction of the  $A_d$ -recurrence solution space.

*Proof.* The  $N_d^{\text{rel}}$  relations give  $\dim V_d \leq 2M - N_d^{\text{rel}} = d + 2$ ; theorem 3.3 gives  $\Pi_d \subset V_d$ , so  $\dim V_d \geq d + 1$ ; and since some active atom (e.g.  $\psi_0$ ) is not polynomial on  $(0, 1)$  – the non-elementarity import of section 1 – the containment is strict, forcing  $d + 2$ . If  $V_d$  contained a polynomial outside  $\Pi_d$ , then  $V_d \subseteq \mathbb{R}[t]$  would make every atom polynomial, the same contradiction; and a kernel larger than the recurrence space would push  $\dim V_d$  below  $d + 2$ .  $\square$

**Proposition 3.6** (Universal fixed-dictionary lower bound). *If finitely many nontrivial affine up-atoms  $f_r(x) = u((x - \tau_r)/\sigma_r)$  span a space containing  $\Pi_d$  on a nondegenerate interval  $I$ , then their number is at least  $d + 2$ . Consequently a generic  $P \in \Pi_d$  cannot be represented by  $d + 1$  or fewer atoms of the dyadic dictionary.*

*Proof.* No  $f_r|_I$  is polynomial (same non-elementarity fact). With  $N \leq d$  the dimension is insufficient; with  $N = d + 1$ , containment of the  $(d + 1)$ -dimensional  $\Pi_d|_I$  forces equality of spans, making every  $f_r|_I$  polynomial – impossible. For genericity, the union over all  $\leq (d + 1)$ -subsets of (proper) subspaces  $\text{span}\{f_r\} \cap \Pi_d$  is a finite union of proper subspaces, of measure zero in coefficient space.  $\square$

### 3.4 The endpoint basis

**Theorem 3.7** (Unique  $(d + 2)$ -atom endpoint synthesis). *The atoms with the  $d + 2$  rightmost centers,  $\mathcal{B}_d = \{\psi_k|_{[0,1]} : M - d - 1 \leq k \leq M\}$ , form a basis of  $V_d$ . Explicitly: extend the first  $N_d^{\text{rel}}$  values  $q_k = Q(k)$ ,  $k = -M + 1, \dots, -M + N_d^{\text{rel}}$ , by the monic integer recurrence  $A_d(E)n = 0$ , and put  $b_k = q_k - n_k$  on the last  $d + 2$  centers; then*

$$P(t) = \frac{1}{M} \sum_{k=M-d-1}^M b_k u\left(\frac{t-k}{M}\right), \quad 0 \leq t \leq 1, \quad (3.7)$$

*uniquely in this basis, with rational  $b_k$  for rational  $P$ . Together with [proposition 3.6](#),  $d + 2$  is the least size of any fixed affine up-dictionary spanning  $\Pi_d$  on an interval.*

*Proof.* The extension  $n$  lies in the complete nullspace, so its synthesis vanishes;  $q - n$  is supported on the last  $2M - N_d^{\text{rel}} = d + 2$  centers, so  $\mathcal{B}_d$  spans  $V_d$ , and [theorem 3.5](#) makes it a basis. Integer recurrence coefficients preserve rationality. Reflection  $t \mapsto 1 - t$  gives the mirror-image left basis.  $\square$

The first monomial instances (coefficients multiply raw atoms  $u((t - k)/M)$ ):

$$1 = u(t) + u(t - 1), \quad (3.8)$$

$$t = -\frac{1}{2}u\left(\frac{t}{2}\right) + u\left(\frac{t-1}{2}\right) + \frac{1}{2}u\left(\frac{t-2}{2}\right), \quad (3.9)$$

$$t^2 = \frac{13}{9}u\left(\frac{t-1}{4}\right) - \frac{31}{9}u\left(\frac{t-2}{4}\right) + \frac{49}{9}u\left(\frac{t-3}{4}\right) + \frac{5}{9}u\left(\frac{t-4}{4}\right), \quad (3.10)$$

$$t^3 = -14u\left(\frac{t-4}{8}\right) + \frac{136}{3}u\left(\frac{t-5}{8}\right) - \frac{208}{3}u\left(\frac{t-6}{8}\right) + \frac{280}{3}u\left(\frac{t-7}{8}\right) - \frac{106}{3}u\left(\frac{t-8}{8}\right), \quad (3.11)$$

all on  $[0, 1]$ ; the six coefficients for  $t^4$  at scale 16 are

$$\left(\frac{257824}{675}, -\frac{358624}{225}, \frac{2010496}{675}, -\frac{2528896}{675}, \frac{661024}{135}, -\frac{2567968}{675}\right)$$

at centers  $11, \dots, 16$ .

### 3.5 The output-sensitive algorithm

The recurrence route touches  $N_d^{\text{rel}} \asymp 2^{d+1}$  values to produce  $d + 2$  numbers. The exponential detour can be eliminated entirely: the annihilator acts on the deconvolution sequence as a *bounded* differential operator.

**Theorem 3.8** (Compressed residual identity). *Define the Bernoulli differential symbol*

$$H_d(z) = \left(\frac{z}{2 \sinh(z/2)}\right)^d \frac{1}{M_u(z)}, \quad \log H_d(z) = - \sum_{m \geq 1} \frac{B_{2m}}{2m (2m)!} \left(d + \frac{1}{1 - 4^{-m}}\right) z^{2m}. \quad (3.12)$$

*Then, with  $Q = M_u(MD)^{-1}P$ ,  $K = -M + 1$ , and  $g_r = (A_d(E)Q)(K + r)$ ,*

$$g_r = A_d(1) [H_d(D)P] \left(r - \frac{d}{2}\right) \quad (r \in \mathbb{Z}). \quad (3.13)$$

*Proof.* By [theorem 2.3](#),  $a_\ell/A_d(1)$  is the law of  $L = \sum_m U_m$ , so  $(A_d(E)Q)(x) = A_d(1) \mathbb{E} Q(x + L)$ . Centering  $L$  at its mean  $N_d^{\text{rel}}/2$  and using  $K + N_d^{\text{rel}}/2 = -d/2$ ,

$$g_r = A_d(1) \left[ \mathbb{E} e^{(L - N_d^{\text{rel}}/2)D} M_u(MD)^{-1} P \right] \left( r - \frac{d}{2} \right),$$

and the two product formulas

$$\mathbb{E} e^{z(L - N_d^{\text{rel}}/2)} = \prod_{m=1}^d \frac{\sinh(2^{m-1}z)}{2^m \sinh(z/2)},$$

$$M_u(Mz) = M_u(z) \prod_{r=0}^{d-1} \frac{\sinh(2^r z)}{2^r z}$$

cancel factor by factor to [\(3.12\)](#). The logarithm follows from the classical expansion

$$\log[2 \sinh(z/2)/z] = \sum_m B_{2m} z^{2m} / (2m (2m)!)$$

together with the cumulant series of  $M_u$ . □

**Algorithm 3.9** (Output-sensitive exact synthesis). Given  $P \in \Pi_d$ : truncate  $H_d$  at degree  $\deg P$  and form  $G = H_d(D)P$ ; compute only  $a_0, \dots, a_{d+1}$  of  $A_d$  by truncated sliding-window convolution of the uniform blocks; set  $g_r = A_d(1) G(r - d/2)$  for  $0 \leq r \leq d+1$ ; solve the unit-diagonal triangular system  $g_r = \sum_{s \leq r} a_{r-s} b_s$  forward; output [\(3.7\)](#). Total cost:  $\mathcal{O}(d^2)$  exact rational operations and  $\mathcal{O}(d)$  stored rationals – polynomial in the output, not exponential in the scale. On  $J$  cells the same procedure with  $r \leq J + d$  yields the multi-cell basis below at cost  $\mathcal{O}(d^2 + (J + d)^2)$ .

For  $P \in \mathbb{Z}[t]$  the denominators of all  $b_k$  divide  $\text{lcm}(\text{den } \gamma_0, \dots, \text{den } \gamma_d)$  (the recurrence is integral), and the raw amplitudes  $b_k/M$  add only the dyadic factor.

### 3.6 Several cells: localization knob

**Theorem 3.10** (Multi-cell dimension and basis). For  $J \geq 1$ , the space  $V_{d,J} = \text{span}\{\psi_k|_{[0,J]} : -M+1 \leq k \leq J+M-1\}$  has

$$\dim V_{d,J} = J + d + 1, \quad V_{d,J} \cap \mathbb{R}[t] = \Pi_d, \quad (3.14)$$

with the  $J + d + 1$  consecutive centers  $M - d - 1, \dots, M + J - 1$  as a basis. On a physical interval  $[a, b]$  with  $h = (b - a)/J$  and  $P(t) = p(a + ht)$ ,

$$p(x) = \frac{1}{M} \sum_{r=0}^{J+d} b_r u \left( \frac{x - a - h(M - d - 1 + r)}{Mh} \right), \quad a \leq x \leq b, \quad (3.15)$$

with support footprint  $[a - (d+1)h, b + (2M-1)h]$ . Larger  $J$  buys a smaller physical scale  $Mh = 2^d(b - a)/J$  at one extra atom per cell.

*Proof.* The same  $N_d^{\text{rel}}$  global relations stay independent, so  $\dim \leq J + 2M - 1 - N_d^{\text{rel}} = J + d + 1$ ; the first-cell endpoint basis plus the atoms centered at  $M + 1, \dots, M + J - 1$  are triangularly independent (each new atom vanishes on  $[0, r]$  and not immediately to the right), giving equality. A polynomial in  $V_{d,J}$  restricts on the first cell into  $V_d \cap \mathbb{R}[t] = \Pi_d$ . Recurrence elimination leaves exactly the stated centers; the affine substitution gives [\(3.15\)](#). □

## 4 The polynomiality certificate and the canonical defect

The one-cell space exceeds  $\Pi_d$  by exactly one dimension. Both the membership test for that codimension and the extra direction itself are completely explicit – and both connect this volume to the spectral arithmetic of the companion volume *Dyadic-Comb Frontiers*.

### 4.1 One scalar decides polynomiality

**Theorem 4.1** (Exact certificate, Thue–Morse form). *Order the active coefficients  $q = (q_0, \dots, q_{2M-1})$  by increasing center ( $q_j$  multiplies  $\psi_{-M+1+j}$ ), and set*

$$\mathfrak{C}_d(q) = [A_d(E) \Delta^{d+1} q]_{-M+1} \stackrel{(2.6)}{=} (-1)^{d+1} \sum_{r=0}^{M-1} \varepsilon_r (q_{2r} - q_{2r+1}). \quad (4.1)$$

*Then  $\sum_j q_j \psi_{-M+1+j}$  is polynomial on  $[0, 1]$  iff  $\mathfrak{C}_d(q) = 0$ ; and for the test sequence  $q_j = (-M + 1 + j)^{d+1}$ ,  $\mathfrak{C}_d(q) = A_d(1) (d + 1)!$ .*

*Proof.* Polynomial syntheses are precisely (deconvolution polynomial sequences of degree  $\leq d$ ) + (null sequences):  $\Delta^{d+1}$  kills the former and  $A_d(E)$  the latter, so  $\mathfrak{C}_d$  vanishes on the  $(2M - 1)$ -dimensional polynomial preimage. It is nonzero on the test sequence by  $\Delta^{d+1} k^{d+1} = (d + 1)!$  and  $A_d(E)1 = A_d(1)$ , so its kernel is exactly that preimage. The Thue–Morse form is the factorization (2.6) applied to the composite operator.  $\square$

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#### Thue–Morse bridge

The same finite Prouhet product that annihilates low polynomial moments (Part I of the companion volume) is the complete codimension-one obstruction to a local up-spline being polynomial: the  $q$ -integer alias annihilator and the signed Thue–Morse first-difference filter are two factorizations of one operator.

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### 4.2 The canonical defect and its Fourier residue

Set  $m = d + 1$  and split the (locally finite) power synthesis  $S_m(t) = \sum_k k^m \psi_k(t)$  into its Poisson zero-frequency part  $\mathcal{P}(t) = M_u(MD) t^m$  and the periodic remainder  $\mathcal{E}_d := S_m - \mathcal{P}$ .

**Theorem 4.2** (Explicit periodic residue).  *$\mathcal{E}_d$  is  $C^\infty$ , one-periodic, mean zero, nonzero, with the absolutely convergent odd-frequency series*

$$\boxed{\mathcal{E}_d(t) = -\frac{(d+1)!}{(2\pi i)^{d+1}} \sum_{\substack{n \in \mathbb{Z} \\ n \text{ odd}}} \frac{\widehat{u}(n/2)}{n^{d+1}} e^{2\pi i n t}}, \quad \mathcal{E}_d(t + \tfrac{1}{2}) = -\mathcal{E}_d(t); \quad (4.2)$$

*a cosine series for odd  $d$ , a sine series for even  $d$ . The canonical local defect is  $\mathcal{D}_d(t) := t^{d+1} + \mathcal{E}_d(t) \in V_d$ , and*

$$V_d = \Pi_d \oplus \text{span}\{\mathcal{D}_d\}, \quad c_{\text{def}}(q) = \frac{\mathfrak{C}_d(q)}{A_d(1) (d + 1)!} \quad (4.3)$$

*is the defect coordinate of an arbitrary synthesis.*

*Proof.* In the Poisson expansion of  $S_m$ , even nonzero frequencies die (zero order  $1 + \nu_2(Mn) > m$ ) and at odd  $n$  only the leading jet of  $\widehat{u}(M\xi)$  survives; by the corpus’s leading-jet formula it equals  $-m! \widehat{u}(n/2)/n^m$ , giving (4.2). Odd-frequency support gives the half-period antisymmetry, and non-vanishing follows from  $\widehat{u}(n/2) \neq 0$  for odd  $n$ .  $\mathcal{E}_d$  itself is *not* in  $V_d$  (else  $t^{d+1} \in V_d$ , contradicting  $V_d \cap \mathbb{R}[t] = \Pi_d$ ); the corrected  $\mathcal{D}_d = S_m - (\mathcal{P} - t^m)$  is, since  $\mathcal{P} - t^m \in \Pi_d$ . Its certificate value is that of the sequence  $k^{d+1}$ , i.e.  $A_d(1)(d + 1)! \neq 0$ , so  $\mathcal{D}_d \notin \Pi_d$  and (4.3) follows from [theorem 3.5](#).  $\square$

### 4.3 Cross-volume identities: the defect is the quadrature first-failure

The series (4.2) is not new to the corpus: it is, up to an explicit constant, the degree- $(m+1)$  error of the *shifted dyadic-comb quadrature* studied in Part I of the companion volume *Dyadic-Comb Frontiers*, where

$$Q_{m,p}(\theta) = 2^{-m} \sum_{k \in \mathbb{Z}} (2^{-m}(k + \theta))^p u(2^{-m}(k + \theta))$$

is exact for  $p \leq m$  and first fails at  $p = m + 1$  through the same odd-alias leading jets.

**Theorem 4.3** (Defect–quadrature duality). *For every  $d \geq 0$  and  $t \in \mathbb{R}$ ,*

$$\mathcal{E}_d(t) = (-1)^{d+1} 2^{d(d+1)} \left( Q_{d,d+1}(t) - \int_{-1}^1 x^{d+1} u(x) dx \right). \quad (4.4)$$

*Proof.* Both sides are the odd-frequency alias series with leading jets of  $\widehat{u}$  at the order- $(d+1)$  zeros: the companion volume’s first-failure theorem gives

$$Q_{d,d+1}(\theta) - \mu_{d+1} = - \left( -\frac{1}{2\pi i} \right)^{d+1} \frac{(d+1)!}{2^{d(d+1)}} \sum_{\ell \text{ odd}} \frac{\widehat{u}(|\ell|/2)}{\ell^{d+1}} e^{2\pi i \ell \theta},$$

and comparing with (4.2) term by term (the  $(-1)^{d+1}$  absorbs the sign of  $(-1)^{-(d+1)}$ ) gives (4.4).  $\square$

Two families of consequences flow across the volumes.

*Phase zeros.* The quadrature’s parity-forced superconvergent phases become the trivial zeros of the defect: for even  $d$  (sine series)  $\mathcal{E}_d(0) = \mathcal{E}_d(1/2) = 0$ , matching the centered/half-shifted superconvergence at even level; for odd  $d$  the quarter-period zeros  $\mathcal{E}_d(1/4) = \mathcal{E}_d(3/4) = 0$  match the quarter-shift superconvergence at odd level.

*Special values.* For odd  $d$ , evaluation at  $t = 0$  gives, with the spectral Dirichlet series  $D(s) = \sum_{q \text{ odd}} \widehat{u}(q/2) q^{-s}$  of the companion volume,

$$\mathcal{E}_d(0) = (-1)^{(d-1)/2} \frac{2(d+1)!}{(2\pi)^{d+1}} D(d+1) \in \mathbb{Q}, \quad (4.5)$$

rational because  $D(2r)/\pi^{2r} \in \mathbb{Q}$  is proved there. The proved exact values  $D(2), \dots, D(8)$  yield

$$\mathcal{E}_1(0) = \frac{1}{18}, \quad \mathcal{E}_3(0) = -\frac{23}{1350}, \quad \mathcal{E}_5(0) = \frac{1543}{119070}, \quad \mathcal{E}_7(0) = -\frac{1196113}{65063250}, \quad (4.6)$$

and  $\mathcal{E}_d(1/2) = -\mathcal{E}_d(0)$  in each case. As a consistency check across independently computed tables: the companion volume’s exact phase-error experiments give  $Q_{1,2}(0) - \mu_2 = \frac{1}{72}$  and  $Q_{3,4}(0) - \mu_4 = -\frac{23}{5529600}$ , and (4.4) converts them to  $\mathcal{E}_1(0) = 2^2 \cdot \frac{1}{72} = \frac{1}{18}$  and  $\mathcal{E}_3(0) = 2^{12} \cdot \left(-\frac{23}{5529600}\right) = -\frac{23}{1350}$  — in exact agreement with (4.5).

**Research question 4.4** (Defect values at other dyadic rationals). *Beyond  $t \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}\}$ , the values  $\mathcal{E}_d(a/2^s)$  are character-twisted versions of  $D(d+1)$  (finite combinations of  $\sum_{q \text{ odd}} \widehat{u}(q/2) \chi(q) q^{-(d+1)}$  over  $2^s$ -periodic characters  $\chi$ ). Do they admit closed forms in Fabius values, finite Thue–Morse sums, or  $q$ -binomial data, as the principal value (4.5) does? This refines the open “defect special values” question of the absorbed source and inherits the machinery of the companion volume’s first-defect chapter.*

## 5 The scale-ladder windows

The second construction inverts the refinement equation itself. It needs no alias analysis: one telescoping identity, iterated, produces polynomial tails whose truncation is exact.

### 5.1 The antiderivative train

Write  $u_{h,c}(x) = u((x-c)/h)$  and  $(Jf)(x) = \int_{-\infty}^x f$ .

**Proposition 5.1** (One-step identity).  $Ju(x) = \sum_{k \geq 0} u\left(\frac{x-(2k+1)}{2}\right)$ , a locally finite sum; hence  $Ju_{h,c}(x) = h \sum_{k \geq 0} u\left(\frac{x-c-h(2k+1)}{2h}\right)$ .

*Proof.* By the refinement derivative,  $\frac{d}{dx}u\left(\frac{x-(2k+1)}{2}\right) = u(x-2k) - u(x-2k-2)$ ; the locally finite series telescopes to  $u(x)$ , and both sides vanish for  $x \ll 0$ .  $\square$

**Theorem 5.2** (Iterated train). For every  $m \geq 1$ ,

$$J^m u_{h,c}(x) = h^m 2^{m(m-1)/2} \sum_{N \geq 0} p_m(N) u\left(\frac{x-c-h(2^m-1+2N)}{2^m h}\right), \quad (5.1)$$

locally finite, and the atom indexed by  $N$  has support  $[c-h+2hN, c+h(2^{m+1}-1+2N)]$ : its left endpoint is independent of the order  $m$ .

*Proof.* Induct with [proposition 5.1](#): integrating the  $N$ -th atom produces atoms at scale  $2^{m+1}h$  and centers  $c+h(2^{m+1}-1+2(N+2^m k))$ , so the new multiplicity is  $\sum_{k \geq 0} p_m(N'+2^m k) = p_{m+1}(N')$  – exactly the recurrence (2.3) – while the prefactor gains  $2^m h$ . The support bookkeeping is immediate from  $\text{supp } u_{\rho,c'} = [c'-\rho, c'+\rho]$ .  $\square$

The weights are thus the Green sequence of [proposition 2.2](#): repeated *continuous* antidifferentiation of the refinement derivative and the *discrete* inversion of the Thue–Morse difference stencil are the same computation.

### 5.2 Polynomial tails and the finite window

**Lemma 5.3** (Tail polynomial). For  $n \geq 0$  and  $x \geq c+h$ ,  $J^{n+1}u_{h,c}(x) = \frac{h^{n+1}}{n!} G_n\left(\frac{x-c}{h}\right)$ , with  $G_n$  the moment Appell family (0.1).

*Proof.*  $J^{n+1}u_{h,c}(x) = \frac{1}{n!} \int (x-t)^n u\left(\frac{t-c}{h}\right) dt$  over the full support once  $x \geq c+h$ ; substitute  $t = c+hz$ .  $\square$

Combining [theorem 5.2](#) and [lemma 5.3](#) at order  $m = n+1$  gives the locally finite half-line identity

$$G_n\left(\frac{x-c}{h}\right) = n! 2^{n(n+1)/2} \sum_{N \geq 0} p_{n+1}(N) u\left(\frac{x-c-h(2^{n+1}-1+2N)}{2^{n+1}h}\right) \quad (x \geq c+h), \quad (5.2)$$

and the support-edge invariance makes one truncation index serve all degrees simultaneously:

**Theorem 5.4** (Exact finite polynomial window). Let  $a < b$ ,  $L = b-a$ ,  $h > 0$ ,  $K = \lceil L/(2h) \rceil$ , and for  $n \geq 0$

$$\Psi_{n,h}(x) = n! 2^{n(n+1)/2} \sum_{N=0}^K p_{n+1}(N) u\left(\frac{x-a-h(2^{n+1}-2+2N)}{2^{n+1}h}\right). \quad (5.3)$$

Then  $\Psi_{n,h}(x) = G_n\left(\frac{x-a+h}{h}\right)$  for all  $x \in [a, b]$ ; in particular  $\{\Psi_{0,h}, \dots, \Psi_{d,h}\}$  restricts to a monic triangular basis of  $\Pi_d$  on  $[a, b]$ , using  $(d+1)(K+1)$  atoms at the scales  $2h, 4h, \dots, 2^{d+1}h$ .

*Proof.* Take  $c = a-h$  in (5.2), so  $x \geq c+h$  on all of  $[a, b]$ . Every omitted atom  $N \geq K+1$  has left support endpoint  $a-2h+2hN \geq a+2hK \geq b$ , hence vanishes on  $[a, b]$  (at a boundary touch the argument is  $-1$  and  $u(-1) = 0$ ).  $\square$

**Corollary 5.5** (Two atoms per degree). *For  $h \geq L/2$ ,  $K = 1$  and  $p_{n+1}(0) = p_{n+1}(1) = 1$ : each block is a pair of equal-amplitude atoms, and with the canonical choice  $h = L/2$ ,*

$$\boxed{\Psi_n(x) = n! 2^{n(n+1)/2} \left[ u\left(\frac{x-a-(2^n-1)L}{2^n L}\right) + u\left(\frac{x-a-2^n L}{2^n L}\right) \right] = G_n\left(1 + \frac{2(x-a)}{L}\right)} \quad (5.4)$$

on  $[a, b]$ . Every  $p \in \Pi_d$  therefore uses at most  $2d + 2$  raw atoms; the case  $n = 0$  is the partition of unity (1.2).

*Proof.* If  $h \geq L/2$ , then  $0 < L/(2h) \leq 1$ , hence  $K = 1$ . The binary-partition series has  $p_{n+1}(0) = p_{n+1}(1) = 1$ , so theorem 5.4 has precisely the displayed two summands. Substituting  $h = L/2$  in its polynomial argument gives  $(x - a + h)/h = 1 + 2(x - a)/L$ . Expanding  $p$  in the resulting  $d + 1$  triangular blocks uses two atoms per block; for  $n = 0$  the formula is exactly (1.2).  $\square$

### 5.3 Exact coefficients: the Appell transform

**Theorem 5.6** (Coefficient transform). *Let  $p \in \Pi_d$ ,  $c = a - h$ ,  $\tilde{p}(y) = p(c + hy)$ . The unique expansion  $\tilde{p} = \sum_{n \leq d} b_n G_n$  has*

$$b_n = [y^n] M_u(D)^{-1} \tilde{p} = \sum_{r=0}^{d-n} \frac{\gamma_r h^{n+r}}{n! r!} p^{(n+r)}(c) = \sum_{r=0}^{d-n} \binom{n+r}{r} \gamma_r c_{n+r}, \quad (5.5)$$

the last form for  $\tilde{p} = \sum_j c_j y^j$ ; equivalently the monic-triangular descending recurrence  $b_n = c_n - \sum_{k>n} b_k \binom{k}{n} \mu_{k-n}$  holds. Then

$$p(x) = \sum_{n=0}^d b_n \Psi_{n,h}(x) \quad (a \leq x \leq b), \quad (5.6)$$

with raw atom amplitudes  $\alpha_{n,N} = n! 2^{n(n+1)/2} b_n p_{n+1}(N)$  – in the canonical two-atom form, both atoms of block  $n$  share the amplitude  $\alpha_n = n! 2^{n(n+1)/2} b_n$ . For monomials the coefficients are closed Bernoulli–Bell forms:  $p(x) = (x - a)^m$  has  $b_n = h^m \binom{m}{n} \mathcal{A}_{m-n}(-1)$ . Everything is rational for rational data, affine-covariant, and computable in  $\mathcal{O}(d^2 + dK)$  exact operations.

*Proof.*  $G_n = M_u(D) y^n$  makes  $M_u(D)$  triangular-invertible on  $\Pi_d$ ; apply  $M_u(D)^{-1}$  to the expansion, extract coefficients, and use (5.3). For the monomial form: with  $c = a - h$ ,  $(x - a)^m = h^m (y - 1)^m$ , Appell operators commute with shifts, so  $M_u(D)^{-1} (y - 1)^m = \mathcal{A}_m(y - 1) = \sum_n \binom{m}{n} \mathcal{A}_{m-n}(-1) y^n$  by the Appell binomial theorem.  $\square$

**Example 5.7** (Worked quartic). For  $p(x) = 3x^4 - 2x^3 + 5x - 7$  on  $[0, 1]$  with  $h = \frac{1}{2}$  ( $c = -\frac{1}{2}$ ):  $\tilde{p}(y) = \frac{3}{16}y^4 - y^3 + \frac{15}{8}y^2 + y - \frac{145}{16}$ , the Appell coordinates are  $(b_0, \dots, b_4) = (-\frac{2084}{225}, \frac{4}{3}, \frac{7}{4}, -1, \frac{3}{16})$  (re-verified exactly for this consolidation via the triangular recurrence), the block amplitudes are  $(\alpha_0, \dots, \alpha_4) = (-\frac{2084}{225}, \frac{8}{3}, 28, -384, 4608)$ , and

$$p(x) = \sum_{n=0}^4 \alpha_n \left[ u\left(\frac{x-(2^n-1)}{2^n}\right) + u\left(\frac{x-2^n}{2^n}\right) \right] \quad (0 \leq x \leq 1), \quad (5.7)$$

a ten-atom exact window. The absorbed source verifies the identity symbolically (residual exactly zero) and independently by FFT inversion of a 44-factor sinc product ( $10^{-8}$ -level numerical residuals measuring only the Fourier grid); see figure 1.



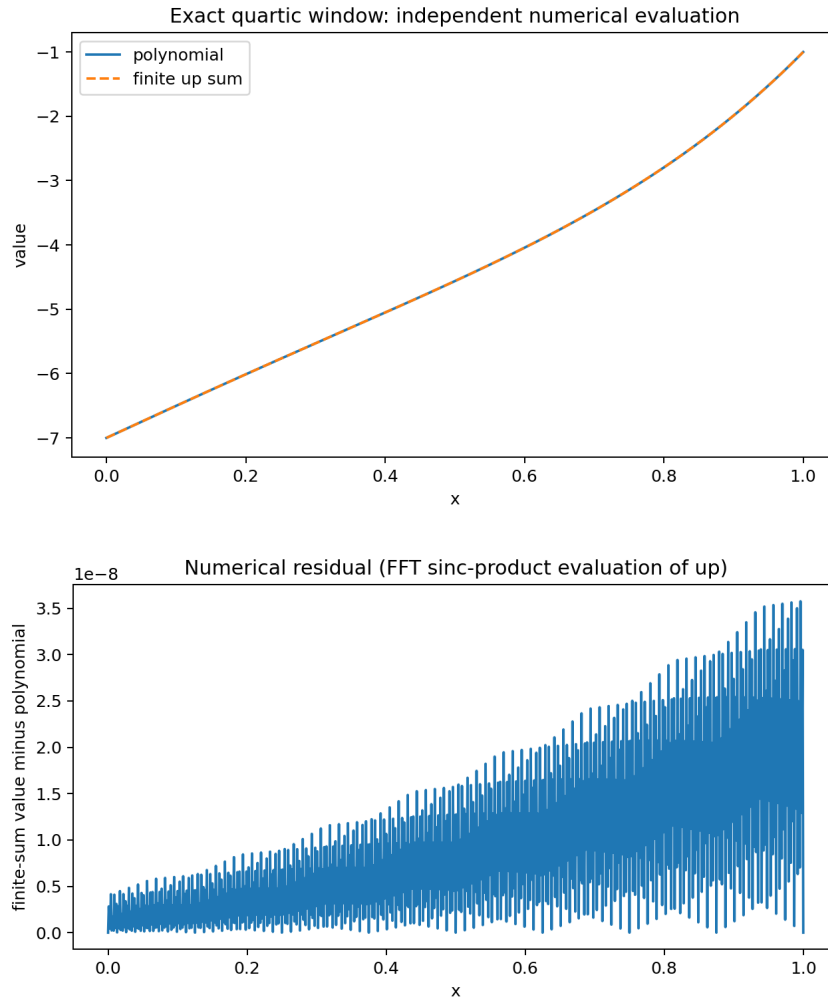


Figure 1: The worked quartic (5.7): the ten-atom window (top, visually indistinguishable from the polynomial on  $[0, 1]$ ) and the residual of the independent FFT check (bottom); the exact symbolic residual is zero.



## 5.4 Structure: Chebyshev system, extension operator, collars

**Proposition 5.8** (Derivative ladder, Wronskian, interpolation). *On  $[a, b]$ ,  $\Psi_{n,h}^{(k)} = n^k h^{-k} \Psi_{n-k,h}$ ; the Wronskian is constant,  $W(\Psi_{0,h}, \dots, \Psi_{d,h}) = h^{-d(d+1)/2} \prod_{n \leq d} n!$ ; and for distinct nodes  $x_0, \dots, x_d \in [a, b]$ ,*

$$\det[\Psi_{j,h}(x_i)] = h^{-d(d+1)/2} \prod_{i < j} (x_j - x_i). \quad (5.8)$$

*The block family is thus an extended complete Chebyshev system on every subinterval, nonsingular for interpolation at every node set.*

*Proof.* On the interval the blocks are the monic polynomials  $G_n((x - a + h)/h)$ ; a monic triangular change of basis has determinant one, the monomial Wronskian is  $\prod n!$ , each derivative row scales by  $h^{-k}$ , and the interpolation determinant is the Vandermonde in  $y_i = (x_i - a + h)/h$ .  $\square$

**Proposition 5.9** (Extension operator and support). *The map  $E_{d,a,b,h}: p \mapsto \sum_{n,N} \alpha_{n,N}(p) u((x - C_{n,N})/2^{n+1}h)$ , with centers  $C_{n,N} = a + h(2^{n+1} - 2 + 2N)$ , is a linear injection  $\Pi_d \rightarrow C_c^\infty(\mathbb{R})$  that restricts to the identity on  $[a, b]$  — an explicit compactly supported smooth extension operator for polynomial data, flat at its outer support endpoints, with*

$$\text{supp } E_{d,a,b,h} p \subseteq [a - 2h, a + h(2^{d+2} - 2 + 2K)] \subset [a - 2h, b + 2^{d+2}h]. \quad (5.9)$$

*Proof.* The coefficients in [theorem 5.6](#) depend linearly on  $p$ , so  $E_{d,a,b,h}$  is linear, and [theorem 5.4](#) gives  $E_{d,a,b,h} p = p$  on  $[a, b]$ . This restriction identity also proves injectivity. Every summand is a translate and dilation of the compactly supported  $C^\infty$  function  $u$ , hence the finite sum is compactly supported, smooth, and flat at the two outer endpoints. The leftmost support edge occurs for  $(n, N) = (0, 0)$  and equals  $a - 2h$ ; the rightmost occurs for  $(n, N) = (d, K)$  and equals  $a + h(2^{d+2} - 2 + 2K)$ . Finally  $2hK < L + 2h$  gives the strict second inclusion in [\(5.9\)](#).  $\square$

**Theorem 5.10** (Collar–atom tradeoff). *For any collar  $\rho > 0$ , the choice  $h = \rho/2^{d+2}$  gives an exact window supported in  $[a - \rho, b + \rho]$  with at most  $(d + 1)(1 + \lceil 2^{d+1}L/\rho \rceil)$  atoms. Conversely the sparse  $h = L/2$  choice has linear atom count  $2d + 2$  but an exponentially long one-sided scale ladder,  $\text{supp} \subseteq [a - L, a + 2^{d+1}L]$ . A reflected anticausal train (right-tail integration  $J_+$ ) mirrors the overhang, and orientations may be mixed degree by degree, since any sequence of monic degree- $n$  blocks remains a triangular basis.*

*Proof.*  $2h = \rho/2^{d+1} \leq \rho$  bounds the left overhang and [\(5.9\)](#) the right one; the count is  $(d + 1)(K + 1)$ . The reflection statement is  $J_+ u(x) = \sum_{k \geq 0} u(\frac{x+2k+1}{2})$  by evenness.  $\square$

**Remark 5.11** (Conditioning). The level normalization  $n! 2^{n(n+1)/2}$  grows superexponentially and the paired atoms are sampled near flat support edges, so the polynomial emerges from cancellation among large terms: construct coefficients exactly, evaluate in extended precision, sum blockwise with compensation. The quasipolynomial law [\(2.4\)](#) shows why shrinking  $h$  does not uniformly worsen coefficients (normalized weights grow like  $N^n$  while the Appell coordinates carry powers of  $h$ ): a nontrivial conditioning-optimal  $h$  should exist ([problem 12.5](#)). The same warning applies to the endpoint basis of [section 3](#), whose minimal atom count also comes at the price of large signed coefficients.

## 6 The oversampled lattice: free scales, ghost antiderivatives, and the Thue–Morse stencil

The third of the three exact-polynomial sources, `Rvachev_Up_Exact_Polynomial_Representation_Report`, detaches the dictionary construction from the special lattice  $M = 2^d$  at unit spacing: the reproduction order depends only on the *ratio* of atom radius to center spacing, and the physical scale is free. This adds an interval algorithm with prescribed support, a real-space antiderivative calculus dual to the ladder of [section 5](#), and the exact dyadic derivative stencil.

## 6.1 Twisted Poisson summation and general ratios

**Lemma 6.1** (Twisted Poisson identity). *For  $m \in \mathbb{N}_{>0}$ ,  $t \in \mathbb{R}$ ,  $z \in \mathbb{C}$ , with  $M_u$  the moment generating function,*

$$\sum_{k \in \mathbb{Z}} e^{kz} u\left(\frac{t-k}{m}\right) = m \sum_{n \in \mathbb{Z}} e^{(z-2\pi i n)t} M_u(-mz + 2\pi i m n), \quad (6.1)$$

both sides entire in  $z$ .

*Proof.* Poisson summation for the smooth compactly supported  $x \mapsto e^{zx} u((t-x)/m)$ , whose  $n$ -th Fourier coefficient is  $m e^{(z-2\pi i n)t} M_u(-mz + 2\pi i m n)$ .  $\square$

At a nonzero frequency  $n$ , the alias factor vanishes at  $z = 0$  to order  $1 + \nu_2(mn) \geq \nu_2(m) + 1$ . Dividing (6.1) by  $m M_u(mz)$  and comparing Taylor coefficients gives, in one stroke, the general-ratio version of the global synthesis: with the scaled deconvolution Appell samples

$$A_r^{(m)}(k) := m^{r-1} \mathcal{A}_r(k/m) = \frac{1}{m} \sum_{j=0}^r \binom{r}{j} k^{r-j} \beta_j^{(m)}, \quad \frac{1}{M_u(mz)} = \sum_n \beta_n^{(m)} \frac{z^n}{n!}, \quad (6.2)$$

one has, for every  $m \in \mathbb{N}_{>0}$  and  $0 \leq r \leq \nu_2(m)$ ,

$$t^r = \sum_{k \in \mathbb{Z}} A_r^{(m)}(k) u\left(\frac{t-k}{m}\right) \quad (t \in \mathbb{R}), \quad (6.3)$$

locally finite, and the order  $\nu_2(m)$  is *sharp* (theorem 3.1); the same argument works for any  $C_c^\infty$  generator of unit mass whose transform vanishes to order  $d+1$  at the nonzero dual lattice. The probabilistic content of the deconvolution is exact:  $q^\sharp = M_u(mD)^{-1}q$  is the unique polynomial with  $\mathbb{E} q^\sharp(t+mZ) = q(t)$  – the coefficients undo smoothing by a random up-distributed displacement.

The first failure at degree  $L = \nu_2(m) + 1$  generalizes the defect series (4.2) beyond dyadic ratios: for  $m = 2^\nu q$  with  $q$  odd,

$$\sum_{k \in \mathbb{Z}} A_L^{(m)}(k) u\left(\frac{t-k}{m}\right) - t^L = \frac{(-1)^{L+1} L!}{(2\pi i)^L} \sum_{\substack{n \in \mathbb{Z} \\ n \text{ odd}}} \frac{\widehat{u}(qn/2)}{n^L} e^{-2\pi i n t}, \quad (6.4)$$

nonzero because  $\widehat{u}(q/2) \neq 0$ : an odd factor  $q$  in the ratio replaces the half-integer values  $\widehat{u}(n/2)$  by  $\widehat{u}(qn/2)$  – the arithmetic face of the odd-overscaling phenomenon of conjecture 12.2.

## 6.2 The physical-scale interval algorithm

Affine reduction makes the lattice fully physical: for  $p \in \Pi_d$ , an interval  $I = [A, B]$ , an origin  $x_0$ , a spacing  $h > 0$ , and any ratio  $m$  with  $\nu_2(m) \geq d$  (canonically  $m = 2^d$ ), put  $\rho = mh$  and  $p_\rho^\sharp = M_u(\rho D)^{-1}p$ . Then

$$p(x) = \frac{1}{m} \sum_{k=\lceil (A-x_0)/h-m \rceil}^{\lfloor (B-x_0)/h+m \rfloor} p_\rho^\sharp(x_0 + kh) u\left(\frac{x-x_0-kh}{\rho}\right) \quad (x \in [A, B]), \quad (6.5)$$

a finite  $C^\infty$  plateau supported in  $[A - 2\rho, B + 2\rho]$  with at most  $\lfloor m((B-A)/\rho + 2) \rfloor + 1$  atoms, rational for rational data, matching *every* endpoint jet of  $p$  (all derivatives agree on the interior; take one-sided limits). Two practical features distinguish it from both normal forms above: the representation is determined by  $\mathcal{O}(d)$  field elements (the polynomial  $p_\rho^\sharp$  and the lattice parameters), however long the interval; and evaluation at a point touches at most  $2m+1$  atoms. The atom count  $2^d(B-A)/\rho + 2^{d+1} + \mathcal{O}(1)$  exposes the radius–density tradeoff: shrinking the support collar costs interior atoms, and the  $2^{d+1}$  boundary-overlap term never disappears – the lattice-side counterpart of the ladder’s collar–atom law (theorem 5.10), with the dictionary’s  $d+2$  and the ladder’s  $2d+2$  as the atom-minimal and scale-sparse extremes.

### 6.3 Ghost antiderivatives and the exact derivative stencil

The refinement derivative also supports a purely real-space route, dual to the antiderivative train of [section 5](#). Writing  $U_{a,c}(x) = u((x - c)/a)$ , the *parent–child identity*

$$D U_{2a,c} = \frac{1}{a} (U_{a,c-a} - U_{a,c+a}) \quad (6.6)$$

converts differentiation into a two-term difference one scale down. A finite child sum  $\sum_{j=L}^R b_j U_{a,c+(2j+1)a}$  has a finite parent antiderivative iff  $\sum_j b_j = 0$ , in which case the parent coefficients are the prefix sums  $\sum_{r \leq j} b_r$  – and the zero-sum obstruction is repaired by *ghost atoms*: children whose supports miss the working interval carry the compensating mass, and the split of that mass between the two ends is a free constant of integration, canonically fixed by the  $\ell^2$ - or  $\ell^\infty$ -optimal choices  $\gamma_2 = -\frac{1}{N} \sum P_j$ ,  $\gamma_\infty = -\frac{1}{2}(\max P_j + \min P_j)$ . Iterating from a partition-of-unity representation of  $p^{(d)}$  downward gives a local, sparse, boundary-adaptable alternative to (6.5) in which scales may change per stage.

Iterating (6.6) upward instead gives the exact dyadic derivative stencil:

$$D^d U_{2^d h, c}(x) = h^{-d} 2^{-\binom{d}{2}} \sum_{k=0}^{2^d-1} \varepsilon_k U_{h, c+(2k-(2^d-1))h}(x), \quad (6.7)$$

with the signed Thue–Morse signs  $\varepsilon_k$ : each differentiation doubles the leaves and appends a binary digit carrying the sign, the chain-rule factors multiplying to  $h^{-d} 2^{-\binom{d}{2}}$ . Together with [theorem 5.2](#) this closes a perfect duality: *integration* spreads one atom into a train weighted by the partition numbers  $p_m(N)$  (the causal inverse of the Thue–Morse block, [proposition 2.2](#)), while *differentiation* contracts a coarse atom into a block weighted by the Thue–Morse signs themselves – the two convolution inverses of one stencil, realized by up-atoms at dyadic scales. The Gaussian-binomial deformation  $\prod_{j < d} (1 - yq^j) = \sum_r (-1)^r q^{\binom{r}{2}} \begin{bmatrix} d \\ r \end{bmatrix}_q y^r$  suggests geometrically scaled stencils ([research question 12.15](#)).

### 6.4 Finite sinc prefixes

Truncating the Fourier product at  $N$  factors gives the piecewise-polynomial generator  $u_N$  (an  $N$ -fold uniform convolution, support radius  $1 - 2^{-N}$ ), whose lattice has exact reproduction degree  $\min(N - 1, \nu_2(m))$ , whose cumulants obey the exact truncation law  $\kappa_{2^r}^{(N)} = (1 - 2^{-2rN}) \kappa_{2^r}$ , and whose Appell coefficients converge to the up-versions at the dyadic rate

$$A_{r,N}^{(m)} = A_r^{(m)} + \binom{r}{2} \kappa_2 m^2 2^{-2N} A_{r-2}^{(m)} + \mathcal{O}_{r,m}(2^{-4N}) : \quad (6.8)$$

exact spline plateaux of any prescribed finite order, converging coefficientwise to the  $C^\infty$  theory. (The eighth-wave report absorbed into the exponents volume as its Part III develops the prefix generators themselves in depth: an exact truncated-power formula with Thue–Morse top-derivative jumps, the exact uniform error law  $\|u_N - u\|_\infty = \frac{4}{9} 4^{-N}$  with all derivative orders, and positive quadrature tail surrogates of arbitrary order.)

### 6.5 Verification and arithmetic

The absorbed package verifies the interval algorithm in exact rational arithmetic: the worked degree-four example ( $p = \frac{7}{5} - 3x + 2x^2 - \frac{1}{2}x^3 + \frac{3}{7}x^4$  on  $[-\frac{1}{2}, \frac{3}{2}]$ ,  $m = 16$ ,  $\rho = 2$ ,  $p_2^\sharp = \frac{1301}{1575} - \frac{7}{3}y + \frac{6}{7}y^2 - \frac{1}{2}y^3 + \frac{3}{7}y^4$ , 49 atoms) has residual *exactly zero* at 129 dyadic points, and seven generic tests of degrees 0–6 likewise return exact zero (assets/Rvachev\_Up\_Exact\_Polynomial\_Representation\_Report/). Arithmetically, every reduced cumulant  $\kappa_{2^r}$  has *odd* denominator (the single factor of two in the reduced denominator of  $B_{2^r}$  cancels against  $2^{2r-1}/r$ ), which yields explicit 2-adic denominator bounds for all output coefficients: for data in  $2^{-s}\mathbb{Z}[1/O]$  ( $O$  odd) the coefficients lie in  $2^{-[s(d+1)+d+\nu]}\mathbb{Z}[1/O']$ , all further growth odd ([research question 12.13](#)).

## 7 Dictionary versus ladder, and the minimal atom count

The two constructions solve the same problem with complementary economies:

|                              | Common-scale dictionary (section 3)                       | Scale ladder (section 5)                                |
|------------------------------|-----------------------------------------------------------|---------------------------------------------------------|
| Scales used                  | one: $2^d h$                                              | $d + 1$ : $2h, 4h, \dots, 2^{d+1}h$                     |
| Atoms for $\Pi_d$ , one cell | $d + 2$ (optimal for fixed dictionaries)                  | $2d + 2$ (two per degree, equal amplitudes per block)   |
| Coefficients                 | recurrence residual / $H_d(D)$ symbol, $\mathcal{O}(d^2)$ | triangular Appell transform, $\mathcal{O}(d^2)$         |
| Uniqueness                   | unique in the endpoint basis                              | unique in the block basis                               |
| Support overhang             | $[a - (d + 1)h, b + (2M - 1)h]$ , knob $J$                | collar $\rho$ at will via $h = \rho/2^{d+2}$ , knob $h$ |
| Certificate / defect         | one-scalar Thue–Morse test; explicit $\mathcal{D}_d$      | exactness by construction (blocks are $G_n$ )           |
| Discrete engine              | alias annihilator $A_d = \prod [2^m]_z$                   | Green inverse $1/\Theta_m$ , weights $p_m(N)$           |

The two discrete engines are the two directions of one convolution identity ( $\Theta$  against its causal inverse), and the two coefficient calculi are the two members of the Appell pair (0.1). The oversampled lattice of section 6 is the third normal form, extremal in a third sense: many atoms ( $2^d(B - A)/\rho + 2^{d+1} + \mathcal{O}(1)$ ) but an  $\mathcal{O}(d)$ -element *description* with  $2m + 1$ -local evaluation, a free physical scale, and full endpoint-jet matching — with the ghost-antiderivative route as its sparse, per-stage adaptable variant. The comparison also settles more of the minimal-atom question than any source alone.

**Theorem 7.1** (Minimal atom counts). *Let  $N_d$  be the least integer such that every real polynomial of degree at most  $d$  admits a representation (1.1) on every nondegenerate interval with at most  $N_d$  atoms. Then*

$$N_0 = 2, \quad \boxed{N_d \leq d + 2} \quad (d \geq 0), \quad (7.1)$$

*the upper bound achieved by the single fixed endpoint dictionary of theorem 3.7; and every fixed dictionary whose span contains  $\Pi_d$  has at least  $d + 2$  atoms (proposition 3.6).*

*Proof.*  $N_0 \geq 2$  is the strict monotonicity of section 1;  $N_0 \leq 2$  is the partition (1.2). For general  $d$  the endpoint basis represents every  $P \in \Pi_d$  with the same  $d + 2$  atoms.  $\square$

The bound  $N_d \leq d + 2$  sharpens the ladder count  $2d + 2$  of corollary 5.5 — one of the dividends of consolidating the two sources. What remains open is whether *adaptivity* can beat the dictionary:

**Conjecture 7.2** (Target-adaptive rigidity). *For Lebesgue-almost every  $P \in \Pi_d$  there is no identity (1.1) with  $N \leq d + 1$  atoms, even when amplitudes, shifts, and scales all depend on  $P$ ; hence  $N_d = d + 2$ . The fixed-dictionary case is proposition 3.6; the adaptive case replaces a finite union of proper subspaces by an uncountable parametrized family and appears to require a rigidity theorem for cancellation among nonanalytic affine copies of  $u$ . Within the causal-ladder class (degree-independent left support edge, exact degree- $n$  polynomial tails), at least two atoms per active level are conjectured necessary for generic targets, making  $2d + 2$  minimal in that class.*

**Remark 7.3** (Strang–Fix reconciliation). There is no tension with the classical no-go: a single fixed-scale integer-shift  $u$ -system reproduces only constants globally (theorem 3.1 at  $\nu_2(\rho) = 0$ ). Both constructions bypass it lawfully — the dictionary by paying scale ( $\rho = 2^d$  restores order  $d$  globally before localizing), the ladder by paying scales *and* asking only for a window (boundary truncation on a bounded interval). The classical  $UP_n$  polynomial inclusions of Rvachev theory [3, 5] are thereby made explicit in two normal forms.

## 8 Lagrange cardinal synthesis and exact interpolation loops

The preceding chapters solve polynomial synthesis independently of any choice of interpolation nodes. Lagrange interpolation adds a finite analysis map to that synthesis map and thereby closes an exact loop. This section collects the six reports that developed that observation. Their repeated foundations are stated once in the parent volume; the genuinely different conclusions are organized here as consequences of one diagram.

Fix pairwise distinct nodes

$$X = (x_0, \dots, x_d) \subset [-1, 1], \quad \ell_j(x) = \prod_{r \neq j} \frac{x - x_r}{x_j - x_r},$$

and write

$$\mathcal{D}_u := M_u(D)^{-1}, \quad m \in \mathbb{N}_{>0}, \quad h = m^{-1}, \quad K_m := \{k \in \mathbb{Z} : |k| < 2m\}.$$

The mesh is called *admissible through degree  $d$*  when  $\nu_2(m) \geq d$ . The canonical choice is  $m = 2^d$ . Notice that  $\mathcal{D}_u x^r = \mathcal{A}_r(x)$  in the notation of (0.1).

### 8.1 Cardinal polynomials as literal translates

**Theorem 8.1** (Universal Lagrange–Rvachev cardinal formula). *For every admissible  $m$ , every  $0 \leq j \leq d$ , and every  $x \in [-1, 1]$ ,*

$$\ell_j(x) = h \sum_{k \in K_m} (\mathcal{D}_u \ell_j)(kh) u(x - kh). \quad (8.1)$$

Consequently, for arbitrary nodal data  $y = (y_0, \dots, y_d)^T$  and  $I_X y := \sum_j y_j \ell_j$ ,

$$(I_X y)(x) = h \sum_{k \in K_m} [\mathcal{D}_u (I_X y)](kh) u(x - kh). \quad (8.2)$$

Both sums are finite identities, not approximations. The atoms in these formulas are literal translates of the original radius-one function  $u$ ; there is no dilation.

*Proof.* Use the physical-scale identity (6.5) with target interval  $[-1, 1]$ , origin 0, spacing  $h = m^{-1}$ , and radius  $\rho = mh = 1$ . It applies because  $\deg \ell_j = d \leq \nu_2(m)$  and gives (8.1); compact support leaves exactly the indices  $|k| < 2m$ . Multiply by  $y_j$ , sum over the finite node set, and use linearity of  $\mathcal{D}_u$  to obtain (8.2).  $\square$

The coefficients are closed algebraic expressions in the nodes. Put

$$\lambda_j = \left( \prod_{r \neq j} (x_j - x_r) \right)^{-1},$$

and let  $e_s^{(j)}$  be the elementary symmetric polynomial of degree  $s$  in the  $d$  nodes with  $x_j$  omitted.

**Corollary 8.2** (Nodes-only Bernoulli–Bell amplitudes). *For every  $0 \leq j \leq d$  and every  $t \in \mathbb{R}$ ,*

$$(\mathcal{D}_u \ell_j)(t) = \sum_{r=0}^{\lfloor d/2 \rfloor} \gamma_{2r} \frac{\ell_j^{(2r)}(t)}{(2r)!} \quad (8.3)$$

$$= \lambda_j \sum_{n=0}^d (-1)^{d-n} e_{d-n}^{(j)} \mathcal{A}_n(t). \quad (8.4)$$

Thus rational nodes give rational amplitudes. More explicitly, the  $\gamma_{2r}$  are complete Bell polynomials in the negated Bernoulli cumulants of (0.1), so no implicit linear solve is hidden in (8.1).

*Proof.* The reciprocal moment series is even, and it terminates on a degree- $d$  polynomial; this gives (8.3). Expanding

$$\ell_j(x) = \lambda_j \sum_{n=0}^d (-1)^{d-n} e_{d-n}^{(j)} x^n$$

and applying  $\mathcal{D}_u x^n = \mathcal{A}_n(x)$  term by term gives (8.4). The Bell-cumulant description is exactly the coefficient identity following (0.1).  $\square$

## 8.2 The exact right inverse and its projector

Let  $q_m = |K_m| = 4m - 1$  and define

$$U_{ik} = u(x_i - kh), \quad B_{kj} = h(\mathcal{D}_u \ell_j)(kh). \quad (8.5)$$

Thus  $U \in \mathbb{R}^{(d+1) \times q_m}$  samples the atoms and  $B \in \mathbb{R}^{q_m \times (d+1)}$  synthesizes the cardinals. For a coefficient row  $c \in \mathbb{R}^{K_m}$  and a continuous function  $f$ , write

$$\text{Synth } c := \sum_{k \in K_m} c_k u(\cdot - kh), \quad I_X f := \sum_{j=0}^d f(x_j) \ell_j.$$

Thus  $\text{Synth}: \mathbb{R}^{K_m} \rightarrow C[-1, 1]$  and  $I_X: C[-1, 1] \rightarrow \Pi_d$  have compatible types in the intertwining law below.

**Theorem 8.3** (Nodal closure, basis independence, and determinant identity). *The matrices in (8.5) satisfy*

$$\boxed{UB = I_{d+1}}. \quad (8.6)$$

Consequently

$$\mathcal{P}_X := BU, \quad \mathcal{P}_X^2 = \mathcal{P}_X, \quad \text{ran } \mathcal{P}_X = \text{ran } B, \quad \ker \mathcal{P}_X = \ker U, \quad (8.7)$$

and

$$\text{rank } \mathcal{P}_X = \text{tr } \mathcal{P}_X = d + 1.$$

Its characteristic polynomial and synthesis intertwining law are

$$\chi_{\mathcal{P}_X}(t) = t^{q_m-d-1}(t-1)^{d+1}, \quad \text{Synth } \mathcal{P}_X = I_X \text{Synth}. \quad (8.8)$$

More generally, let  $p_0, \dots, p_d$  be any basis of  $\Pi_d$  and put

$$V_{in} = p_n(x_i), \quad \mathbf{D}_{kn} = h(\mathcal{D}_u p_n)(kh). \quad (8.9)$$

Then

$$\boxed{U\mathbf{D} = V, \quad B = \mathbf{D}V^{-1}, \quad \mathbf{D} = BV}. \quad (8.10)$$

For  $0 \leq r \leq d$  one also has

$$B(x_0^r, \dots, x_d^r)^T = h(\mathcal{A}_r(kh))_{k \in K_m}, \quad (8.11)$$

and Cauchy–Binet gives the exact partition of one

$$1 = \sum_{\substack{S \subset K_m \\ |S|=d+1}} \det U_{:,S} \det B_{S,:}. \quad (8.12)$$



*Proof.* Evaluate (8.1) at  $x = x_i$  to obtain  $(UB)_{ij} = \ell_j(x_i) = \delta_{ij}$ . Hence  $\mathcal{P}_X^2 = B(UB)U = BU$ ; the range, kernel, rank, and trace assertions are the standard consequences of a right inverse.

The  $n$ th column of  $UD$  consists of the nodal values of the exact synthesis of  $p_n$ , hence equals the  $n$ th column of  $V$ . Since the columns of  $V^{-1}$  are the coordinates of the cardinal polynomials in the basis  $(p_n)$ ,  $B = DV^{-1}$ , and multiplication by  $V$  gives  $D = BV$ . Taking  $p_r(x) = x^r$  proves (8.11). An idempotent is diagonalizable with eigenvalues zero and one, so its rank gives the characteristic polynomial. Moreover  $\text{Synth } BU = I_X \text{ Synth}$  because  $BU$  first samples the synthesized function and then synthesizes its cardinal interpolant. Finally apply Cauchy–Binet to  $\det(UB) = 1$ .  $\square$

The factorization has an exact probabilistic normalization. Put

$$\tilde{U} = hU, \quad \tilde{B} = h^{-1}B.$$

**Proposition 8.4** (Positive Markov encoder and forced signed decoder). *The matrix  $\tilde{U}$  is nonnegative and row-stochastic,  $\tilde{B}\mathbf{1} = \mathbf{1}$ , and*

$$\tilde{U}\tilde{B} = I_{d+1}. \quad (8.13)$$

*If two distinct rows of  $\tilde{U}$  overlap in a strictly positive column, then  $\tilde{B}$  necessarily has a negative entry.*

*Proof.* Nonnegativity is  $u \geq 0$ , and the partition of unity at admissible mesh gives  $h \sum_k u(x_i - kh) = 1$ . Since  $\sum_j \ell_j = 1$  and  $\mathcal{D}_u \mathbf{1} = \mathbf{1}$ , every row of  $\tilde{B}$  sums to one. Equation (8.13) is (8.6).

Suppose instead that  $\tilde{B} \geq 0$ . Its rows are then probability vectors. The equation  $e_i = \tilde{U}_{i,:} \tilde{B}$  expresses the extreme point  $e_i$  of the probability simplex as a convex combination of those rows. Every row used with positive weight must therefore equal  $e_i$ . A column positive in two different encoder rows would force the same decoder row to equal two different extreme points, a contradiction.  $\square$

The sign obstruction has a quantitative form. For a row-unital right inverse  $E$  of a nonnegative row-stochastic matrix  $A$ , write

$$V(E) = \max_k \sum_j |E_{kj}|, \quad \text{TV}(a, b) = \frac{1}{2} \|a - b\|_1.$$

**Theorem 8.5** (Decoder variation and the Lebesgue barrier). *If  $AE = I$  and  $E\mathbf{1} = \mathbf{1}$ , then for distinct rows  $i, j$ ,*

$$V(E) \geq \frac{2}{\|A_{i,:} - A_{j,:}\|_1} = \frac{1}{\text{TV}(A_{i,:}, A_{j,:})}. \quad (8.14)$$

*Some row of  $E$  has total negative mass at least*

$$\frac{1}{\|A_{i,:} - A_{j,:}\|_1} - \frac{1}{2}. \quad (8.15)$$

*For the Lagrange pair  $(A, E) = (\tilde{U}, \tilde{B})$ ,*

$$V(\tilde{B}) \geq \Lambda_X, \quad (8.16)$$

*where  $\Lambda_X$  is the Lebesgue constant of the node set.*

*Proof.* From  $(A_{i,:} - A_{j,:})E = e_i - e_j$ ,

$$2 \leq \sum_k |A_{ik} - A_{jk}| \|E_{k,:}\|_1 \leq \|A_{i,:} - A_{j,:}\|_1 V(E),$$

which proves (8.14). A row summing to one has negative mass  $(\|E_{k,:}\|_1 - 1)/2$ , giving (8.15).

For nodal data  $y$  with  $\|y\|_\infty \leq 1$ , the synthesized interpolant is  $\sum_k (By)_k u(x - kh)$ . Since  $\sum_k u(x - kh) = h^{-1}$ ,

$$\|I_X y\|_\infty \leq h^{-1} \|By\|_\infty = \|\tilde{B}y\|_\infty \leq V(\tilde{B}).$$

Taking the supremum over such  $y$  gives the operator norm  $\Lambda_X$  and proves (8.16).  $\square$

Thus changing from cardinal coordinates to smooth compactly supported atoms does not regularize interpolation. It preserves the interpolating polynomial exactly and must carry any Runge amplification into signed decoder coordinates.

### 8.3 The local shift space and its exact sequence

Take the canonical  $m = M = 2^d$  and use all literal translates meeting  $[-1, 1]$ :

$$\mathcal{V}_d^{\text{lit}} := \text{span}\{u(\cdot - k/M)|_{[-1,1]} : k \in K_M\}.$$

Let  $\mathcal{T}$  synthesize coefficient vectors into this space and let  $\mathcal{E}_X$  evaluate at the  $d + 1$  nodes. Finally put

$$\mathcal{Z}_X = \{f \in \mathcal{V}_d^{\text{lit}} : f(x_j) = 0 \text{ for all } j\}.$$

**Theorem 8.6** (Alias/sample exact sequence). *One has*

$$\dim \mathcal{V}_d^{\text{lit}} = 2M + d + 1, \quad \mathcal{V}_d^{\text{lit}} \cap \mathbb{R}[x] = \Pi_d, \quad (8.17)$$

$$\dim \ker \mathcal{T} = 2M - d - 2, \quad \dim \mathcal{Z}_X = 2M. \quad (8.18)$$

Moreover,

$$\boxed{\mathcal{V}_d^{\text{lit}} = \Pi_d \oplus \mathcal{Z}_X}, \quad (8.19)$$

and synthesis induces the short exact sequence

$$0 \longrightarrow \ker \mathcal{T} \longrightarrow \ker(\mathcal{E}_X \mathcal{T}) \xrightarrow{\mathcal{T}} \mathcal{Z}_X \longrightarrow 0. \quad (8.20)$$

Hence the zero eigenspace of the coefficient projector has the exact split

$$4M - d - 2 = (2M - d - 2) + 2M : \quad (8.21)$$

*genuine Fourier aliases plus nonzero local splines invisible at the nodes.*

*Proof.* Apply [theorem 3.10](#) to  $[-1, 1]$  with physical cell width  $M^{-1}$ , so the interval contains  $J = 2M$  cells. It gives  $\dim \mathcal{V}_d^{\text{lit}} = J + d + 1 = 2M + d + 1$  and the polynomial intersection  $\Pi_d$ . The coefficient space has dimension  $4M - 1$ , which gives the first kernel dimension.

Evaluation is onto because its restriction to  $\Pi_d$  is the usual Vandermonde isomorphism. Rank-nullity gives  $\dim \mathcal{Z}_X = 2M$ . The intersection  $\Pi_d \cap \mathcal{Z}_X$  is zero: a degree- $d$  polynomial with  $d + 1$  distinct roots vanishes. Dimensions prove (8.19).

If a coefficient vector is killed by  $\mathcal{E}_X \mathcal{T}$ , its synthesis lies in  $\mathcal{Z}_X$ . Conversely every member of  $\mathcal{Z}_X$  has a coefficient preimage, necessarily killed by the samples. The kernel of this restricted synthesis is exactly  $\ker \mathcal{T}$ , which proves exactness and the dimension split.  $\square$

The complete  $2M - d - 2$ -dimensional alias sector has the explicit confluent Fourier basis of [theorem 3.4](#). The complementary  $2M$ -dimensional nodal-zero sector depends on  $X$ ; it is the precise part that interpolation removes without synthesizing the zero function.

### 8.4 The minimal dictionary and its rank-one ghost

Transport the endpoint basis  $\mathcal{B}_d$  of [theorem 3.7](#) from  $[0, 1]$  to  $[-1, 1]$  and denote its  $d + 2$  normalized atoms by  $\Phi_0, \dots, \Phi_{d+1}$ . Let

$$\widehat{U}_{ir} = \Phi_r(x_i)$$

and let  $\widehat{B}$  contain the unique endpoint-basis coordinates of the cardinals. Then  $\widehat{U} \widehat{B} = I_{d+1}$ .

For  $0 \leq r \leq d + 1$ , delete column  $r$  from  $\widehat{U}$  and row  $r$  from  $\widehat{B}$ , and set

$$v_r = (-1)^r \det \widehat{U}_{\widehat{r}}, \quad \zeta_r = (-1)^r \det \widehat{B}_{\widehat{r}}. \quad (8.22)$$



**Theorem 8.7** (Rank-one complement and canonical nodal ghost). *The cofactor vectors satisfy*

$$\widehat{U}v = 0, \quad \zeta^T \widehat{B} = 0, \quad \zeta^T v = 1, \quad (8.23)$$

and

$$\boxed{I_{d+2} - \widehat{B}\widehat{U} = v\zeta^T.} \quad (8.24)$$

For  $a \in \mathbb{R}^{d+2}$  put

$$F_a = \sum_r a_r \Phi_r, \quad G_X = \sum_r v_r \Phi_r.$$

Then  $G_X(x_i) = 0$  for every node and

$$\boxed{F_a = I_X F_a + (\zeta^T a) G_X.} \quad (8.25)$$

In particular,  $F_a$  is a degree- $d$  polynomial on the interval if and only if  $\zeta^T a = 0$ .

*Proof.* Alternating maximal minors give a right-null vector of a full-row-rank matrix, hence  $\widehat{U}v = 0$ ; applied to  $\widehat{B}^T$  they give  $\zeta^T \widehat{B} = 0$ . Cauchy–Binet and  $\widehat{U}\widehat{B} = I$  yield

$$1 = \sum_r \det \widehat{U}_{\widehat{r}} \det \widehat{B}_{\widehat{r}} = \zeta^T v.$$

The complementary idempotent  $I - \widehat{B}\widehat{U}$  has rank one, range  $\ker \widehat{U} = \text{span}\{v\}$ , and left range  $\ker \widehat{B}^T = \text{span}\{\zeta\}$ . The normalization  $\zeta^T v = 1$  proves (8.24).

The endpoint coefficient vector of  $I_X F_a$  is  $\widehat{B}\widehat{U}a$ . Subtracting it from  $a$  and applying (8.24) proves (8.25). The ghost is nonpolynomial: if it belonged to  $\Pi_d$ , its  $d+1$  nodal zeros would make it zero, contradicting  $v \neq 0$  and independence of the endpoint basis. Thus the scalar defect (ghost) coefficient  $\zeta^T a$  vanishes exactly for polynomial synthesis.  $\square$

This reconciles two apparently different “defects.” The universal polynomiality functional of [theorem 4.1](#) is node independent; after normalization it is the same one-dimensional quotient coordinate as  $\zeta^T$ . The ghost  $G_X$  is a node-dependent representative of that universal nonpolynomial quotient, chosen to vanish at the interpolation nodes.

**Corollary 8.8** (Defect-completed square inverse). *The square matrix*

$$\mathcal{M}_X = \begin{pmatrix} \widehat{U} \\ \zeta^T \end{pmatrix} \in \mathbb{R}^{(d+2) \times (d+2)} \quad (8.26)$$

is invertible and

$$\boxed{\widehat{B} = \mathcal{M}_X^{-1} \begin{pmatrix} I_{d+1} \\ 0 \end{pmatrix}.} \quad (8.27)$$

In the normalized cell coordinate  $t \in [0, 1]$ , let

$$\mathcal{D}_d(t) = t^{d+1} + \mathcal{E}_d(t)$$

be the canonical defect of [theorem 4.2](#) and let  $I_\Theta$  interpolate at the affinely transported nodes  $\Theta = (\theta_0, \dots, \theta_d)$ . The defect-one representative of the evaluation kernel inside the transported  $(d+2)$ -dimensional endpoint-basis space is

$$\boxed{\mathcal{G}_{d,\Theta} = \mathcal{D}_d - I_\Theta \mathcal{D}_d = \omega_\Theta + \mathcal{E}_d - I_\Theta \mathcal{E}_d,} \quad \omega_\Theta(t) = \prod_{j=0}^d (t - \theta_j). \quad (8.28)$$

Thus the algebraic ghost  $v$  of [theorem 8.7](#) and the Fourier defect of [section 4](#) are the same quotient direction with two different normalizations.

*Proof.* If  $\mathcal{M}_X a = 0$ , then  $\zeta^T a = 0$ , so  $F_a$  is a polynomial of degree at most  $d$ . The equation  $\widehat{U}a = 0$  says that this polynomial has the  $d + 1$  distinct zeros  $x_j$ , hence it is zero. Independence of the endpoint atoms gives  $a = 0$ , proving invertibility. The two identities  $\widehat{U}\widehat{B} = I$  and  $\zeta^T \widehat{B} = 0$  are exactly

$$\mathcal{M}_X \widehat{B} = \begin{pmatrix} I \\ 0 \end{pmatrix},$$

which proves (8.27).

Both  $\mathcal{D}_d$  and  $I_\Theta \mathcal{D}_d$  belong to the normalized local shift space. Their difference vanishes at every  $\theta_j$ , and it is nonzero because its defect coordinate is one whereas the interpolating polynomial has defect coordinate zero. The evaluation kernel inside the transported  $(d + 2)$ -dimensional endpoint-basis space is one-dimensional, so this is its normalized generator. The full nodal-zero sector from (8.20) is much larger. Finally,  $t^{d+1} - I_\Theta t^{d+1} = \omega_\Theta$  because both sides are monic of degree  $d + 1$  and have precisely the nodes  $\theta_j$  as zeros.  $\square$

### 8.5 Nested grids and exact multiresolution algebra

Fix an integer  $s \geq 0$ . Let  $X^{(0)} \subset X^{(1)} \subset \dots \subset X^{(s)}$  be nested node sets, with  $|X^{(q)}| = d_q + 1$ , and let  $I_q$  be the corresponding interpolation projector. Let  $C_s$  be the coefficient space of a fixed terminal up-dictionary. Choose once and for all a synthesis map  $T_s : C_s \rightarrow C[-1, 1]$  and its canonical coefficient map  $E_s : \Pi_{d_s} \rightarrow C_s$  at the terminal degree, so  $T_s E_s = I$  on  $\Pi_{d_s}$ .

**Theorem 8.9** (Nested projector semilattice and coefficient flag). *For all  $0 \leq q, r \leq s$ ,*

$$\boxed{I_q I_r = I_r I_q = I_{\min(q,r)}}. \quad (8.29)$$

*The lifted coefficient operators*

$$\widehat{\mathcal{P}}_q^{(s)} := E_s I_q T_s \quad (8.30)$$

*satisfy the same multiplication law and  $\text{rank } \widehat{\mathcal{P}}_q^{(s)} = d_q + 1$ . Hence the details*

$$\widehat{\Delta}_0^{(s)} = \widehat{\mathcal{P}}_0^{(s)}, \quad \widehat{\Delta}_q^{(s)} = \widehat{\mathcal{P}}_q^{(s)} - \widehat{\mathcal{P}}_{q-1}^{(s)} \quad (q \geq 1) \quad (8.31)$$

*are pairwise-annihilating idempotents with ranks  $d_0 + 1$  and  $d_q - d_{q-1}$ , respectively.*

*Proof.* Assume  $q \leq r$ . The polynomial  $I_r f$  agrees with  $f$  on the smaller node set, so  $I_q I_r f = I_q f$ . Conversely  $I_q f \in \Pi_{d_q} \subseteq \Pi_{d_r}$ , so  $I_r I_q f = I_q f$ . This proves (8.29).

Since  $I_r T_s$  takes values in  $\Pi_{d_s}$ ,

$$\widehat{\mathcal{P}}_q^{(s)} \widehat{\mathcal{P}}_r^{(s)} = E_s I_q (T_s E_s) I_r T_s = E_s I_q I_r T_s.$$

The semilattice law follows. The range is exactly  $E_s(\Pi_{d_q})$ , proving the rank. Expanding products of the differences with the semilattice law proves idempotence, pairwise annihilation, and the rank differences.  $\square$

**Corollary 8.10** (Level-grouped self-series and dyadic absolute convergence). *For any function  $f$  for which  $I_q f \rightarrow f$  in a chosen function-space topology, put  $\Delta_0 = I_0$  and  $\Delta_q = I_q - I_{q-1}$ . For each  $q$  let  $\mathcal{U}_q p$  denote any finite up-atom synthesis supplied by theorem 8.1 for  $p \in \Pi_{d_q}$ , so that  $\mathcal{U}_q p = p$  on  $[-1, 1]$ . Then*

$$\boxed{\sum_{q=0}^N \mathcal{U}_q(\Delta_q f) = I_N f \quad \text{on } [-1, 1]} \quad (8.32)$$

*for every  $N$ . Therefore the level-grouped atom series converges exactly in the same topology, with exactly the same error, as the original interpolation sequence.*

Let  $J_n$  denote interpolation at the  $n + 1$  Chebyshev–Lobatto nodes. For every  $R > 0$ ,

$$\|u - J_n u\|_\infty = \mathcal{O}_R(n^{-R}). \quad (8.33)$$

The dyadic subsequence  $I_q^{\text{CL}} := J_{2^q}$  is nested. With  $\Delta_0^{\text{CL}} = I_0^{\text{CL}}$  and  $\Delta_q^{\text{CL}} = I_q^{\text{CL}} - I_{q-1}^{\text{CL}}$ , choose for each  $q$  a finite up-atom synthesis  $\mathcal{U}_q^{\text{CL}} : \Pi_{2^q} \rightarrow C[-1, 1]$  supplied by [theorem 8.1](#), with  $\mathcal{U}_q^{\text{CL}} p = p$  on  $[-1, 1]$ . Then

$$\boxed{\sum_{q=0}^{\infty} \|\Delta_q^{\text{CL}} u\|_\infty < \infty, \quad u = \sum_{q=0}^{\infty} \mathcal{U}_q^{\text{CL}}(\Delta_q^{\text{CL}} u)} \quad (8.34)$$

absolutely and uniformly on  $[-1, 1]$ . Despite that summability, no uniform geometric rate in the polynomial degree follows.

*Proof.* Every synthesized block equals the polynomial  $\Delta_q f$  on the target interval, so the finite sum telescopes to  $I_N f$ . This proves the exact error transfer in any topology where the expressions are defined.

For Chebyshev–Lobatto nodes, Lebesgue’s lemma gives  $\|u - J_n u\|_\infty \leq (1 + \Lambda_n)E_n(u)$  with  $\Lambda_n = \mathcal{O}(\log n)$ . Since the zero extension of  $u$  is  $C^\infty$ , Jackson estimates give  $E_n(u) = \mathcal{O}_r(n^{-r})$  for every  $r$ ; choosing  $r > R + 1$  proves (8.33).

The Chebyshev–Lobatto grids at degrees  $2^q$  are nested, and for  $q \geq 1$ ,

$$\|\Delta_q^{\text{CL}} u\|_\infty \leq \|u - J_{2^q} u\|_\infty + \|u - J_{2^{q-1}} u\|_\infty = \mathcal{O}_R(2^{-qR}).$$

Any  $R > 1$  proves summability. On the target interval every synthesized block equals its polynomial detail, so the Weierstrass test and finite telescoping identity give (8.34). A uniform geometric rate would extend  $u$  holomorphically through a Bernstein ellipse, contradicting flatness and nontrivial compact support.  $\square$

At each fixed level there is no second, hidden truncation error: the differential series defining  $\mathcal{D}_u$  terminates at the polynomial degree; compact support makes the lattice sum finite; and cardinal substitution is a finite node sum. Floating-point cancellation is a numerical phenomenon, not an additional mathematical approximation.

## 8.6 Geometric nodes and the Gaussian-binomial lift

The interpolation geometry and the dyadic alias geometry can be varied independently. To see this algebraically, take geometric nodes  $1, q, \dots, q^d$  over a characteristic-zero field, with distinct powers, and write

$$\Omega_{d+1}^{(q)}(x) = \prod_{r=0}^d (x - q^r) = \sum_{s=0}^{d+1} g_s x^s, \quad (8.35)$$

$$g_s = (-1)^{d+1-s} q^{\binom{d+1-s}{2}} \begin{bmatrix} d+1 \\ s \end{bmatrix}_q. \quad (8.36)$$

Synthetic division gives

$$\frac{\Omega_{d+1}^{(q)}(x)}{x - q^j} = \sum_{n=0}^d h_{n,j} x^n, \quad h_{n,j} = \sum_{s=n+1}^{d+1} g_s q^{j(s-n-1)}, \quad (8.37)$$

and

$$(\Omega_{d+1}^{(q)})'(q^j) = (-1)^j q^{jd-j(j+1)/2} (q; q)_j (q; q)_{d-j}. \quad (8.38)$$

For the literal up-function realization below, specialize to  $q \in [-1, 1]$  with the powers  $1, q, \dots, q^d$  distinct, and let  $m \in \mathbb{N}_{>0}$  be an admissible integer mesh.

**Proposition 8.11** (Gaussian-binomial/Appell coefficient matrix). *For this real specialization,  $k \in K_m$ , and  $0 \leq j \leq d$ , the coefficient of the  $j$ th geometric-node cardinal is*

$$B_{kj}^{(q)} = \frac{h}{(\Omega_{d+1}^{(q)})'(q^j)} \sum_{n=0}^d h_{n,j} \mathcal{A}_n(kh). \quad (8.39)$$

Thus a Gaussian-binomial row from node geometry is convolved, finitely, with the Rvachev–Appell row from moment deconvolution. The separate dyadic annihilator  $\prod_{r=1}^d [2^r]_z$  controls compression of the resulting atom row; it is not the same  $q$ -product.

*Proof.* The cardinal polynomial is  $\Omega_{d+1}^{(q)}(x)/[(x - q^j)(\Omega_{d+1}^{(q)})'(q^j)]$ . Insert (8.37), apply  $\mathcal{D}_u x^n = \mathcal{A}_n(x)$  term by term, sample at  $kh$ , and multiply by the quadrature weight  $h$  from (8.1).  $\square$

The proposition separates the two uses of  $q$ -algebra cleanly: geometric nodes create Gaussian binomial coefficients, whereas the binary sinc product creates the  $q$ -integer/Thue–Morse alias law. Their interaction is a genuine two-parameter finite transform, not an identification of the two products.

### 8.7 Appell–Vandermonde inversion and unavoidable amplification

The scale-ladder construction of section 5 gives a second, sparse cardinal realization. Let  $y_0, \dots, y_d$  be distinct, let  $L_j(y)$  be their cardinal polynomials, and expand

$$L_j(y) = \sum_{n=0}^d b_{nj} G_n(y).$$

Set

$$A_{in} = G_n(y_i), \quad B = (b_{nj})_{0 \leq n, j \leq d}, \quad D_n = n! 2^{\binom{n+1}{2}}. \quad (8.40)$$

**Theorem 8.12** (Appell–Vandermonde inverse and volume growth). *One has*

$$B = A^{-1}, \quad \det A = \prod_{0 \leq i < j \leq d} (y_j - y_i). \quad (8.41)$$

The raw two-atom amplitude map

$$T = \text{diag}(D_0, \dots, D_d) A^{-1}$$

therefore satisfies

$$\det T = \frac{2^{\binom{d+2}{3}} \prod_{n=0}^d n!}{\prod_{i < j} (y_j - y_i)}. \quad (8.42)$$

If the nodes lie in an interval of length two, then

$$\|T\|_2 \geq 2^{d(d-1)/6} \left( \prod_{n=0}^d n! \right)^{1/(d+1)}. \quad (8.43)$$

Thus even optimal node placement cannot remove the quadratic-exponential volume expansion caused by the dyadic integration ladder.

*Proof.* Evaluation of the cardinal expansion at  $y_i$  gives  $AB = I$ . Each  $G_n$  is monic of degree  $n$ , so replacing the monomial columns  $(1, y, \dots, y^d)$  by  $(G_0, \dots, G_d)$  is a unit-triangular column operation. It preserves the ordinary Vandermonde determinant and proves (8.41).

Multiply that determinant by

$$\prod_{n=0}^d D_n = \left( \prod_{n=0}^d n! \right) 2^{\sum_{n=0}^d \binom{n+1}{2}} = \left( \prod_{n=0}^d n! \right) 2^{\binom{d+2}{3}}$$

to obtain (8.42). Finally,  $\|T\|_2^{d+1} \geq |\det T|$  and  $\prod_{i < j} |y_j - y_i| \leq 2^{\binom{d+1}{2}}$ . Taking the  $(d+1)$ st root and using

$$\binom{d+2}{3} - \binom{d+1}{2} = \binom{d+1}{3}$$

gives (8.43).  $\square$

**Proposition 8.13** (Parity-adapted four-atom Appell ladder). *For  $n, r \geq 0$ , define on  $[-1, 1]$  the causal and anticausal Appell blocks*

$$\Phi_n^L(x) := D_n \left[ u \left( -1 + \frac{x+3}{2^{n+1}} \right) + u \left( -1 + \frac{x+1}{2^{n+1}} \right) \right] = G_n(x+2), \quad (8.44)$$

$$\Phi_n^R(x) := \Phi_n^L(-x) = G_n(2-x), \quad (8.45)$$

$$\mathcal{E}_r^{\text{sym}}(x) := \frac{1}{2} (\Phi_{2r}^L(x) + \Phi_{2r}^R(x)). \quad (8.46)$$

Then, for every  $r \geq 0$ ,  $\mathcal{E}_r^{\text{sym}}$  is an even monic polynomial of degree  $2r$ . Consequently, for every  $N \geq 0$ ,  $\mathcal{E}_0^{\text{sym}}, \dots, \mathcal{E}_N^{\text{sym}}$  form a basis of the even polynomials of degree at most  $2N$ : every such polynomial  $p$  has a unique expansion

$$p(x) = \sum_{r=0}^N \eta_r \mathcal{E}_r^{\text{sym}}(x), \quad x \in [-1, 1]. \quad (8.47)$$

Each  $\mathcal{E}_r^{\text{sym}}$  is a symmetric four-atom up-block. For  $r = 0$  the repeated atoms combine by the partition identity, so no artificial multiplicity is required. In particular, every even Fourier–Legendre cutoff  $S_N$  has this exact parity-adapted realization.

*Proof.* The two-atom identity (5.4), specialized to  $[a, b] = [-1, 1]$ , is exactly the first equality in (8.44); reflection gives the right block. Hence  $\mathcal{E}_r^{\text{sym}}(-x) = \mathcal{E}_r^{\text{sym}}(x)$ . The moment Appell polynomial  $G_{2r}$  is monic, and both  $G_{2r}(x+2)$  and  $G_{2r}(2-x)$  have leading term  $x^{2r}$ ; their average is therefore monic of degree  $2r$ . Relative to  $1, x^2, \dots, x^{2N}$ , the ordered family  $\mathcal{E}_0^{\text{sym}}, \dots, \mathcal{E}_N^{\text{sym}}$  has a unit-triangular coefficient matrix. It is therefore a basis, proving existence and uniqueness in (8.47). Expanding the two definitions of  $\Phi^L$  and  $\Phi^R$  displays the four atoms explicitly.  $\square$

## 8.8 Sharp interpolation rate and endpoint amplitude barrier

**Theorem 8.14** (Exact Chebyshev root-rate obstruction). *Let  $I_d^{\text{CL}}$  be interpolation at the  $d+1$  Chebyshev–Lobatto nodes. For the nonzero compactly supported  $C^\infty$  function  $u$ ,*

$$\limsup_{d \rightarrow \infty} \|u - I_d^{\text{CL}} u\|_\infty^{1/d} = 1. \quad (8.48)$$

Consequently there are no constants  $C > 0$ ,  $\rho > 1$ , and  $d_0$  for which

$$\|u - I_d^{\text{CL}} u\|_\infty \leq C \rho^{-d} \quad (d \geq d_0).$$

This assertion concerns an eventual uniform bound; it does not rule out faster exceptional subsequences.

*Proof.* The superalgebraic convergence in (8.33) implies the limsup in (8.48) is at most one. If it were less than one, choose  $\rho > 1$  below its reciprocal. The interpolation errors would then be eventually bounded by  $C\rho^{-d}$ , and the best degree- $d$  approximation errors would be no larger. Bernstein's converse theorem would extend  $u$  holomorphically to a Bernstein ellipse. Its restriction to the real axis would be analytic through an endpoint, impossible for a nonzero compactly supported function that is flat there. Hence the limsup is one.  $\square$

Using the constant-extension convention for  $F$  from the parent notation, put

$$\delta_d = (d+2)2^{-d} = 2^{-t_d}, \quad t_d = d - \log_2(d+2).$$

Define

$$c_{d,r} = 2^{d+1} - 2d - 3 + 2r, \quad 0 \leq r \leq d+1.$$

In the raw endpoint basis, write

$$P(x) = \sum_{r=0}^{d+1} a_r u\left(\frac{x - c_{d,r}}{2^{d+1}}\right), \quad x \in [-1, 1]. \quad (8.49)$$

The endpoint centers are chosen so that every displayed atom argument belongs to  $[-1, -1 + \delta_d]$ .

**Theorem 8.15** (Endpoint amplitude and quadratic-log barriers). *Every endpoint representation (8.49) satisfies*

$$\boxed{\|a\|_1 \geq \frac{\|P\|_\infty}{F_{\text{bd}}(\delta_d)}}. \quad (8.50)$$

Consequently, for any sequence  $P_d \in \Pi_d$  with  $\|P_d\|_\infty \geq 1$ ,

$$\boxed{\liminf_{d \rightarrow \infty} \frac{\log \|a^{(d)}\|_1}{d^2} \geq \frac{\log 2}{2}}. \quad (8.51)$$

The same leading lower bound holds after multiplying all amplitudes by  $2^d$ , as in the normalized endpoint basis.

*Proof.* For  $d \geq 2$ , every active argument is  $-1 + s$  with  $0 \leq s \leq \delta_d \leq 1$ , so monotonicity and  $u(-1 + s) = F_{\text{bd}}(s)$  give

$$0 \leq u\left(\frac{x - c_{d,r}}{2^{d+1}}\right) \leq F_{\text{bd}}(\delta_d)$$

for every active atom and  $x \in [-1, 1]$ . For  $d = 0, 1$  one instead uses  $0 \leq u \leq 1 = F_{\text{bd}}(\delta_d)$ , because  $\delta_d \geq 1$  under the stated constant extension. Hence in every degree  $|P(x)| \leq F_{\text{bd}}(\delta_d)\|a\|_1$ ; taking the supremum proves (8.50). The established endpoint law

$$-\log F_{\text{bd}}(2^{-t}) = \frac{\log 2}{2}t^2 + o(t^2) \quad (t \rightarrow \infty)$$

is equation dyadic-quadratic-law in Fabius\_Function\_and\_Rvachev\_Up.tex; its normalized limit is kernel-verified as fabiusLogProfile\_normalized\_tendsto in FabiusLogSquaredAsympotic.lean. This law and  $t_d/d \rightarrow 1$  now give (8.51). A factor  $2^d$  adds only  $d \log 2 = o(d^2)$  to the logarithm.  $\square$

### 8.9 Denominator support on an equispaced grid

Let  $d \geq 1$ , take the canonical synthesis mesh  $m = 2^d$ ,  $h = 2^{-d}$ , and put  $x_j = -1 + 2j/d$  for  $0 \leq j \leq d$ . Define

$$\mathcal{P}_d = \{p \text{ prime} : p \leq d+1\} \cup \bigcup_{1 \leq r \leq \lfloor d/2 \rfloor} \{p \text{ prime} : p \mid 2^{2r} - 1\}. \quad (8.52)$$

**Theorem 8.16** (Prime support of rational cardinal amplitudes). *For the equispaced grid and canonical mesh above, every prime occurring in the reduced denominator of any literal cardinal amplitude*

$$h(\mathcal{D}_u \ell_j)(kh), \quad 0 \leq j \leq d, \quad k \in K_m,$$

*belongs to  $\mathcal{P}_d$ .*

*Proof.* After clearing the common spacing, the  $j$ th cardinal is an integer polynomial divided by  $2^d j!(d-j)!$ . Its coefficients and derivatives therefore introduce only primes at most  $d$ . Multiplication by  $h = 2^{-d}$  and evaluation at  $kh \in \mathbb{Z}[1/2]$  introduce only the already listed prime 2.

The even cumulant

$$\kappa_{2r} = \frac{B_{2r}}{2r(1-4^{-r})}$$

can have denominator primes only from  $r$ , from  $2^{2r} - 1$ , and from  $B_{2r}$ . By von Staudt–Clausen, a prime in the last denominator satisfies  $p-1 \mid 2r$ , hence  $p \leq 2r+1 \leq d+1$ . Finally the reciprocal moments obey the explicit recurrence

$$\gamma_0 = 1, \quad \gamma_n = - \sum_{r=1}^n \binom{n-1}{r-1} \kappa_r \gamma_{n-r}. \quad (8.53)$$

Induction shows that addition and multiplication by the displayed integers introduce no denominator prime absent from the cumulants already used. Substitution in (8.3) proves the claim.  $\square$

**Problem 8.17** (Exact denominator valuations). *Determine the exact  $p$ -adic valuations of the least common denominator of the whole cardinal-amplitude matrix for equispaced, dyadic, and Chebyshev-rational nodes. Theorem 8.16 determines the possible primes but not the exponents; the latter mix factorial valuations, von Staudt–Clausen denominators, the factors  $2^{2r} - 1$ , and substantial cancellation.*

## 9 Legendre synthesis, spectral closure, and self-reconstruction

The Legendre reports compose two exact transforms: Fourier–Legendre analysis of the up-function and finite synthesis of each polynomial mode by up-atoms. Their principal notational ambiguity is removed once and for all by the box

$$\boxed{\begin{aligned} Q_n^- &:= M_u(D)^{-1} P_n, & R_n &:= M_u(D) P_n, \\ X_- &:= M_u(D)^{-1} x M_u(D), & X_+ &:= M_u(D) x M_u(D)^{-1}. \end{aligned}} \quad (9.1)$$

Thus  $Q_n^-$  is the deconvolved polynomial whose samples are synthesis coefficients, whereas  $R_n$  is the distinct moment-smoothed polynomial used in the Appell connection of section 10.4. If  $K(z) = \log M_u(z)$ , the commutator identity  $g(D)x = xg(D) + g'(D)$  gives

$$X_- = x - K'(D), \quad X_+ = x + K'(D). \quad (9.2)$$

We also write  $\mathcal{D}_u := M_u(D)^{-1}$  for the deconvolution operator.

Use the classical Legendre normalization  $P_n(1) = 1$ , for which

$$h_n = \int_{-1}^1 P_n(x)^2 dx = \frac{2}{2n+1} \quad (9.3)$$



and define the even Fourier–Legendre coefficients

$$a_n := \frac{4n+1}{2} \int_{-1}^1 u(x) P_{2n}(x) dx. \quad (9.4)$$

The primary exposition’s section on the Fourier–Legendre expansion of up in `Fabius_Function_and_d_Rvachev_Up.tex` proves

$$u(x) = \sum_{n=0}^{\infty} a_n P_{2n}(x) \quad (9.5)$$

absolutely and uniformly on  $[-1, 1]$ , with rational  $a_n$ ; the exact Lean anchors are `summable_abs_rvachevLegendreCoefficient`, `hasSum_rvachevLegendreSeries`, and `hasSum_rvachevLegendreSeries_uniform` in `FabiusLegendreSeries.lean`.

### 9.1 Finite mode synthesis and the legal block loop

**Theorem 9.1** (Literal Legendre synthesis and Jacobi closed form). *Let  $\ell \geq 0$  and let  $M \in \mathbb{N}_{>0}$  satisfy  $\nu_2(M) \geq \ell$ . Then, for  $x \in [-1, 1]$ ,*

$$P_\ell(x) = \frac{1}{M} \sum_{|k| < 2M} Q_\ell^-(k/M) u(x - k/M). \quad (9.6)$$

The deconvolved polynomial has the finite Jacobi expansion

$$Q_\ell^-(x) = \sum_{r=0}^{\lfloor \ell/2 \rfloor} \frac{\gamma_{2r}}{(2r)!} \frac{(\ell+2r)!}{2^{2r} \ell!} P_{\ell-2r}^{(2r, 2r)}(x). \quad (9.7)$$

In particular  $Q_\ell^-(-x) = (-1)^\ell Q_\ell^-(x)$  and all its coefficients are rational.

*Proof.* Equation (9.6) is [theorem 8.1](#) applied to the polynomial  $P_\ell$ , or equivalently (6.5) with unit atom radius. Expand  $M_u(D)^{-1} = \sum_r \gamma_r D^r / r!$ . Odd coefficients vanish by evenness, and the classical derivative formula

$$D^{2r} P_\ell(x) = 2^{-2r} \frac{(\ell+2r)!}{\ell!} P_{\ell-2r}^{(2r, 2r)}(x)$$

gives (9.7). Jacobi polynomials with integral parameters have rational coefficients; every summand has the parity of  $\ell$ .  $\square$

For every integer  $n \geq 0$ , put  $m_n = 4^n$  and define its finite even-mode train

$$B_n(x) := \frac{a_n}{m_n} \sum_{|k| < 2m_n} Q_{2n}^-(k/m_n) u(x - k/m_n). \quad (9.8)$$

Then  $B_n = a_n P_{2n}$  on the target interval.

**Theorem 9.2** (Rank-one atomized spectral details). *For every integer  $n \geq 0$  and  $f \in L^2([-1, 1])$ , define*

$$\Delta_n f := \frac{4n+1}{2} \langle f, P_{2n} \rangle P_{2n}. \quad (9.9)$$

*Then  $\Delta_n$  is the self-adjoint orthogonal rank-one projector onto  $\text{span}\{P_{2n}\}$ , and it has the finite atomized form*

$$(\Delta_n f)(x) = \frac{4n+1}{2m_n} \langle f, P_{2n} \rangle \sum_{|k| < 2m_n} Q_{2n}^-(k/m_n) u(x - k/m_n) \quad (x \in [-1, 1]). \quad (9.10)$$

For all  $m, n \geq 0$ ,

$$\boxed{\Delta_n \Delta_m = \delta_{nm} \Delta_n, \quad \Delta_n u = B_n.} \quad (9.11)$$

Thus the block loop is exactly the orthogonal spectral resolution of  $u$ , with each spectral detail itself realized by a finite up-train.

*Proof.* The Legendre norm is  $\langle P_{2n}, P_{2n} \rangle = 2/(4n+1)$ , so (9.9) is the usual formula for the orthogonal projector onto a one-dimensional subspace. Legendre orthogonality gives  $\Delta_n \Delta_m = \delta_{nm} \Delta_n$ . Substituting the finite synthesis (9.6) with  $M = m_n$  proves (9.10). Finally,  $\langle u, P_{2n} \rangle = 2a_n/(4n+1)$  by (9.4), whence  $\Delta_n u = a_n P_{2n} = B_n$ .  $\square$

**Theorem 9.3** (Blockwise self-loop and common-mesh partial sums). *For every  $x \in [-1, 1]$ ,*

$$\boxed{u(x) = \sum_{n=0}^{\infty} B_n(x),} \quad (9.12)$$

where the series converges absolutely and uniformly when each finite train is kept as one block.

For an integer cutoff  $N \geq 0$ , put

$$S_N = \sum_{n=0}^N a_n P_{2n}, \quad C_N^- = \mathcal{D}_u S_N = \sum_{n=0}^N a_n Q_{2n}^-. \quad (9.13)$$

Then  $\deg S_N \leq 2N$ , and the sufficient common mesh  $M_N = 4^N$  gives

$$\boxed{S_N(x) = \frac{1}{M_N} \sum_{|k| < 2M_N} C_N^-(k/M_N) u(x - k/M_N).} \quad (9.14)$$

The right side converges uniformly to  $u$ . The mesh  $4^N$  is least for universal exactness on the whole space  $\Pi_{2N}$ ; no target-specific minimality for  $S_N$  is asserted.

*Proof.* Substituting (9.6) into (9.5) gives (9.12); its  $n$ th block is exactly  $a_n P_{2n}$ . Since  $\|P_j\|_{L^\infty[-1,1]} \leq 1$  and  $\sum_n |a_n| < \infty$ , the Weierstrass test proves absolute uniform block convergence.

The degree inequality in (9.13) is unconditional; equality would require  $a_N \neq 0$ . Since  $\nu_2(4^N) = 2N$ , applying (9.6) to the single polynomial  $S_N$  gives (9.14). Its convergence is precisely  $S_N \rightarrow u$ . The universal least-mesh statement is `isLeast_rvachevCombExactThrough_even` from `CompositeMeshSharpness.lean`, with the claim boundary already stated after [theorem 3.1](#).  $\square$

**Warning 9.4** (Blockwise does not mean atomwise). Finite regrouping within  $B_n$  is always legal, and the outer series may be rearranged as a series of blocks. It does not follow that the individual atoms can be flattened into an absolutely or unconditionally convergent double series. The repeated central coefficient in [theorem 10.23](#) proves the contrary.

## 9.2 Orthogonal blocks and positive energy series

**Theorem 9.5** (Orthogonal translate blocks and exact tails). *For  $m, n \geq 0$ ,*

$$\boxed{\int_{-1}^1 B_n(x) B_m(x) dx = \delta_{nm} \frac{2a_n^2}{4n+1}.} \quad (9.15)$$

Consequently

$$\|u - S_N\|_2^2 = \sum_{n > N} \frac{2a_n^2}{4n+1}, \quad (9.16)$$

$$\|u\|_2^2 = \sum_{n=0}^{\infty} \frac{2a_n^2}{4n+1}. \quad (9.17)$$

For the Legendre Sturm–Liouville operator

$$\mathcal{L} = -D(1 - x^2)D, \quad \lambda_n = 2n(2n + 1),$$

one further has

$$\int_{-1}^1 (1 - x^2) B'_n B'_m dx = \delta_{nm} \lambda_n \frac{2a_n^2}{4n + 1}, \quad (9.18)$$

and, for every real  $s \geq 0$  in the spectral domain,

$$\|(I + \mathcal{L})^{s/2} u\|_2^2 = \sum_{n=0}^{\infty} (1 + \lambda_n)^s \frac{2a_n^2}{4n + 1}. \quad (9.19)$$

*Proof.* On  $[-1, 1]$ ,  $B_n = a_n P_{2n}$ , so (9.15) is ordinary Legendre orthogonality. Orthogonal convergence gives (9.16) and (9.17). Since  $\mathcal{L}P_{2n} = \lambda_n P_{2n}$ , integration by parts gives

$$\int (1 - x^2) B'_n B'_m = \langle \mathcal{L}B_n, B_m \rangle,$$

which proves (9.18). The last identity is the spectral theorem for the nonnegative self-adjoint Legendre operator.  $\square$

The individual translates inside the blocks are not orthogonal. Expanding (9.15) instead gives an exact cross-scale atom-Gram cancellation identity; this is the literal content formalized by `sum_rvac hevLegendreAtomCoefficient_mul_gram`.

Let

$$\mathcal{E}_F := \int_0^1 F_{\text{bd}}(t)^2 dt. \quad (9.20)$$

Because  $u(x) = F_{\text{bd}}(1 - |x|)$  on  $[-1, 1]$ ,  $\|u\|_2^2 = 2\mathcal{E}_F$ .

**Theorem 9.6** (Fabius square energy: rational series and sinc integrals). *One has the three exact representations*

$$\boxed{\mathcal{E}_F = \sum_{n=0}^{\infty} \frac{a_n^2}{4n + 1}} \quad (9.21)$$

$$= \int_0^{\infty} \prod_{j=0}^{\infty} \text{sinc}_{\pi}^2(\xi/2^j) d\xi \quad (9.22)$$

$$= \frac{1}{2\pi} \int_0^{\infty} \prod_{r=1}^{\infty} \left( \frac{\sin(t/2^r)}{t/2^r} \right)^2 dt. \quad (9.23)$$

Every finite partial sum in (9.21) is a nonnegative rational number, and the sequence of partial sums is nondecreasing. It is strict at the  $n$ th step exactly when  $a_n \neq 0$ .

*Proof.* Divide (9.17) by two and use  $\|u\|_2^2 = 2\mathcal{E}_F$ . Rationality follows from  $a_n \in \mathbb{Q}$ . Parseval for the Fourier convention of section 2 gives  $\|u\|_2^2 = \int_{\mathbb{R}} |\widehat{u}(\xi)|^2 d\xi$ . Evenness, division by two, and the sinc product give (9.22); the substitution  $t = 2\pi\xi$  gives (9.23). Monotonicity and its equality criterion follow because the added term is  $a_n^2/(4n + 1)$ .  $\square$

**Corollary 9.7** (Nested dyadic-Fabius square series). *For every integer  $n \geq 0$ , define the finite dyadic quantity*

$$\mathcal{R}_n^{\text{dy}} := \sum_{j=0}^n (-1)^{n+j} \binom{2n}{n+j} \binom{2n+2j}{2n} (2j)! 2^{\binom{2j+1}{2}} F_{\text{bd}}(2^{-2j-1}). \quad (9.24)$$

Then, for every  $n \geq 0$ , the coefficient identity in the following box holds, and its substitution into (9.21) gives the displayed energy series:

$$a_n = 4^{-n}(4n+1)\mathcal{R}_n^{\text{dy}}, \quad \mathcal{E}_F = \sum_{n=0}^{\infty} (4n+1)16^{-n}(\mathcal{R}_n^{\text{dy}})^2. \quad (9.25)$$

Every inner quantity  $\mathcal{R}_n^{\text{dy}}$  is rational. Hence this is a positive outer series whose  $n$ th term is the square of a finite rational combination of the nested values  $F_{\text{bd}}(2^{-1}), F_{\text{bd}}(2^{-3}), \dots, F_{\text{bd}}(2^{-2n-1})$ .

*Proof.* The coefficient identity in (9.25) is kernel-verified as `rvachevLegendreCoefficient_eq_fabius_sum` in `FabiusLegendreCoefficients.lean`. Here is its finite human-readable derivation. Put  $\mu_{2j} = \int_{-1}^1 x^{2j} u(x) dx$ . Expanding  $P_{2n}$  in monomials in (9.4) gives

$$a_n = \frac{4n+1}{2^{2n+1}} \sum_{j=0}^n (-1)^{n+j} \binom{2n}{n+j} \binom{2n+2j}{2n} \mu_{2j}.$$

The exact moment–Fabius bridge from the primary exposition is

$$\mu_{2j} = 2(2j)! 2^{\binom{2j+1}{2}} F_{\text{bd}}(2^{-2j-1});$$

obtained by integrating the dyadic derivative-cell formula; its dyadic values are kernel-verified by `fabiusDyadic_cast`. Substitution proves the formula for  $a_n$ . Insert its square into (9.21) and cancel one factor  $4n+1$  to obtain the second identity. The same exact dyadic-value theorem gives rationality.  $\square$

The four exact cutoffs

$$\frac{1}{4}, \quad \frac{7}{18}, \quad \frac{3271}{8100}, \quad \frac{3246043}{8037225}$$

are kernel-verified in `FabiusLegendreRationalEnergy.lean`; the Fourier and sinc equalities are kernel-verified in `FabiusSquareEnergyFourier.lean`. A decimal evaluation of  $\mathcal{E}_F$  is not needed for any theorem here.

### 9.3 Translated Legendre analysis and exact biorthogonality

**Definition 9.8** (Translated Legendre analysis kernel). For  $m \geq 0$  and  $c \in \mathbb{R}$ , set

$$\Lambda_m(c) = \frac{2m+1}{2} \int_{-1}^1 u(x-c) P_m(x) dx. \quad (9.26)$$

**Theorem 9.9** (Analysis kernel and Fourier–Bessel representation). For every  $m \geq 0$ , the function  $\Lambda_m$  belongs to  $C_c^\infty(\mathbb{R})$ , has support contained in  $[-2, 2]$ , and satisfies

$$\Lambda_m(-c) = (-1)^m \Lambda_m(c), \quad (9.27)$$

$$\Lambda_{2n}(0) = a_n, \quad \Lambda_{2n+1}(0) = 0 \quad (n \geq 0), \quad (9.28)$$

$$\Lambda_m(c) = (2m+1)(-i)^m \int_{\mathbb{R}} \widehat{u}(\xi) j_m(2\pi\xi) e^{2\pi i \xi c} d\xi. \quad (9.29)$$

Here  $j_m$  is the spherical Bessel function.

*Proof.* Let  $\chi_m(x) = P_m(x) \mathbf{1}_{[-1,1]}(x)$ . Evenness of  $u$  gives  $\Lambda_m = (2m+1)(u * \chi_m)/2$ . Convolution of a compactly supported  $C^\infty$  function with a compactly supported integrable function is  $C^\infty$ , and supports add to  $[-2, 2]$ . Reflection gives (9.27); (9.28) is (9.4) and oddness.

The classical integral representation gives  $\widehat{\chi}_m(\xi) = 2(-i)^m j_m(2\pi\xi)$ . The rapid decay of  $\widehat{u}$  makes the product integrable, so Fourier inversion of the convolution proves (9.29).  $\square$

**Lemma 9.10** (Rational dyadic truncated moments). *If  $a, b \in \mathbb{Z}[1/2]$ ,  $a \leq b$ , and  $r \geq 0$ , then*

$$\int_a^b x^r u(x) dx \in \mathbb{Q}. \quad (9.30)$$

Consequently  $\Lambda_m(c) \in \mathbb{Q}$  for every dyadic  $c$ .

*Proof.* Intersect  $[a, b]$  with the support  $[-1, 1]$ , which preserves dyadic endpoints. On  $[-1, 1]$  the differential refinement equation implies

$$u(x) = \frac{1}{2} u' \left( \frac{x-1}{2} \right),$$

because the other child  $u(x-2)$  vanishes. For a polynomial  $p$  and dyadic  $a, b \in [-1, 1]$ , substitute  $t = (x-1)/2$  and integrate by parts:

$$\begin{aligned} \int_a^b p(x) u(x) dx &= [p(2t+1)u(t)]_{(a-1)/2}^{(b-1)/2} \\ &\quad - \int_{(a-1)/2}^{(b-1)/2} 2p'(2t+1)u(t) dt. \end{aligned}$$

The boundary values of  $u$  at dyadic arguments are rational by `fabiusDyadic_cast` in `DyadicAnalytic.lean`, the exact dyadic-value theorem recorded in the primary exposition. Induction on  $\deg p$  therefore proves the first claim, with the constant case terminating after the boundary term.

For dyadic  $c$ , substitute  $x = c + y$  in (9.26). Expand  $P_m(y + c)$  and split at the dyadic endpoints of  $[-1, 1] \cap [-1 - c, 1 - c]$ . The result is a finite rational combination of the moments just proved.  $\square$

**Theorem 9.11** (Legendre-up dyadic biorthogonality). *Let  $\ell \geq 0$ ,  $M \in \mathbb{N}_{>0}$  with  $\nu_2(M) \geq \ell$ , and  $m \geq 0$ . Then*

$$\boxed{\frac{1}{M} \sum_{|k| < 2M} Q_\ell^-(k/M) \Lambda_m(k/M) = \delta_{m\ell}.} \quad (9.31)$$

*Proof.* Multiply (9.6) by  $(2m+1)P_m(x)/2$  and integrate over  $[-1, 1]$ . The left side is  $\delta_{m\ell}$  by Legendre orthogonality, and each atom integral is  $\Lambda_m(k/M)$ .  $\square$

**Corollary 9.12** (Finite biorthogonal matrices and projector identities). *Let  $d \geq 0$  and  $M \in \mathbb{N}_{>0}$  satisfy  $\nu_2(M) \geq d$ . Then the matrices indexed by  $K_M := \{k \in \mathbb{Z} : |k| < 2M\}$ ,*

$$A_{k\ell}^{(d)} = M^{-1} Q_\ell^-(k/M), \quad L_{mk}^{(d)} = \Lambda_m(k/M) \quad (k \in K_M, 0 \leq m, \ell \leq d)$$

*satisfy  $L^{(d)} A^{(d)} = I_{d+1}$ . Consequently  $A^{(d)} L^{(d)}$  is an oblique projector of rank and trace  $d+1$ , and*

$$d+1 = \frac{1}{M} \sum_{|k| < 2M} \sum_{\ell=0}^d Q_\ell^-(k/M) \Lambda_\ell(k/M), \quad (9.32)$$

$$1 = \sum_{\substack{S \subset K_M \\ |S|=d+1}} \det L_{:,S}^{(d)} \det A_{S,:}^{(d)}. \quad (9.33)$$

*Proof.* Because  $\ell \leq d$ , the hypothesis  $\nu_2(M) \geq d$  permits theorem 9.11 for every matrix entry, so  $L^{(d)} A^{(d)} = I_{d+1}$ . Idempotence, rank, and trace follow from this right-inverse identity; taking the trace and applying Cauchy–Binet give (9.32) and (9.33).  $\square$

#### 9.4 Reverse local closure and rational spectral identities

For  $x \in [-1, 1]$  and  $c \in [-2, 2]$  put

$$I(c) = [-1, 1] \cap [c - 1, c + 1].$$

The shifted canonical expansion is

$$u(x - c) = \mathbf{1}_{I(c)}(x) \sum_{r=0}^{\infty} a_r P_{2r}(x - c). \quad (9.34)$$

It is absolutely and uniformly convergent on  $[-1, 1] \times [-2, 2]$  because  $\sum_r |a_r| < \infty$  and  $|P_{2r}| \leq 1$  on  $[-1, 1]$ . For  $\ell, r \geq 0$  and  $M \in \mathbb{N}_{>0}$  with  $\nu_2(M) \geq \ell$ , define

$$H_{\ell r}^{(M)}(x) := \frac{1}{M} \sum_{|k| < 2M} Q_{\ell}^{-}(k/M) \mathbf{1}_{I(k/M)}(x) P_{2r}(x - k/M). \quad (9.35)$$

For  $m, r \geq 0$ , define

$$\mathcal{C}_{m,r}(c) := \frac{2m+1}{2} \int_{I(c)} P_m(x) P_{2r}(x - c) dx. \quad (9.36)$$

**Theorem 9.13** (Reverse local loop and rational spectral closure). *Let  $\ell \geq 0$  and  $M \in \mathbb{N}_{>0}$  satisfy  $\nu_2(M) \geq \ell$ . Then*

$$P_{\ell}(x) = \sum_{r=0}^{\infty} a_r H_{\ell r}^{(M)}(x) \quad (9.37)$$

*absolutely and uniformly on  $[-1, 1]$ . Moreover, for every  $m \geq 0$ ,*

$$\Lambda_m(c) = \sum_{r=0}^{\infty} a_r \mathcal{C}_{m,r}(c) \quad (9.38)$$

*absolutely and uniformly for  $c \in [-2, 2]$ . For  $m, r \geq 0$  define*

$$T_{m\ell r}^{(M)} := \frac{1}{M} \sum_{|k| < 2M} Q_{\ell}^{-}(k/M) \mathcal{C}_{m,r}(k/M), \quad (9.39)$$

*Then every  $T_{m\ell r}^{(M)}$  is rational and*

$$\delta_{m\ell} = \sum_{r=0}^{\infty} a_r T_{m\ell r}^{(M)}. \quad (9.40)$$

*If  $m + \ell$  is odd, then  $T_{m\ell r}^{(M)} = 0$  for every  $r \geq 0$ .*

*Proof.* Insert (9.34) into the finite synthesis (9.6). Interchanging a finite sum with an absolutely uniform series is legitimate and produces (9.37). Insert the same expansion into (9.26); dominated uniform convergence gives (9.38).

Now substitute (9.38) into (9.31). The sum over  $k$  is finite, so its interchange with the absolutely convergent  $r$ -series gives (9.40). At rational  $k/M$ , the polynomial  $Q_{\ell}^{-}$  is rational, while (9.36) is the integral of a rational polynomial over an interval with rational endpoints; hence every  $T$  is rational. Finally,

$$Q_{\ell}^{-}(-c) \mathcal{C}_{m,r}(-c) = (-1)^{m+\ell} Q_{\ell}^{-}(c) \mathcal{C}_{m,r}(c),$$

so odd total parity cancels in the symmetric  $k$ -sum.  $\square$

The functions  $H_{\ell r}^{(M)}$  are only piecewise polynomial, but their infinite weighted sum is the smooth polynomial  $P_{\ell}$ . The cancellation of all boundary jets is made explicit below.

### 9.5 Gram geometry and finite projector frames

Fix an integer  $N \geq 0$ , put  $M = 4^N$ , and write  $K_M := \{k \in \mathbb{Z} : |k| < 2M\}$ . Let

$$\phi_k(x) = u(x - k/M), \quad \mathbf{D}_{kn} = M^{-1}Q_{2n}^-(k/M), \quad 0 \leq n \leq N,$$

and define

$$G_{k\ell} = \int_{-1}^1 \phi_k(x)\phi_\ell(x) dx, \quad H = \text{diag}(h_0, h_2, \dots, h_{2N}).$$

**Theorem 9.14** (Exact Gram isometry and coefficient projector). *The synthesis matrix satisfies*

$$\boxed{\mathbf{D}^T G \mathbf{D} = H.} \quad (9.41)$$

Thus

$$\Pi_{\text{coef}} := \mathbf{D} H^{-1} \mathbf{D}^T G \quad (9.42)$$

is an idempotent, is self-adjoint for the Gram seminorm ( $G\Pi_{\text{coef}} = \Pi_{\text{coef}}^T G$ ), has range  $\text{col } \mathbf{D}$ , and has trace  $N + 1$ .

The even Christoffel–Darboux kernel

$$K_N^e(x, y) = \sum_{n=0}^N \frac{P_{2n}(x)P_{2n}(y)}{h_{2n}} \quad (9.43)$$

has the two-sided atom factorization

$$\boxed{K_N^e(x, y) = \frac{1}{M^2} \sum_{k, \ell \in K_M} \tilde{K}_N^e(k/M, \ell/M) u(x - k/M) u(y - \ell/M),} \quad (9.44)$$

where

$$\tilde{K}_N^e(s, t) = \sum_{n=0}^N \frac{Q_{2n}^-(s)Q_{2n}^-(t)}{h_{2n}}.$$

Its coefficient matrix is positive semidefinite of rank  $N + 1$ .

*Proof.* The  $(m, n)$  entry of  $\mathbf{D}^T G \mathbf{D}$  is the inner product of the finite syntheses of  $P_{2m}$  and  $P_{2n}$ , hence is  $h_{2n}\delta_{mn}$ . The projector identities follow by substituting  $\mathbf{D}^T G \mathbf{D} = H$ ; cyclicity of trace gives  $\text{tr } \Pi_{\text{coef}} = N + 1$ .

Substitute (9.6) in both variables of (9.43) and interchange finite sums. The coefficient matrix is  $\mathbf{D} H^{-1} \mathbf{D}^T$ , hence positive semidefinite;  $\mathbf{D}$  has full column rank, so its rank is  $N + 1$ .  $\square$

There is also a one-sided synthesis/analysis factorization. Put

$$K_N^\sharp(t, y) = \sum_{n=0}^N \frac{Q_{2n}^-(t)P_{2n}(y)}{h_{2n}}, \quad g_k(x) = u(x - k/M), \quad \eta_k(y) = M^{-1}K_N^\sharp(k/M, y).$$

Then

$$\Pi_N^e : L^2([-1, 1]) \longrightarrow \text{span}\{P_0, P_2, \dots, P_{2N}\}$$

denotes the orthogonal projector, and

$$K_N^e(x, y) = \sum_{k \in K_M} g_k(x)\eta_k(y), \quad (\Pi_N^e f)(x) = \sum_{k \in K_M} g_k(x)\langle f, \eta_k \rangle. \quad (9.45)$$



It follows by applying polynomial synthesis in the  $x$  variable. Since  $\Pi_N^\circ$  is an orthogonal projector of rank  $N + 1$ ,

$$\sum_k \langle g_k, \eta_k \rangle = N + 1, \quad (9.46)$$

$$\sum_{k,\ell} \langle g_k, g_\ell \rangle \langle \eta_k, \eta_\ell \rangle = N + 1. \quad (9.47)$$

The first is the trace and the second the squared Hilbert–Schmidt norm of the same projector. These identities constrain any attempt to choose a better-conditioned nullspace gauge.

## 9.6 Refinement null trains and a lifting law

For  $p \in \Pi_d$  put  $p^\sharp = \mathcal{D}_u p$ . At a mesh  $m \in \mathbb{N}_{>0}$  with  $\nu_2(m) \geq d$ , define the canonical finite coefficient row

$$(F_m p)_k = m^{-1} p^\sharp(k/m), \quad |k| < 2m.$$

Let  $E_r$  embed a coarse row into the  $r$ -times finer grid by putting it at indices divisible by  $r$  and zero elsewhere.

**Theorem 9.15** (Interlevel null train and refinement cocycle). *Let  $d \geq 0$ ,  $p \in \Pi_d$ , and  $r, s, m \in \mathbb{N}_{>0}$  with  $\nu_2(m) \geq d$ . Then, on  $[-1, 1]$ ,*

$$\boxed{\frac{1}{rm} \sum_{|k| < 2rm} (1 - r \mathbf{1}_{r|k}) p^\sharp(k/(rm)) u(x - k/(rm)) = 0.} \quad (9.48)$$

Equivalently,  $G_{m;r} p := F_{rm} p - E_r F_m p$  belongs to the local synthesis kernel. These gauge corrections satisfy

$$\boxed{G_{m;rs} p = G_{rm;s} p + E_s G_{m;r} p.} \quad (9.49)$$

*Proof.* Write the fine synthesis of  $p$ . Rewrite the coarse synthesis on the fine index set by setting  $k = rj$ ; its coefficient becomes  $r(rm)^{-1} p^\sharp(k/(rm))$  on divisible indices and zero elsewhere. Subtracting proves (9.48). Both sides of (9.49) telescope to  $F_{rsm} p - E_s E_r F_m p$ .  $\square$

For every integer  $N \geq 0$ , let  $a^{(N)}$  be the common-mesh coefficient row for  $S_N$ :

$$a_k^{(N)} = 4^{-N} C_N^-(k/4^N).$$

**Corollary 9.16** (Exact quarter-grid Legendre lifting). *For every integer  $N \geq 0$ , put  $m = 4^N$ . For  $|k| < 8m$ ,*

$$\boxed{a_k^{(N+1)} - (E_4 a^{(N)})_k = \frac{1 - 4 \mathbf{1}_{4|k}}{4m} C_N^-(k/(4m)) + \frac{a_{N+1}}{4m} Q_{2N+2}^-(k/(4m)).} \quad (9.50)$$

The first term synthesizes zero on  $[-1, 1]$ ; the second synthesizes the new orthogonal block  $B_{N+1}$ .

*Proof.* At  $t = k/(4m)$ ,

$$a_k^{(N+1)} = \frac{1}{4m} (C_N^-(t) + a_{N+1} Q_{2N+2}^-(t)),$$

whereas  $(E_4 a^{(N)})_k = 4 \mathbf{1}_{4|k} C_N^-(t)/(4m)$ . Subtract. Apply [theorem 9.15](#) to  $p = S_N$  for the first term and (9.6) to the new mode for the second.  $\square$

### 9.7 Flatness and finite boundary identities

**Theorem 9.17** (Endpoint flatness and finite jet sum rules). *For every  $s \geq 0$ ,*

$$\sum_{n \geq \lceil s/2 \rceil} a_n \frac{(2n+s)!}{2^s s! (2n-s)!} = 0, \quad (9.51)$$

*and the series is absolutely convergent.*

*For every  $\ell \geq 0$ , every  $M \in \mathbb{N}_{>0}$  with  $\nu_2(M) \geq \ell$ , and every  $s \geq 0$ ,*

$$\frac{1}{M} \sum_{|k| < 2M} Q_\ell^-(k/M) u^{(s)}(1 - k/M) = \begin{cases} \frac{(\ell+s)!}{2^s s! (\ell-s)!}, & s \leq \ell, \\ 0, & s > \ell. \end{cases} \quad (9.52)$$

*At the left endpoint the right side is multiplied by  $(-1)^{\ell+s}$ .*

*Proof.* Repeated self-adjointness of  $\mathcal{L}$  shows that the Legendre coefficients of the  $C^\infty$  flat function  $u$  decay faster than every power. Hence (9.5) may be differentiated any fixed number of times. Evaluate its  $s$ th derivative at  $x = 1$ , use  $u^{(s)}(1) = 0$ , and use the classical formula

$$P_j^{(s)}(1) = \begin{cases} 2^{-s} (j+s)! / [s! (j-s)!], & s \leq j, \\ 0, & s > j. \end{cases}$$

This proves (9.51) and its absolute convergence.

Differentiate the finite identity (9.6); no convergence argument is needed. Evaluation at 1 gives (9.52), and reflection gives the left endpoint.  $\square$

**Corollary 9.18** (Boundary jets as finite Thue–Morse identities). *Let  $\ell, L, s$  be nonnegative integers with  $L \geq \ell$ , and set  $M = 2^L$ . For  $1 \leq k \leq 2M - 1$ , let  $j_{k,s} \in \{0, \dots, 2^s - 1\}$  and  $y_{k,s} \in [-1, 1)$  be the unique cell coordinates satisfying*

$$1 - \frac{k}{M} = -1 + \frac{2j_{k,s} + 1 + y_{k,s}}{2^s}. \quad (9.53)$$

*Then*

$$\frac{2^{\binom{s+1}{2}}}{M} \sum_{k=1}^{2M-1} Q_\ell^-(k/M) (-1)^{\text{wt}_2(j_{k,s})} u(y_{k,s}) = \begin{cases} \frac{(\ell+s)!}{2^s s! (\ell-s)!}, & s \leq \ell, \\ 0, & s > \ell. \end{cases} \quad (9.54)$$

*All values  $y_{k,s}$  are dyadic. Thus every displayed identity is an exact rational cancellation combining the Bell–Bernoulli polynomial samples  $Q_\ell^-(k/M)$  with a level- $s$  Thue–Morse sign row.*

*Proof.* The exact derivative-cell law is

$$u^{(s)}\left(-1 + \frac{2j+1+y}{2^s}\right) = 2^{\binom{s+1}{2}} (-1)^{\text{wt}_2(j)} u(y), \quad 0 \leq j < 2^s, \quad -1 \leq y \leq 1.$$

It is the theorem `iteratedDeriv_rvachev_cell` in the primary Lean development. The half-open cells in (9.53) partition  $[-1, 1)$ , so the stated coordinates exist uniquely. In (9.52), every term with  $k < 0$  is evaluated strictly to the right of the support, while  $k = 0$  is the flat endpoint 1; all those derivatives vanish. Apply the cell law to the remaining indices  $1 \leq k < 2M$  and substitute. At a shared cell boundary the other convention has  $y = 1$  instead of  $y = -1$ ; both values of  $u$  vanish, so the identity is independent of the convention. Dyadicity and rationality follow from the definitions and the exact dyadic-value theorem.  $\square$

**Corollary 9.19** (Central-binomial dyadic cancellation). *For  $n \geq 0$  and  $M = 4^n$ ,*

$$\boxed{Q_{2n}^-(0) + 2 \sum_{k=1}^{M-1} Q_{2n}^-(k/M)u(k/M) = (-1)^n \binom{2n}{n}.} \quad (9.55)$$

*Proof.* Evaluate (9.6) for  $P_{2n}$  at  $x = 0$ . The boundary terms vanish, and parity pairs  $k$  with  $-k$ . Multiply by  $M = 4^n$  and use  $P_{2n}(0) = (-1)^n 4^{-n} \binom{2n}{n}$ .  $\square$

## 10 Legendre operator transmutation, parity, and arithmetic

The preceding chapter describes the finite analysis/synthesis loop. This chapter records what that loop reveals about the polynomial families  $Q_n^- = M_u(D)^{-1}P_n$  and  $R_n = M_u(D)P_n$ . The distinction is essential:  $Q_n^-$  supplies synthesis samples, whereas  $R_n$  is the moment-smoothed Legendre family. Both are triangular transforms of the classical Legendre basis, but their natural Hilbert geometries and conjugated coordinate operators point in opposite directions.

### 10.1 Two pullback geometries and the Favard obstruction

For polynomials define

$$\langle f, g \rangle_- = \int_{-1}^1 (M_u(D)f)(x)(M_u(D)g)(x) dx, \quad \langle f, g \rangle_+ = \int_{-1}^1 (\mathcal{D}_u f)(x)(\mathcal{D}_u g)(x) dx. \quad (10.1)$$

Both forms are positive definite because the two triangular differential operators are invertible on every finite polynomial space.

**Theorem 10.1** (Dual pullback Legendre geometries). *For all  $m, n \geq 0$ ,*

$$\boxed{\langle Q_m^-, Q_n^- \rangle_- = h_n \delta_{mn}, \quad \langle R_m, R_n \rangle_+ = h_n \delta_{mn}.} \quad (10.2)$$

*Consequently either family, truncated at degree  $d$ , is an orthogonal basis of  $\Pi_d$  for its indicated inner product.*

*Proof.* By definition,  $M_u(D)Q_n^- = P_n$  and  $\mathcal{D}_u R_n = P_n$ . Both equalities in (10.2) are therefore ordinary Legendre orthogonality. The differential transforms preserve degree and leading coefficient, so the first  $d + 1$  members of either family form a basis of  $\Pi_d$ .  $\square$

This exact orthogonality is not integration against a hidden scalar weight.

**Proposition 10.2** (No scalar moment measure and the explicit Favard defect). *There is no finite signed measure  $\rho$ , with all polynomial moments, such that*

$$\langle f, g \rangle_- = \int f(x)g(x) d\rho(x)$$

*for all polynomials. Moreover, if  $\widehat{Q}_n^-$  denotes the monic normalization, then*

$$\begin{aligned} \widehat{Q}_2^- &= x^2 - \frac{4}{9}, & \widehat{Q}_3^- &= x^3 - \frac{14}{15}x, \\ \widehat{Q}_4^- &= x^4 - \frac{32}{21}x^2 + \frac{1072}{4725}, \end{aligned} \quad (10.3)$$

*and*

$$\boxed{\widehat{Q}_4^- = x\widehat{Q}_3^- - \frac{62}{105}\widehat{Q}_2^- - \frac{8}{225}\widehat{Q}_0^-} \quad (10.4)$$

*Thus the scalar three-term Favard recurrence fails already at degree four, even for an indefinite quasi-definite scalar functional.*

*Proof.* Every scalar moment form makes multiplication by  $x$  symmetric. Here  $M_u(D)x = x$  and  $M_u(D)x^2 = x^2 + \mu_2 = x^2 + 1/9$ , so

$$\langle x, x \rangle_- = \int_{-1}^1 x^2 dx = \frac{2}{3}, \quad \langle 1, x^2 \rangle_- = \int_{-1}^1 \left( x^2 + \frac{1}{9} \right) dx = \frac{8}{9}.$$

The inequality contradicts scalar symmetry.

Applying the finite reciprocal-moment series  $\mathcal{D}_u = \sum_r \gamma_r D^r / r!$  to  $P_2, P_3, P_4$  and dividing by their leading coefficients gives (10.3). Substitution and collection of powers proves (10.4). A monic parity-alternating scalar orthogonal sequence would have  $\hat{Q}_{n+1} = x\hat{Q}_n - \lambda_n \hat{Q}_{n-1}$ ; the nonzero  $\hat{Q}_0$  term is impossible.  $\square$

There is no contradiction between this proposition and theorem 10.1: the latter transports the coordinate operator along with the inner product. Scalar multiplication by  $x$  is not that transported coordinate.

## 10.2 Conjugated recurrence, Sturm operator, and projection kernel

Let  $K(z) = \log M_u(z)$  and recall

$$X_- = x - K'(D), \quad X_+ = x + K'(D).$$

Define

$$\mathcal{L} = -D(1 - x^2)D, \quad \mathcal{L}_- = \mathcal{D}_u \mathcal{L} M_u(D).$$

**Theorem 10.3** (Deconvolved Legendre operator calculus). *With the convention  $Q_{-1}^- := 0$ , the deconvolved family satisfies, for every  $n \geq 0$ ,*

$$(n+1)Q_{n+1}^- = (2n+1)X_- Q_n^- - nQ_{n-1}^-, \quad (10.5)$$

$$\mathcal{L}_- = -D(1 - X_-^2)D, \quad \mathcal{L}_- Q_n^- = n(n+1)Q_n^-. \quad (10.6)$$

The operator  $\mathcal{L}_-$  is symmetric for  $\langle \cdot, \cdot \rangle_-$ . For every  $d \geq 0$ , put

$$\tilde{K}_d(x, y) = \sum_{n=0}^d \frac{Q_n^-(x)Q_n^-(y)}{h_n}, \quad (10.7)$$

one has, for  $f \in \Pi_d$ ,

$$f(x) = \langle f, \tilde{K}_d(x, \cdot) \rangle_-, \quad (10.8)$$

$$M_u(D_x)M_u(D_y)\tilde{K}_d(x, y) = K_d(x, y), \quad (10.9)$$

$$(X_{-,x} - X_{-,y})\tilde{K}_d(x, y) = \frac{d+1}{2}(Q_{d+1}^-(x)Q_d^-(y) - Q_d^-(x)Q_{d+1}^-(y)). \quad (10.10)$$

*Proof.* Conjugate the classical Legendre recurrence by  $\mathcal{D}_u$ . The commutator identity  $g(D)x = xg(D) + g'(D)$  gives  $\mathcal{D}_u x M_u(D) = X_-$ , proving (10.5). Since the differential transforms commute with  $D$  and are mutual inverses,

$$\mathcal{D}_u x^2 M_u(D) = (\mathcal{D}_u x M_u(D))^2 = X_-^2,$$

which proves the operator formula and eigenvalue equation in (10.6). The intertwining identity  $M_u(D)\mathcal{L}_- = \mathcal{L}M_u(D)$  reduces symmetry to ordinary Legendre integration by parts; the factor  $1 - x^2$  kills the boundary term.

Expand  $f$  in the orthogonal  $Q_n^-$  basis to obtain (10.8). Applying  $M_u(D)$  in both variables gives (10.9). Finally apply  $\mathcal{D}_{u,x}\mathcal{D}_{u,y}$  to the classical Christoffel–Darboux identity

$$(x - y)K_d(x, y) = \frac{d+1}{2}(P_{d+1}(x)P_d(y) - P_d(x)P_{d+1}(y));$$

transporting  $x$  and  $y$  gives (10.10).  $\square$

### 10.3 Gauss cardinal synthesis and weighted singular values

Fix  $d \geq 0$ . Let  $\xi_0, \dots, \xi_d$  be the Gauss–Legendre nodes, let  $\omega_j > 0$  be their quadrature weights, and put

$$K_d(x, y) = \sum_{n=0}^d \frac{P_n(x)P_n(y)}{h_n}, \quad K_d^\sharp(t, y) = \sum_{n=0}^d \frac{Q_n^-(t)P_n(y)}{h_n}. \quad (10.11)$$

Let  $m \in \mathbb{N}_{>0}$  satisfy  $\nu_2(m) \geq d$ , put  $h = m^{-1}$  and  $K_m = \{k \in \mathbb{Z} : |k| < 2m\}$ , and define  $\mathbf{D}_{kn} = hQ_n^-(kh)$  for  $k \in K_m$ ,  $0 \leq n \leq d$ ,  $V_{jn} = P_n(\xi_j)$  for  $0 \leq j, n \leq d$ ,  $W = \text{diag}(\omega_0, \dots, \omega_d)$ , and  $H = \text{diag}(h_0, \dots, h_d)$ .

**Theorem 10.4** (Christoffel–Darboux cardinals and weighted SVD transfer). *The Gauss cardinal polynomials and their literal up-atom coefficients are*

$$\ell_j(x) = \omega_j K_d(x, \xi_j), \quad B_{kj} = h\omega_j K_d^\sharp(kh, \xi_j). \quad (10.12)$$

Moreover

$$O = W^{1/2} V H^{-1/2} \quad (10.13)$$

is orthogonal and

$$\boxed{B W^{-1/2} = \mathbf{D} H^{-1/2} O^T.} \quad (10.14)$$

Thus  $B W^{-1/2}$  and  $\mathbf{D} H^{-1/2}$  have exactly the same singular values.

*Proof.* Gauss quadrature with  $d + 1$  nodes is exact through degree  $2d + 1$ , hence on products of two polynomials of degree at most  $d$ . Therefore  $V^T W V = H$  and (10.13) is orthogonal. The reproducing property of  $K_d$  gives  $\omega_j K_d(\xi_i, \xi_j) = \delta_{ij}$ ; hence  $\ell_j = \omega_j K_d(\cdot, \xi_j)$ . Apply the general polynomial synthesis theorem in the first variable of this finite kernel to get the coefficient formula in (10.12).

In matrix form the basis-independent factorization of theorem 8.3 is  $B = \mathbf{D} V^{-1}$ . From  $V^T W V = H$ ,  $V^{-1} = H^{-1} V^T W$ . Therefore

$$B W^{-1/2} = \mathbf{D} H^{-1} V^T W^{1/2} = \mathbf{D} H^{-1/2} (W^{1/2} V H^{-1/2})^T,$$

which is (10.14). Right multiplication by an orthogonal matrix preserves singular values.  $\square$

### 10.4 Moment-smoothed Legendre/Appell connection

For every integer  $m \geq 0$ , write

$$P_{2m}(x) = \sum_{j=0}^m p_{m,j} x^{2j}, \quad p_{m,j} = (-1)^{m-j} \frac{(2m+2j)!}{2^{2m}(m-j)!(m+j)!(2j)!}. \quad (10.15)$$

Since  $\mathcal{A}_n = \mathcal{D}_u x^n$ , the smoothed polynomial  $R_{2m} = M_u(D) P_{2m}$  gives the connection from  $P_{2m}$  to the Rvachev–Appell basis.

**Theorem 10.5** (Forward and inverse Legendre–Appell transforms). *For every integer  $m \geq 0$  and every  $0 \leq r \leq m$ , set*

$$C_{m,r} = \sum_{j=r}^m p_{m,j} \frac{(2j)!}{(2r)!} \frac{\mu_{2(j-r)}}{(2(j-r))!}. \quad (10.16)$$

Then, for every  $m \geq 0$ ,

$$\boxed{R_{2m}(x) = \sum_{r=0}^m C_{m,r} x^{2r}, \quad P_{2m}(x) = \sum_{r=0}^m C_{m,r} \mathcal{A}_{2r}(x).} \quad (10.17)$$

Conversely, for every integer  $r \geq 0$ ,

$$\mathcal{A}_{2r}(x) = \sum_{m=0}^r D_{r,m} P_{2m}(x), \quad (10.18)$$

where, for  $0 \leq m \leq r$ ,

$$D_{r,m} = \sum_{s=0}^{r-m} \binom{2r}{2s} \gamma_{2s} \frac{(4m+1)2^{2m}(2r-2s)!(r-s+m)!}{(r-s-m)!(2r-2s+2m+1)!}. \quad (10.19)$$

For every integer  $N \geq 0$ , let  $C^{(N)}$  and  $D^{(N)}$  be the square matrices indexed by  $0, \dots, N$ , with entries  $C_{m,r}$  and  $D_{r,m}$  in their respective lower triangles and zeros above them. These finite matrices are mutual inverses, and

$$\det C^{(N)} = \prod_{m=0}^N 2^{-2m} \binom{4m}{2m}. \quad (10.20)$$

The determinant is unchanged for every normalized even formal moment series: no subleading moment enters it.

*Proof.* Apply  $M_u(D) = \sum_s \mu_s D^s / s!$  to (10.15). Odd moments vanish, and the coefficient of  $x^{2r}$  is exactly (10.16). Applying  $\mathcal{D}_u$  to the first identity in (10.17) gives the second.

The direct Appell expansion

$$\mathcal{A}_{2r}(x) = \sum_{s=0}^r \binom{2r}{2s} \gamma_{2s} x^{2r-2s}$$

and the classical monomial formula

$$x^{2q} = \sum_{m=0}^q \frac{(4m+1)2^{2m}(2q)!(q+m)!}{(q-m)!(2q+2m+1)!} P_{2m}(x)$$

give (10.19) after collecting  $P_{2m}$ . The two displayed expansions are inverse changes of basis, so their finite matrices are mutual inverses. Finally  $M_u(D)$  preserves leading coefficients; hence  $C_{m,m} = 2^{-2m} \binom{4m}{2m}$ . Multiplication of the triangular diagonal proves (10.20) and its stated universality.  $\square$

**Corollary 10.6** (Smoothed transmutation and its dual geometry). *With the convention  $R_{-1} := 0$ , the moment-smoothed family satisfies, for every  $n \geq 0$ ,*

$$(n+1)R_{n+1} = (2n+1)X_+ R_n - nR_{n-1}, \quad (10.21)$$

$$[(1 - X_+^2)D^2 - 2X_+ D + n(n+1)]R_n = 0. \quad (10.22)$$

It is orthogonal for the dual pullback form  $\langle \cdot, \cdot \rangle_+$  of (10.1).

*Proof.* Apply  $M_u(D)$  to the classical recurrence and differential equation. Conjugation is an algebra homomorphism,  $D$  commutes with  $M_u(D)$ , and  $M_u(D)x\mathcal{D}_u = X_+$ . Orthogonality is the second identity of theorem 10.1.  $\square$

## 10.5 Legendre moments completed by one Thue–Morse row

Fix an integer  $d \geq 0$ . For  $0 \leq r \leq d+1$  put

$$c_{d,r} = 2^{d+1} - 2d - 3 + 2r, \quad \Phi_{d,r}(x) = 2^{-d} u\left(\frac{x - c_{d,r}}{2^{d+1}}\right), \quad (10.23)$$

the normalized endpoint basis on  $[-1, 1]$ . Its polynomiality certificate may be normalized as

$$\tau_{d,r} = (-1)^{r+1} \varepsilon_{\lfloor (2^{d+1}-d-2+r)/2 \rfloor}, \quad 0 \leq r \leq d+1. \quad (10.24)$$

Thus  $\sum_r b_r \Phi_{d,r}$  belongs to  $\Pi_d$  if and only if  $\tau_d^T b = 0$ , by [theorem 4.1](#). Define

$$(\mathbb{L}_d)_{n,r} = \frac{2n+1}{2} \int_{-1}^1 \Phi_{d,r}(x) P_n(x) dx \quad (0 \leq n \leq d), \quad \mathcal{Q}_d = \begin{pmatrix} \mathbb{L}_d \\ \tau_d^T \end{pmatrix}. \quad (10.25)$$

**Theorem 10.7** (Legendre–Thue–Morse coordinate completion). *The matrix  $\mathcal{Q}_d$  is invertible. Its first  $d+1$  inverse columns are the endpoint coordinates of  $P_0, \dots, P_d$ . Its last inverse column is the unique coefficient vector whose first  $d+1$  Legendre moments vanish and whose defect coordinate  $\tau_d^T b$  equals one.*

*Proof.* Suppose  $\mathcal{Q}_d b = 0$ . The last row makes the endpoint synthesis a polynomial in  $\Pi_d$ . The preceding rows say that all of its Legendre coefficients through degree  $d$  vanish; therefore the polynomial is zero. Endpoint-basis independence gives  $b = 0$ , proving invertibility. For the coordinate vector of  $P_n$ , Legendre orthogonality produces the  $n$ th standard basis vector in the first block and polynomiality produces zero in the last row. The final-column characterization is the remaining standard basis vector under the inverse.  $\square$

**Theorem 10.8** (Compressed one-scale Legendre closure). *For every integer  $N \geq 0$ , set  $d = 2N$ . Let  $\beta_{d;n,r}$  be the unique endpoint coordinates*

$$P_n(x) = \sum_{r=0}^{d+1} \beta_{d;n,r} \Phi_{d,r}(x), \quad -1 \leq x \leq 1, \quad 0 \leq n \leq d, \quad (10.26)$$

and define

$$b_{N,r}^{\text{end}} := \sum_{m=0}^N a_m \beta_{2N;2m,r} \quad (0 \leq r \leq 2N+1). \quad (10.27)$$

Then the Fourier–Legendre cutoff has the exact  $2N+2$ -atom representation

$$S_N(x) = 4^{-N} \sum_{r=0}^{2N+1} b_{N,r}^{\text{end}} u\left(\frac{x - c_{2N,r}}{2 \cdot 4^N}\right), \quad -1 \leq x \leq 1, \quad (10.28)$$

and its coefficient row obeys the finite Thue–Morse certificate

$$\sum_{r=0}^{2N+1} \tau_{2N,r} b_{N,r}^{\text{end}} = 0. \quad (10.29)$$

Consequently the right side of (10.28) converges uniformly to  $u$  as  $N \rightarrow \infty$ .

*Proof.* The first  $d+1$  inverse columns in [theorem 10.7](#) are precisely the vectors  $\beta_{d;0}, \dots, \beta_{d;d}$ , proving existence and uniqueness in (10.26). Take  $d = 2N$ , multiply the representation of  $P_{2m}$  by  $a_m$ , and sum over  $0 \leq m \leq N$ . Interchanging these two finite sums gives (10.27); inserting (10.23) and  $2^{-2N} = 4^{-N}$  gives (10.28). Each column  $\beta_{2N;2m}$  lies in  $\ker \tau_{2N}^T$ , so linearity proves (10.29). Finally the finite train is identically  $S_N$ , and (9.5) converges uniformly to  $u$ .  $\square$

**Warning 10.9** (The endpoint dictionary changes with the cutoff). Equation (10.28) complements the raw common-mesh train (9.14): the latter displays  $4 \cdot 4^N - 1$  sampled centers, whereas endpoint reduction leaves exactly  $2N+2$  atoms. In the compressed limit the scale, centers, and coefficients all depend on  $N$ . Uniform convergence therefore makes no coordinatewise or atomwise limiting claim.



### 10.6 Parity anatomy and a central reflected-pair basis

Return to the normalized cell  $t \in [0, 1]$ . Reflection  $(\mathcal{R}f)(t) = f(1 - t)$  sends the local atom  $\psi_k$  to  $\psi_{1-k}$ . Write  $V_d^\pm$  and  $\Pi_d^\pm$  for the two reflection eigenspaces. Let

$$\mathcal{D}_d(t) = t^{d+1} + \mathcal{E}_d(t)$$

be the canonical defect of [theorem 4.2](#).

**Lemma 10.10** (Reflection character of the canonical defect). *The periodic residue and the one-dimensional quotient satisfy*

$$\mathcal{E}_d(1 - t) = (-1)^{d+1} \mathcal{E}_d(t), \quad \mathcal{R}|_{V_d/\Pi_d} = (-1)^{d+1}. \quad (10.30)$$

*Proof.* Replace  $t$  by  $1 - t$  in the odd-frequency Fourier series for  $\mathcal{E}_d$  and then replace the frequency by its negative. Integrality of the frequencies and evenness of  $\hat{u}$  leave the factor  $(-1)^{d+1}$ . Moreover,

$$\mathcal{R}\mathcal{D}_d - (-1)^{d+1} \mathcal{D}_d = (1 - t)^{d+1} - (-1)^{d+1} t^{d+1} \in \Pi_d,$$

because the leading terms cancel. This proves the quotient character.  $\square$

**Theorem 10.11** (Parity split of the local up-spline space). *For every  $N \geq 0$ ,*

$$\boxed{V_{2N}^+ = \Pi_{2N}^+, \quad V_{2N+1}^- = \Pi_{2N+1}^-}. \quad (10.31)$$

*Thus every symmetric member of  $V_{2N}$  is already an even polynomial, and every antisymmetric member of  $V_{2N+1}$  is already an odd polynomial about the cell midpoint.*

*Proof.* The local decomposition  $V_d = \Pi_d \oplus \text{span}\{\mathcal{D}_d\}$  makes  $V_d/\Pi_d$  one-dimensional. By [lemma 10.10](#), the extra direction is antisymmetric when  $d = 2N$  and symmetric when  $d = 2N + 1$ . Hence the opposite parity sector has the same dimension as its polynomial subspace; containment then gives the two equalities.  $\square$

For the determinant argument put

$$A_d(z) = \prod_{m=1}^d [2^m]_z, \quad N_d^{\text{rel}} = \deg A_d = 2^{d+1} - d - 2. \quad (10.32)$$

The following generating function packages the endpoint reduction used repeatedly in the source reports.

**Lemma 10.12** (Endpoint-tail generating function). *Let  $d$  be a nonnegative integer,  $M = 2^d$ ,  $K = -M + 1$ ,  $R = N_d^{\text{rel}}$ , and  $A_d(z) = \sum_{j=0}^R \alpha_j z^j$ . Suppose an active coefficient row  $q$  is supported on the initial  $R$  centers  $K, \dots, K + R - 1$ . If  $b_s$  are its coordinates on the consecutive endpoint basis beginning at  $K + R$ , then*

$$\boxed{\sum_{s \geq 0} b_s z^s = \frac{\sum_{p=0}^{R-1} q_{K+p} \text{rev}_p A_d(z)}{A_d(z)}}, \quad \text{rev}_p A_d(z) = \sum_{j=0}^p \alpha_{p-j} z^j. \quad (10.33)$$

*Only the first  $d + 2$  coefficients are needed for the endpoint basis.*

*Proof.* Extend each initial impulse by the monic annihilator recurrence. After shifting  $K$  to zero, palindromicity of  $A_d$  writes that recurrence as  $\sum_{j=0}^R \alpha_j n_{m-j} = 0$  for  $m \geq R$ . For the impulse at  $p < R$ , its generating function  $N_p$  satisfies

$$A_d(z)N_p(z) = z^p \sum_{j=0}^{R-p-1} \alpha_j z^j = z^p A_d(z) - z^R \text{rev}_p A_d(z).$$

The endpoint coordinate is the original row minus this null extension. At index  $R + s$  the original row is zero, so its coordinate is the coefficient of  $z^s$  in  $\text{rev}_p A_d/A_d$ . Linearity over the initial impulses proves (10.33).  $\square$

Fix an integer  $N \geq 0$ , put  $d = 2N$  and  $M = 4^N$ , and define the reflected central pairs

$$S_r^{(N)} = \psi_{-r} + \psi_{r+1}, \quad 0 \leq r \leq N. \quad (10.34)$$

Under  $t = (x + 1)/2$ , the corresponding physical pair is

$$\mathcal{S}_r^{(N)}(x) := \frac{1}{4^N} \left[ u\left(\frac{x - (2r + 1)}{2 \cdot 4^N}\right) + u\left(\frac{x + (2r + 1)}{2 \cdot 4^N}\right) \right]. \quad (10.35)$$

They belong to  $\Pi_{2N}^+$  by [theorem 10.11](#). Express them in the endpoint basis, obtaining a  $(2N+2) \times (N+1)$  coordinate matrix; let  $B_N$  be its first  $N + 1$  rows.

**Theorem 10.13** (Central  $q$ -determinant). *For every integer  $N \geq 1$ , define*

$$\sigma_N = [z^N] \frac{1}{(1-z)A_{2N-1}(z)} = [z^N] \frac{(1-z)^{2N-2}}{\prod_{m=1}^{2N-1} (1-z^{2^m})}, \quad C_N = 2A_{2N-1}(1) = 2^{1+N(2N-1)}. \quad (10.36)$$

Then

$$\det B_N = (-1)^N (C_N \sigma_N - 1). \quad (10.37)$$

This determinant is an odd, nonzero integer. For  $N = 0$  the unique minor is one.

*Proof.* For  $N = 0$ , the degree-zero endpoint basis and the partition identity give  $B_0 = (1)$ , so its unique minor is one.

For  $N = 1$ , the endpoint annihilator is  $A_2(z) = (1+z)(1+z+z^2+z^3)$ . Direct endpoint reduction gives the two central-pair columns

$$(3, -2, 2, -3)^T, \quad (2, -1, 1, -2)^T.$$

Thus  $B_1 = \begin{pmatrix} 3 & 2 \\ -2 & -1 \end{pmatrix}$  has determinant 1. Also  $\sigma_1 = [z](1-z^2)^{-1} = 0$  and  $C_1 = 4$ , so  $1 = (-1)(4\sigma_1 - 1)$  as required.

Suppose henceforth that  $N \geq 2$ . In [lemma 10.12](#) take  $d = 2N$  and  $R = N_{2N}^{\text{rel}} = 2M - 2N - 2$ . The two impulses in the  $r$ th central pair have shifted positions

$$p_- = M - r - 1, \quad p_+ = M + r \quad (0 \leq r \leq N).$$

They satisfy  $0 \leq p_-, p_+ < R$ , because  $3N + 2 < 4^N = M$  for  $N \geq 2$ . The endpoint-tail lemma is therefore applicable to every column below.

Write  $A_{2N-1}(z) = \sum c_j z^j$  and  $\ell_j = \sum_{q=0}^j c_q$ . Let  $L_N$  be the unit lower-triangular Toeplitz matrix with entries  $(L_N)_{sr} = \ell_{s-r}$  for  $s \geq r$ . Apply [lemma 10.12](#) to the two impulses at positions  $M - r - 1$  and  $M + r$ . Using

$$A_{2N}(z) = A_{2N-1}(z)[M]_z$$

and convolving by  $A_{2N-1}(z)/(1-z)$  gives  $L_N B_N = F_N$ , where

$$(F_N)_{sr} = \begin{cases} C_N, & r < s, \\ C_N - \ell_{r-s}, & r \geq s. \end{cases} \quad (10.38)$$

Indeed, below degree  $N < M$  the factor  $(1 - z^M)^{-1}$  contributes only its constant term; the two reversed prefixes then sum to  $2A_{2N-1}(1)$ , except for the displayed truncated tail.

Since  $\det L_N = 1$ , it remains to evaluate  $\det F_N$ . Subtract row  $s+1$  from row  $s$  for  $0 \leq s < N$ . The leading  $N$ -by- $N$  block becomes the upper-triangular Toeplitz matrix  $-(c_{r-s})$ , with determinant  $(-1)^N$ . If  $A_{2N-1}^{-1} = \sum g_j z^j$ , inversion of that block is convolution by  $g_j$ . Its Schur complement is

$$C_N \sum_{j=0}^N g_j - 1 = C_N [z^N] \frac{1}{(1-z)A_{2N-1}(z)} - 1 = C_N \sigma_N - 1.$$

This proves (10.37). The inverse series has integer coefficients because  $A_{2N-1}(0) = 1$  and the polynomial is monic over  $\mathbb{Z}$ . Thus  $\sigma_N \in \mathbb{Z}$ ; as  $C_N$  is even, the final factor is odd and cannot vanish.  $\square$

**Corollary 10.14** (Central reflected-pair basis and exact Gram duality). *For every integer  $N \geq 0$ , the  $N+1$  pairs  $S_0^{(N)}, \dots, S_N^{(N)}$  form a basis of  $\Pi_{2N}^+$ . Hence every even Legendre mode  $P_{2m}$ ,  $0 \leq m \leq N$ , has a unique rational expansion*

$$P_{2m}(x) = \sum_{r=0}^N c_{m,r}^{(N)} \mathcal{S}_r^{(N)}(x) \quad (10.39)$$

in the affinely transported physical pair blocks  $\mathcal{S}_r^{(N)}$ . Let  $\mathbf{S}_N$  collect those blocks,  $\mathbf{P}_N = (P_0, P_2, \dots, P_{2N})^T$ ,  $\mathbf{P}_N = \mathbf{U}_N \mathbf{S}_N$ , and

$$\mathcal{G}_N = \int_{-1}^1 \mathbf{S}_N \mathbf{S}_N^T dx, \quad H_N^+ = \text{diag}(h_0, h_2, \dots, h_{2N}).$$

Then

$$\boxed{\mathbf{U}_N \mathcal{G}_N \mathbf{U}_N^T = H_N^+, \quad \mathcal{G}_N^{-1} = \mathbf{U}_N^T (H_N^+)^{-1} \mathbf{U}_N,} \quad (10.40)$$

and the even projection kernel has the pair-minimal factorization

$$K_N^+(x, y) = \mathbf{S}_N(x)^T \mathcal{G}_N^{-1} \mathbf{S}_N(y). \quad (10.41)$$

*Proof.* The nonzero minor in [theorem 10.13](#) proves independence. The pairs already lie in the  $(N+1)$ -dimensional space  $\Pi_{2N}^+$ , hence they form a basis. Their endpoint coordinates are integral and the minor is a nonzero integer, so rational targets have rational coordinates.

Integrate  $\mathbf{P}_N = \mathbf{U}_N \mathbf{S}_N$  against its transpose and use Legendre orthogonality to obtain the first identity in (10.40); invert it for the second. Finally substitute the same change of basis into  $K_N^+(x, y) = \mathbf{P}_N(x)^T (H_N^+)^{-1} \mathbf{P}_N(y)$ .  $\square$

*Remark 10.15* (What the central determinant does not prove). The reflected-pair basis spans the even sector at degree  $2N$ . It is not the same assertion as the open central-consecutive-row-block conjecture [12.1](#), which concerns a full local basis in both parities. The two integer determinants must therefore remain distinct.

## 10.7 Smooth convergence, log-square decay, and moment units

**Theorem 10.16** (Superalgebraic convergence in every derivative). *For all integers  $r, L \geq 0$ ,*

$$\boxed{\|u - S_N\|_{C^r[-1,1]} = \mathcal{O}_{r,L}(N^{-L}).} \quad (10.42)$$

*Proof.* Let  $\mathcal{L} = -D(1 - x^2)D$ . Since  $u$  is smooth and flat at both endpoints, repeated integration by parts gives, for every  $s \geq 0$  and  $\ell \geq 1$ ,

$$\int_{-1}^1 u P_\ell = [\ell(\ell + 1)]^{-s} \int_{-1}^1 (\mathcal{L}^s u) P_\ell.$$

The last integral is bounded uniformly in  $\ell$ , so the Legendre coefficient  $a_n$  is  $\mathcal{O}_s(n^{1-2s})$ . For fixed  $r$ ,  $\|P_\ell^{(r)}\|_\infty = \mathcal{O}_r(\ell^{2r})$ . Sum the differentiated tail and choose  $s$  larger than  $L + r + 2$ .  $\square$

The primary Fourier theory gives, for  $|\xi| \geq 1$ ,

$$|\widehat{u}(\xi)| \leq C_0 \exp\left(-\frac{(\log |\xi|)^2}{2 \log 2}\right) |\xi|^{-\kappa_\infty}, \quad \kappa_\infty = \frac{3}{2} + \frac{\log(\pi/\sqrt{3})}{\log 2}. \quad (10.43)$$

This is equation `fourier-two-sided-envelope` in `Fabius_Function_and_Rvachev_Up.tex`; the exact Lean bound is `norm_rvachevFourier_le_decayGauge_abs` in `RvachevFourierDecay.lean`.

**Theorem 10.17** (Log-square decay inherited from the sinc product). *There exist constants  $C > 0$  and  $n_0 \in \mathbb{N}_0$  such that, for every  $n \geq n_0$ ,*

$$|a_n| \leq C \frac{n^\sigma}{\log n} \exp\left(-\frac{(\log n)^2}{2 \log 2}\right), \quad \sigma = 2 - \kappa_\infty + \log_2(2\pi). \quad (10.44)$$

Hence

$$\liminf_{\substack{n \rightarrow \infty \\ a_n \neq 0}} \frac{-\log |a_n|}{(\log n)^2} \geq \frac{1}{2 \log 2}. \quad (10.45)$$

*Proof.* Use (9.29) at  $m = 2n, c = 0$  and split the integral at  $X_n = n/(2\pi)$ . The power series for  $J_\nu$ , followed by Stirling, gives constants  $C_1 > 0$  and  $q < 1$  such that  $|j_{2n}(t)| \leq C_1 q^{2n}$  for  $|t| \leq n$ . Thus the inner integral is  $\mathcal{O}(n^2 q^{2n})$ . On the complement use  $|j_{2n}| \leq 1$  and (10.43). With  $a = \log 2$  and  $\kappa > 1$ , convexity of  $t^2/(2a) + (\kappa - 1)t$  after the substitution  $t = \log x$  yields

$$\int_X^\infty \exp\left(-\frac{(\log x)^2}{2a}\right) x^{-\kappa} dx \leq \frac{aX^{1-\kappa}}{\log X + (\kappa - 1)a} \exp\left(-\frac{(\log X)^2}{2a}\right).$$

Set  $X = X_n$ . Expanding  $\log X_n = \log n - \log(2\pi)$  produces the exponent  $\sigma$  and the denominator  $\log n$  in (10.44); the exponentially small inner contribution is absorbed. Taking logarithms gives (10.45).  $\square$

**Theorem 10.18** (Every even up-moment is a two-adic unit). *For every  $m \geq 0$ ,*

$$\boxed{\nu_2(\mu_{2m}) = 0, \quad \mu_{2m} \equiv 1 \pmod{2}} \quad (10.46)$$

inside  $\mathbb{Q}_2$ .

*Proof.* The distributional fixed point for  $u$  gives

$$(4^m - 1)\mu_{2m} = \sum_{r=1}^m \binom{2m}{2r} \frac{\mu_{2m-2r}}{2r+1}. \quad (10.47)$$

Induct on  $m$ . The initial moment is  $\mu_0 = 1$ . Every denominator  $2r + 1$  and the factor  $4^m - 1$  are odd, and the inductive moments are all one modulo two. The right side of (10.47) is therefore congruent to

$$\sum_{r=1}^m \binom{2m}{2r} = 2^{2m-1} - 1 \equiv 1 \pmod{2}.$$

Division by the odd unit  $4^m - 1$  preserves the congruence.  $\square$

### 10.8 Exterior anchors and one-sided ladder instability

Let  $\rho_0 = 2 + \sqrt{3}$ , the exterior growth base of Legendre polynomials at  $-2$ , and let  $A_{2N,N}$  denote the raw top-degree amplitude obtained by expanding the partial Legendre sum in the one-sided Appell ladder  $G_n(x+2)$ .

**Theorem 10.19** (Exterior divergence and exact top-amplitude growth). *The Legendre coefficients satisfy*

$$\limsup_{N \rightarrow \infty} |a_N|^{1/N} = 1. \quad (10.48)$$

Consequently the terms  $a_N P_{2N}(-2)$  are unbounded along a subsequence. The top raw amplitude is

$$A_{2N,N} = (2N)! 2^{\binom{2N+1}{2}} a_N 2^{-2N} \binom{4N}{2N}, \quad (10.49)$$

and

$$\limsup_{\substack{N \rightarrow \infty \\ a_N \neq 0}} \frac{\log |A_{2N,N}|}{N^2} = 2 \log 2. \quad (10.50)$$

*Proof.* Since  $a_N \rightarrow 0$ , the limsup in (10.48) is at most one. If it were below one, the Legendre series would converge locally uniformly in a Bernstein ellipse and analytically continue  $u$  through its flat endpoints. The only analytic function with the zero endpoint Taylor series is locally zero, contradicting  $u(0) = 1$ . This proves (10.48).

Equation (10.48) says that no geometric bound  $|a_N| = \mathcal{O}(\rho^{-2N})$  with  $\rho > 1$  can hold. Fix  $1 < \rho < \rho_0$ . Infinitely many  $N$  have  $|a_N| > \rho^{-2N}$ , while

$$|P_{2N}(-2)| \asymp \frac{\rho_0^{2N}}{\sqrt{N}}.$$

Their products are unbounded.

The deconvolution operator preserves leading coefficients, and  $[x^{2N}]P_{2N} = 2^{-2N} \binom{4N}{2N}$ . Multiplying that coordinate by the ladder amplitude  $D_{2N} = (2N)! 2^{\binom{2N+1}{2}}$  proves (10.49). The coefficient bound  $|a_N| \leq (4N+1)/2$  and (10.48) show that its logarithm contributes  $o(N^2)$  along the limsup. Stirling shows that the factorial and central binomial contribute only  $\mathcal{O}(N \log N)$ . The sole quadratic term is  $\binom{2N+1}{2} \log 2 = (2N^2 + N) \log 2$ , proving (10.50).  $\square$

**Theorem 10.20** (No exponentially weighted normal Appell closure). *Suppose  $\sum_{n \geq 0} |c_n| R^n < \infty$  for some  $R > 4$ . Then*

$$f(z) = \sum_{n \geq 0} c_n G_n(z+2)$$

*converges normally in a complex neighborhood of  $[-1, 1]$  and is analytic there. Consequently it cannot agree with  $u$  on any nondegenerate subinterval of  $[-1, 1]$ .*

*Proof.* If  $Z$  has density  $u$ , then  $G_n(z+2) = \mathbb{E}(z+2-Z)^n$  and  $|Z| \leq 1$ . Choose  $\varepsilon > 0$  with  $4 + \varepsilon < R$ . On  $|z| \leq 1 + \varepsilon$ ,

$$|G_n(z+2)| \leq (|z+2| + 1)^n \leq (4 + \varepsilon)^n.$$

The Weierstrass test gives normal convergence and analyticity. Equality with  $u$  on a nondegenerate subinterval of  $[-1, 1]$  would make  $u$  analytic at its interior points, contradicting its nowhere-analyticity. The exact primary anchor is `eq:nonanalytic-up-support` in `Fabius_Function_and_Rvachev_Up.tex`; the corresponding Lean declaration is `canonical_rvachev_not_analyticAt` in `NowhereAnalytic.lean`.  $\square$

### 10.9 Factorial deconvolution, Lambert inversion, and non-flattening

Put

$$C = M_u(\pi i) = \prod_{j \geq 1} \frac{\sin(\pi/2^j)}{\pi/2^j}. \quad (10.51)$$

As  $m \rightarrow \infty$ , the nearest-pole expansion of  $1/M_u$  gives

$$\gamma_{2m} = \frac{2(-1)^m (2m)!}{C(2\pi)^{2m}} (1 + \mathcal{O}(m4^{-m})). \quad (10.52)$$

The imported estimate is the nearest-pole specialization of the three-shell expansion in `Invers e_and_Sampling_Frontiers.tex`. Its source theorem and equation labels are, respectively, `p2:thm:three-shell` and `p2:eq:alpha-three-shell`.

**Theorem 10.21** (Central deconvolution asymptotic). *As  $n \rightarrow \infty$ ,*

$$Q_{2n}^-(0) = \frac{2(-1)^n}{C} \frac{(4n)!}{(2n)!(4\pi)^{2n}} \left( 1 + \frac{2\pi^2}{4n-1} + \mathcal{O}(n^{-2}) \right). \quad (10.53)$$

*In particular,*

$$Q_{2n}^-(0) = \frac{2\sqrt{2}}{C} (-1)^n \left( \frac{2n}{\pi e} \right)^{2n} (1 + \mathcal{O}(n^{-1})), \quad |Q_{2n}^-(0)|^{1/(2n)} \sim \frac{2n}{\pi e}. \quad (10.54)$$

*Proof.* Write

$$P_{2n}(x) = \sum_{j=0}^n c_{n,j} x^{2n-2j}, \quad c_{n,j} = 2^{-2n} (-1)^j \frac{(4n-2j)!}{j!(2n-j)!(2n-2j)!}.$$

Since the constant term of  $\mathcal{D}_u x^r$  is  $\gamma_r$ ,  $Q_{2n}^-(0) = \sum_j c_{n,j} \gamma_{2n-2j}$ . Let  $T_{n,j} = c_{n,j} \gamma_{2n-2j}$ . The exact first ratio and (10.52) give

$$\frac{T_{n,1}}{T_{n,0}} = \frac{2\pi^2}{4n-1} + \mathcal{O}(4^{-n}).$$

Uniform two-sided factorial bounds from the same pole expansion give, for  $j \geq 2$ ,

$$\left| \frac{T_{n,j}}{T_{n,0}} \right| \leq \frac{C_1}{j!} \left( \frac{2\pi^2}{n} \right)^j.$$

Their sum is  $\mathcal{O}(n^{-2})$ . Now  $c_{n,0} = 2^{-2n} \binom{4n}{2n}$ ; multiplying by (10.52) proves (10.53). Stirling gives (10.54).  $\square$

For  $X > 0$  let

$$n_Q(X) = \min\{n \geq 1 : |Q_{2n}^-(0)| \geq X\}.$$

**Corollary 10.22** (Lambert- $W$  threshold with discrete monotonicity). *As  $X \rightarrow \infty$ ,*

$$n_Q(X) = \frac{\log X}{2W(\log X/(\pi e))} + \mathcal{O}(1). \quad (10.55)$$

*Proof.* The ratio of consecutive absolute values in (10.54) tends to infinity; hence  $|Q_{2n}^-(0)|$  is strictly increasing for all sufficiently large  $n$ . This justifies inversion of the least-integer threshold. Put  $H = \log X$  and  $w = W(H/(\pi e))$ . Then  $n_0 = H/(2w)$  solves  $2n_0 \log(2n_0/(\pi e)) = H$ . The logarithm of the prefactor in (10.54) changes the continuous solution by  $o(1)$  because the derivative of the leading logarithm tends to infinity. Eventual monotonicity and integer rounding contribute only  $\mathcal{O}(1)$ .  $\square$

**Theorem 10.23** (Sharp central growth and failure of atomwise flattening). *For the coefficient of the repeated central atom*

$$c_n^{(0)} = \frac{a_n Q_{2n}^-(0)}{4^n}$$

one has

$$\limsup_{n \rightarrow \infty} \frac{|c_n^{(0)}|^{1/n}}{n^2} = \frac{1}{\pi^2 e^2}. \quad (10.56)$$

Therefore the flattened atom series is neither absolutely nor unconditionally convergent; the exact self-reconstruction must remain grouped into its finite Legendre blocks.

*Proof.* Equation (10.54) gives

$$\frac{|Q_{2n}^-(0)|^{1/n}}{n^2} \rightarrow \frac{4}{\pi^2 e^2}.$$

By (10.48), multiplication by  $|a_n|^{1/n}$  preserves this value along the limsup, while division by  $4^n$  divides it by four, proving (10.56). In particular the individual central terms do not tend to zero. Absolute or unconditional convergence of a scalar series would force every term to vanish, so both forms of flattening fail.  $\square$

### 10.10 Exact root transition at degree twelve

**Proposition 10.24** (First nonreal zeros; exact computer-assisted certificate). *For  $1 \leq n \leq 11$ ,  $Q_n^-$  has exactly  $n$  real roots. The polynomial  $Q_{12}^-$  has exactly eight real roots and therefore four nonreal roots. The exact nonreal-root counts for  $n = 12, \dots, 20$  are*

$$4, 4, 4, 8, 8, 8, 8, 12, 12. \quad (10.57)$$

*Proof.* The retained exact verifier constructs every  $Q_n^-$  over  $\mathbb{Q}$  from the reciprocal moments and applies the Euclidean Sturm-chain algorithm, with no floating-point decisions. For  $Q_{12}^-$  the chain degrees are 12, 11,  $\dots$ , 0, and the leading-coefficient signs are

$$+, +, +, +, +, +, +, +, -, +, +, +, +.$$

At  $+\infty$  this has two sign variations. At  $-\infty$ , multiplying by the degree parities gives

$$+, -, +, -, +, -, +, -, -, -, +, -, +,$$

with ten variations. Sturm's theorem gives  $10 - 2 = 8$  real roots. The same exact sign-variation calculation gives  $n$  real roots for  $n \leq 11$  and the counts in (10.57); the canonical certificate, full table, and verifier are recorded under `assets/evidence/legendre/root-geometry/`, as documented in section B. Approximate root locations are diagnostic only and are not used in the proposition.  $\square$

### 10.11 Consolidated precise conjectures

**Conjecture 10.25** (Exact two-adic valuations). *For every  $m, n \geq 1$ ,*

$$\nu_2(\mu_{2m} - 1) = 3 + \nu_2(m), \quad \nu_2(a_n) = -1.$$

*Exact rational computations verify both assertions through index 80.*

**Conjecture 10.26** (Sharp log-square coefficient decay).

$$\liminf_{\substack{n \rightarrow \infty \\ a_n \neq 0}} \frac{-\log |a_n|}{(\log n)^2} = \frac{1}{2 \log 2}.$$

*This is the precise sharpness question left by theorem 10.17; no unspecified log-periodic correction is included.*



**Conjecture 10.27** (Central determinant arithmetic). *With  $\sigma_N$  from (10.36),*

$$(-1)^N \sigma_N > 0 \quad (N \geq 2), \quad \log \left( \frac{(-1)^N \sigma_N}{2^{2N-3}} \right) = o(N),$$

*and the normalized factor tends to infinity. The sign assertion has exact evidence for  $2 \leq N \leq 128$ ; the exceptional value is  $\sigma_1 = 0$ .*

**Conjecture 10.28** (Endpoint signs and conditioning). *With the even-degree endpoint coordinates from (10.26), for every  $m \geq 1$  and  $0 \leq r \leq 2m + 1$ ,  $(-1)^r \beta_{2m;2m,r} > 0$ . Separately,*

$$\log \kappa_2(\mathcal{G}_N) = \Theta(N^2), \quad \lim_{\substack{d \rightarrow \infty \\ 2|d}} \frac{\log \max_r |\beta_{d;d,r}|}{d^2} = \log 2.$$

*These are statements about two different bases and are not identified.*

**Conjecture 10.29** (Complex zeros). *The number of nonreal zeros of  $Q_n^-$  tends to infinity, and every  $Q_n^-$  has only simple zeros. The population and simplicity assertions are logically separate.*

Equation (10.48) proves that the set of indices with  $a_n \neq 0$  is infinite.

**Conjecture 10.30** (Exterior side lobes and square energy). *Regard the finite train  $B_n$  of (9.8) as a function on all of  $\mathbb{R}$ . Along the nonzero coefficient indices,*

$$\lim_{\substack{n \rightarrow \infty \\ a_n \neq 0}} \frac{\|B_n\|_{L^2(\mathbb{R} \setminus [-1,1])}}{\|B_n\|_{L^2[-1,1]}} = \infty.$$

*Also the exact square energy  $\mathcal{E}_F = \int_0^1 F_{\text{bd}}(x)^2 dx$  is irrational.*

**Problem 10.31** (Make the remaining normalizations mathematical). *Find explicit diagonal entries for the proposed augmented determinant normalization, and an explicit affine normalization for the proposed two-regular spectral-closure sequences. Until those data are specified, phrases such as natural normalization do not define conjectures.*

## 11 Extensions

### 11.1 Odd overscaling

For  $\rho = 2^d q$  with  $q$  odd, reproduction order stays  $d$  (theorem 3.1) while the alias argument yields the larger annihilator  $A_{\rho,d}(z) = \prod_{r=0}^d (z^{\rho/2^r} - 1)/(z - 1)$  of degree  $2\rho - q - d - 1$ , from the coset multiplicities  $1 + \min\{\nu_2(j), d\}$ ; the one-cell dimension is  $\leq q + d + 1$ , conjecturally with equality (conjecture 12.2): each odd factor buys  $q - 1$  extra *defect* directions and no polynomial order.

### 11.2 Tensor products and multivariate compression

On a box in  $\mathbb{R}^s$ , both constructions tensorize: coordinatewise degrees  $(d_1, \dots, d_s)$  cost  $\prod_j (d_j + 2)$  dictionary atoms or  $\prod_j 2(d_j + 1)$  ladder atoms, exactly on the tensor polynomial space. For *total* degree  $d$  the tensor counts far exceed  $\binom{d+s}{s}$ ; a multivariate alias-ideal compression (the analogue of theorem 3.4 for  $\mathbb{Z}^s$ -cosets) is an attractive frontier problem.

### 11.3 Hermite jet stitching and analytic data

The derivative ladder of proposition 5.8 exposes all endpoint jets of the blocks, so finite Hermite systems stitch different polynomial plateaus on disjoint intervals into one compactly supported  $C^\infty$  function with automatic smooth transitions — an atomic-function analogue of splines with infinitely smooth knots. For  $f$  analytic near  $[a, b]$ , truncating the Appell expansion of  $f(c + hy)$  at degree  $d$  converts any polynomial-approximation remainder into a finite-atom approximation bound: the exact theory doubles as a constructive approximation scheme.

### 11.4 Base $b$ and $q$ -ary ladders

Both engines are dyadic only through the zero set of  $\widehat{u}$ . For an integer base  $b \geq 2$ , the compactly supported density with transform  $\prod_{j \geq 0} \text{sinc}_\pi(\xi/b^j)$  supports the same two constructions – dictionary scale  $b^d$ , and ladders weighted by  $b$ -ary restricted partition numbers  $[z^N] \prod_{j < m} (1 - z^{b^j})^{-1}$  whenever the derivative refinement has an integral translation stencil – with a generalized Prouhet digit character as certificate. The dyadic case is the most arithmetic because the reciprocal stencil is the Thue–Morse block.

### 11.5 Positive synthesis and dual functionals

Sign-changing polynomials force signed amplitudes, but for strictly positive targets, nonnegative-amplitude synthesis in an overcomplete dictionary is a finite-dimensional cone-membership problem after fixing scale and interval ([research question 12.8](#)). Compactly supported dual functionals  $\lambda_n$  with  $\lambda_n(\Psi_m) = \delta_{nm}$  – ideally finite dyadic samples or moments of  $u$  – would upgrade the block basis to a local projection and quadrature calculus ([problem 12.9](#)).

## 12 Conjectures, problems, and the Lean program

Besides [conjecture 7.2](#) and [research question 4.4](#):

**Conjecture 12.1** (Central consecutive basis). *The consecutive central block  $\{\psi_0, \dots, \psi_{d+1}\}|_{[0,1]}$  is a basis of  $V_d$  for every  $d$ ; equivalently the integer determinant  $\Delta_d^{\text{cen}}$  of its reduction to the endpoint basis is nonzero. Exact values through  $d = 12$  support this:*

$$1, 1, 3, -1, 255, -6145, 524287, -62914561, 17716740095, -9620726743041, \\ 11047892835893247, -25076042725198921729, 113484369541461161541631,$$

*a sequence with visibly rich divisibility (note  $255 = 2^8 - 1$ ,  $524287 = 2^{19} - 1$ ) inviting a product, resultant, or cyclotomic interpretation ([research question 12.7](#)). Nonvanishing is an algebraic claim only; some central coordinates are worse conditioned than endpoint ones.*

**Conjecture 12.2** (Odd overscaling dimension). *For  $\rho = 2^d q$ ,  $q$  odd, the one-cell local space has dimension exactly  $q + d + 1$ .*

**Conjecture 12.3** (Bit growth). *The lowest-terms numerators and denominators of the endpoint coefficients of  $t^d$  have bit length  $\mathcal{O}(d^2)$ . (Plausible since  $\log A_d(1) = \Theta(d^2)$  and all operator series are Bernoulli-rational; cancellation makes a proof nontrivial.)*

**Problem 12.4** (Make support–atom Pareto optimality precise). *For fixed  $d$  and collar ratio  $\rho/L \rightarrow 0$ , any causal positive-scale construction with uniformly controlled block condition numbers needs  $\Omega_d(L/\rho)$  atoms; the bound of [theorem 5.10](#) is optimal up to a degree-dependent constant. Problem: specify the admissible causal construction class, the block norm and condition-number bound, and then prove or refute this proposed lower bound.*

**Problem 12.5** (Define and optimize the ladder scale). *Natural coefficient and evaluation condition functionals of the ladder have a finite minimizer in  $h > 0$ , unique up to a short list of adjacent  $K$ -regimes. Problem: choose an explicit operator norm and target normalization, define the treatment of the discontinuous truncation index  $K$ , and determine the minimizers for that precise functional.*

**Research question 12.6** (Condition-optimal  $(d+2)$ -basis). *Among all  $(d+2)$ -subsets of the active dyadic dictionary, which basis minimizes a chosen stability functional ( $L^\infty$  coefficient norm over unit-norm targets, smallest sampled singular value, Lebesgue-type constant, support overhang)?*

**Research question 12.7** (Central determinant sequence). *Find a recurrence, product formula, or resultant interpretation for  $\Delta_d^{\text{cen}}$  and explain its Mersenne-type prime content.*

**Research question 12.8** (Positive synthesis). *Characterize the polynomials representable with nonnegative amplitudes in an overcomplete up-dictionary.*

**Problem 12.9** (Dual functionals). *Construct compactly supported duals of the block (or endpoint) basis from finite dyadic samples or moments of  $u$ , and derive the induced local projection and quadrature formulas.*

**Problem 12.10** (Weight arithmetic). *Expand  $p_m(N)$  and the normalized monic weights in bases adapted to Gaussian  $q$ -binomial coefficients and Prouhet cancellation; make the lower-order quasipolynomial coefficients explicit as Bernoulli polynomials on residue classes modulo  $2^{m-1}$ , and connect them to Mahler-type functional equations.*

**Conjecture 12.11** (Lattice minimality and atom complexity). *(i) There is a length threshold  $L_d$  such that any exact one-scale lattice representation of a degree- $d$  polynomial on an interval of length  $\geq L_d h$ , with no atom identically zero there, forces  $\nu_2(m) \geq d$ : long plateaux cannot beat the stationary binary order by boundary engineering. (ii) For intervals comparable to the atom radius and bounded support enlargement, every one-scale exact representation of all of  $\Pi_d$  needs  $\Omega(2^d)$  atoms in the worst case, making the canonical  $2^{d+1} + \mathcal{O}(1)$  boundary count order-optimal. (The fixed-dictionary count of [proposition 3.6](#) bounds dimension; these bound lattice atom numbers.)*

**Problem 12.12** (Specify stable radii, ghost limits, and local Chebyshev data). *For fixed  $p$  and interval, the coefficient norm of the canonical lattice representation has a finite minimizing radius  $\rho_* > 0$ , governed by the nearest complex critical scale of  $1/M_u(\rho z)$  (the lattice sibling of [problem 12.5](#)). The  $\ell^2$ -optimal ghost-antiderivative iterates converge, after matching scale and lattice, to the canonical Appell samples. And for generic ordered centers on a sufficiently short interval avoiding support endpoints, finite equal-scale up-translate families form an extended Chebyshev system. Problem: define the coefficient norm and critical-scale rule, the scale-matching topology for ghost iterates, and the quantitative genericity and interval-length hypotheses before treating these three proposals as conjectures.*

**Research question 12.13** (Sharp odd-denominator law). *Every reduced cumulant  $\kappa_{2r}$  has odd denominator ([section 6](#)); determine the exact 2-adic valuation of the reduced  $A_r^{(m)}(u)$  at dyadic  $u$  (conjecturally a function only of the dyadic denominator of  $u$  and the binary digit sums of the Bell partitions of  $r$ , all irregular growth confined to odd primes dividing  $2^{2j} - 1$  and Bernoulli numerators).*

**Research question 12.14** (Endpoint asymptotics of the Appell table). *In the boundary scaling  $u = 1 + \delta_d$  with  $\delta_d 2^d/d$  bounded away from 0 and  $\infty$ , the dominant saddle  $z_d$  of  $\mathcal{A}_d(u) = [z^d]d! e^{uz}/M_u(z)$  should obey  $d = \delta_d z_d + \log_2 z_d + \Theta(1)$ , and the full expansion should carry a nonconstant one-periodic lower-Lambert phase — the polynomial-synthesis face of the corpus's Fabius endpoint machinery (fixed- $u$  coefficients are instead governed by the nearest imaginary poles of  $1/M_u$ ).*

**Research question 12.15** ( $q$ -geometric stencils). *Is there a nonstationary atomic family with geometrically varying scales whose exact derivative/reproduction stencils have Gaussian binomial coefficients  $(-1)^r q^{\binom{r}{2}} \begin{bmatrix} d \\ r \end{bmatrix}_q$ , interacting at  $q = 2$  with the Rvachev refinement equation and degenerating at  $q \rightarrow 1$  to ordinary finite differences? (This subsumes the base- $b$  question of [section 11](#) on the stencil side.)*

### Lean formalization roadmap

In dependency order, all finite or formal-power-series layers first: (1) the telescoping identity of [proposition 5.1](#) and the binary-partition induction ([5.1](#)); (2) support bookkeeping and the truncation theorem [theorem 5.4](#); (3) finite-dimensional Appell inversion ([theorem 5.6](#)); (4) polynomial-weighted Poisson summation and the global synthesis [theorem 3.3](#) (the analytic core, using the already formalized integer zero orders); (5) root-of-unity comb derivatives and the alias multiplicities of [theorem 3.4](#); (6) the  $q$ -integer factorization ([2.5](#)); (7) the dimension argument of [theorem 3.5](#) via the existing non-elementarity theorem; (8) recurrence elimination, the Thue–Morse certificate, and the formal cancellation proof of the  $H_d$  symbol ([3.12](#)); (9) twisted Poisson summation ([6.1](#)) and the general-ratio reproduction ([6.3](#)); (10) the parent–child identity, finite prefix-sum primitive, and the Thue–Morse stencil ([6.7](#)) (pure finite algebra plus support reasoning). Steps (1)–(3), (5)–(8), and (10) are finite algebra; only (4) and (9) need Schwartz-class Poisson summation.

### Compiled Poisson and sharpness updates

*Update (28 Aug 2026):* the analytic engine of (4) and (9) is now largely in the library — `Fabius.tsum_shifted_monomial_eq_integral_real` (`MonomialCombExactness.lean`) kernel-verifies the dual (quadrature) face  $\sum_k (\theta + k)^p \text{up}(2^{-m}(\theta + k)) = \int x^p \text{up}(2^{-m}x) dx$  for every real shift and  $p \leq m$ , via Schwartz-class Poisson summation with the nonzero aliases killed by the integer-zero-order theorem (`MonomialCombAlias.lean`); what remains for (4)/(9) is the Appell-weighted synthesis direction over the same machinery. *Update (30 Aug 2026):* `CompositeMeshSharpness.lean` closes the natural-mesh universal exactness threshold itself: one definition and seven theorems give the real first defect, the equivalent valuation and divisibility classifications, canonical membership, the lower bound, and both `IsLeast` forms. This closure is for the complete polynomial space through the cutoff only; target-specific synthesis, local dictionaries, and atom counts remain outside it.

## A Low-order operator data

Truncations of the fast symbol (acting on degrees shown):

$$H_0 = 1, \quad H_1 = 1 + \mathcal{O}(z^2), \quad H_2 = 1 - \frac{5}{36}z^2 + \mathcal{O}(z^4),$$

$$H_3 = 1 - \frac{13}{72}z^2 + \mathcal{O}(z^4), \quad H_4 = 1 - \frac{2}{9}z^2 + \frac{857}{32400}z^4 + \mathcal{O}(z^6);$$

low annihilator coefficients  $(a_0, \dots, a_{d+1})$ :  $(1, 1, 0)$ ,  $(1, 2, 2, 2)$ ,  $(1, 3, 5, 7, 8)$ ,  $(1, 4, 9, 16, 24, 32)$ ,  $(1, 5, 14, 30, 54, 86, 126)$  for  $d = 1, \dots, 5$ ; and the formula sheet of the ladder (all objects rational):

|                |                                                                                     |
|----------------|-------------------------------------------------------------------------------------|
| cumulants      | $\kappa_{2m} = B_{2m}/(2m(1 - 4^{-m}))$ , odd zero                                  |
| reciprocal MGF | $\gamma_0 = 1, \quad \gamma_n = -\sum_r \binom{n-1}{r-1} \kappa_r \gamma_{n-r}$     |
| blocks         | $\Psi_{n,h} = n! 2^{n(n+1)/2} \sum_{N \leq K} p_{n+1}(N) u((x - C_{n,N})/2^{n+1}h)$ |
| centers/scales | $C_{n,N} = a + h(2^{n+1} - 2 + 2N)$ , $2^{n+1}h$ , $K = \lceil (b-a)/2h \rceil$     |
| coordinates    | $b_n = \sum_r \gamma_r h^{n+r} p^{(n+r)}(a-h)/(n! r!)$                              |
| amplitudes     | $\alpha_{n,N} = n! 2^{n(n+1)/2} b_n p_{n+1}(N)$                                     |
| support        | $[a - 2h, a + h(2^{d+2} - 2 + 2K)]$                                                 |

## B Provenance of the absorbed sources

| Source (directory; SHA-256 of main .tex)                                                                                                                             | Contribution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Exact Polynomial Synthesis by Finite Rvachev Up-Function Dictionaries</i> (Rvachev_Up_Polynomial_Representation_Package; a574b30dadbc. . .)                       | The common-scale construction of <a href="#">sections 3 and 4</a> : scale classification, Poisson collapse, deconvolution synthesis, alias nullspace and $q$ -integer annihilator, $\dim V_d = d+2$ , endpoint basis and fixed-dictionary minimality, output-sensitive $H_d$ algorithm, multi-cell theorem, Thue–Morse certificate, canonical defect and Fourier residue, central-basis determinants, odd overscaling, monomial examples and exact data files.                                                                                                                                                                                                                                                                                                             |
| <i>Exact Polynomial Windows from Finite Sums of Shifted and Scaled Rvachev up Functions</i> (rvachev_up_polynomial_representation; 98fe2e5fd256. . .)                | The ladder construction of <a href="#">section 5</a> : antiderivative train, binary-partition weights and Thue–Morse Green pair, quasipolynomial structure, tail polynomials, finite-window and two-atom theorems, Appell coefficient transform with Bernoulli–Bell closed forms, Wronskian/Chebyshev structure, extension operator, collar-atom tradeoff, anticausal mixing, conditioning analysis, worked quartic with figures, classical Rvachev bibliography.                                                                                                                                                                                                                                                                                                          |
| <i>Exact Polynomial Plateaux from Rvachev’s Up-Function</i> (seventh wave; Rvachev_Up_Exact_Polynomial_Representation_Report; SHA in assets/SHA256SUMS-absorbed.txt) | The oversampled lattice of <a href="#">section 6</a> : twisted Poisson identity, general-ratio reproduction with sharp order $\nu_2(m)$ and the general first-defect series (6.4), probabilistic deconvolution characterization, the physical-scale interval algorithm with truncation indices, support/atom-count bounds, jet matching, compressed $\mathcal{O}(d)$ description and $(2m+1)$ -locality; the ghost-atom antiderivative calculus and the Thue–Morse derivative stencil (6.7); finite sinc-prefix generators with the exact cumulant truncation law; generator-independent theorem; odd-cumulant-denominator arithmetic; worked degree-four example with exact-zero verification; the lattice-side conjectures and questions of <a href="#">section 12</a> . |

**Lagrange and Legendre source packages.** The ten later reports contribute 176 labeled proved results. Equivalent polynomial-synthesis, right-inverse, projector, and finite-loop statements are represented once by the stronger canonical forms in [sections 8 and 9](#); the distinct operator, arithmetic, asymptotic, parity, and conditioning results are retained in [section 10](#). Hashes below are SHA-256 over the repository-normalized main source bytes at the consolidation baseline. The directory names are historical identities at pre-retirement commit 443793e846934e7363e314ea01129b9f50197a58, not current navigation.

| Historical source directory, main-source SHA-256                                                        | Distinct contribution retained here                                                                                                                             |
|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rvachev_lagrange_loop_report<br>5d17373e36f6b12576a9add8c46fde57ae2e59b2b934<br>10930fe64fb39f6dd647    | Base cardinal factorization, exact interpolation loop, and nested self-expansion.                                                                               |
| Lagrange_Rvachev_Loop_Package<br>978aeadec8087c5c44881ccc22b87fa2c3c5579aa539<br>a462621c5418d066e4fb   | Minimal defect-completed inverse, nodal ghost, root-rate obstruction, and endpoint amplitude barrier.                                                           |
| lagrange_rvachev_loop_report_v3<br>07587a69794e72f18af9075eae545f0cd5ae1694f318<br>b961d287713a088a1465 | Basis-independent matrix factorization, $q$ -geometric row lift, Legendre deconvolution, weighted kernel geometry, factorial asymptotic, and Lambert threshold. |
| Lagrange_Rvachev_Closed_Loop_Report                                                                     |                                                                                                                                                                 |



| Historical source directory, main-source SHA-256                                                               | Distinct contribution retained here                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 727ccd7491844ffc4f2d0a82f351e052e83f50acbfcb<br>76f7d12aad0059f95099                                           | Appell–Vandermonde inversion, sample-to-atom volume bound, the parity-adapted four-atom ladder, prime support, exterior-anchor instability, and normal-convergence obstruction.                                  |
| Rvachev_Lagrange_Loop_Report_v5<br>118a09539feafbccf24e353e6e6c0c2e647236fb7782<br>2eadba721adc4c38808d        | Markov encoder/signed decoder, variation and Lebesgue lower bounds, alias/sample exact sequence, and coefficient projector flag.                                                                                 |
| Rvachev_Lagrange_Loop_Report_v6<br>e11a2c460ae2b5f33b80217a681dfe874398b8f22c8<br>edcd8d351b5eaeac0b2          | Undilated fixed-radius cardinal synthesis, sharp mesh interpretation, and interlevel dyadic null rows. The source arrived without its advertised scripts or machine-readable data; that limitation is preserved. |
| legendre_rvachev_closed_loop<br>bebd04fe47a308328d717028876e78f93c89d6d33238<br>e54690c16d32bf6a2db            | Translated analysis kernel, exact biorthogonality, reverse spectral closure, flatness and derivative-cell Thue–Morse identities, log-square coefficient bound, and two-adic moment units.                        |
| Legendre_Rvachev_Closed_Loop_Report_v3<br>8bbd922a51dbc6a132892c43c226d57fda220f0b4155<br>0c2d27cd90f97ed324d2 | Pullback geometry and Favard obstruction, operator Christoffel–Darboux identity, parity split, central $q$ -determinant and Gram basis, and the exact degree-twelve Sturm transition.                            |
| Legendre_Rvachev_Closed_Loop_Report_v4<br>7eff3989000e82f72f4b9e9761f75b807ec95bf369f4<br>6b67d96abce207594274 | Moment-smoothed Legendre/Appell matrices and universal determinant, $X_+$ transmutation, Legendre moments completed by a Thue–Morse row, and the compressed $2N + 2$ -atom one-scale closure.                    |
| Legendre_Rvachev_Self_Reconstruction<br>23180ac9e6ed8bfc9c9d0e7e01d97890b2414768fef9<br>82d7a1563083fdb8376    | Orthogonal self-reconstruction blocks and rank-one detail projectors, rational and nested dyadic-Fabius square energies, refinement null trains, exact lifting, derivative energies, and central non-flattening. |

The canonical `assets/` directory retains the historical hashes of the three earlier absorbed `.tex` sources in its `SHA256SUMS-absorbed.txt` ledger. After the ten later reports were completely absorbed, their individual prose/PDF directories were retired. Their 113 selected evidence payloads now have one-to-one canonical destinations and live hashes under `assets/companion-evidence/`, `assets/evidence/`, and `assets/provenance/COMPANION_PAYLOADS.csv`. The ten hashes above, historical source snapshots, companion-asset roles, and the exact computer-assisted certificate boundary are mirrored in the canonical provenance files under `assets/provenance/`. Historical arrival ledgers are not rewritten to masquerade as live checksums.

**Exact replay of the degree-twelve root transition.** The compact certificate, the complete table through degree twenty, and the focused exact verifier used in [proposition 10.24](#) live under `assets/evidence/legendre/root-geometry/`. The verifier reconstructs the rational polynomials and their Sturm chains rather than trusting the text summary; approximate root locations and plots are diagnostic only.

*What was deduplicated:* the corpus background (normalization, probability model, MGF/cumulants, Fourier zeros), the Appell calculus (the third exact-polynomial source’s up-Appell family  $A_r^{(m)}(u) = m^{r-1}\mathcal{A}_r(u/m)$  is the same deconvolution family in a scaled normalization – verified on  $P_2, P_4$ ), the Thue–Morse products, the sharp-order statement (that source’s exact stationary order  $\nu_2(m)$  is [theorem 3.1](#)), rationality and affine covariance, tensor products (that source adds per-coordinate ratios), extension-operator remarks, conditioning warnings, and the Lean roadmaps. *What is new to the consolidation:* the unified Appell-pair and Green-pair framing of [section 2](#), the comparison table and the sharpened bound  $N_d \leq d + 2$  of [theorem 7.1](#), and the cross-volume identities of [section 4.3](#) – the defect–quadrature duality (4.4) with its exact special values (4.6), cross-validated against the companion volume’s independently computed phase-error table. Equivalent source theorems are represented once by their strongest canonical form; distinct proved results are retained, while genuinely open assertions are separated into explicitly labeled conjectures and problems. The first two exact-polynomial sources

audited the repository corpus (67-path inventory) and agreed that the explicit finite constructions were absent from it; that corpus-relative novelty standard is inherited here without being promoted to a claim about the later ten packages.

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